

# A Physical Introduction to Higher Mathematics

Paul Gustafson

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## Part I

# Symmetry

## 1 Superposition

**Definition 1.1** (Vector spaces and linear maps). Let  $k$  be a field. A  $k$ -vector space is an abelian group  $V$  with an action  $k \times V \rightarrow V$  satisfying distributivity, associativity, and  $1v = v$ . A map  $T : V \rightarrow W$  is *linear* if  $T(av + bw) = aT(v) + bT(w)$ .

**Definition 1.2** (Subspaces, kernels, images, and quotients). A *subspace* of  $V$  is a subset closed under linear combinations. For a linear map  $T : V \rightarrow W$ , define

$$\ker T = \{v : T(v) = 0\}, \quad \text{im } T = \{T(v) : v \in V\}.$$

For a subspace  $U \subseteq V$ , the *quotient*  $V/U$  is the vector space of cosets  $v + U$ . The *direct sum*  $V \oplus W$  is  $V \times W$  with componentwise operations.

**Definition 1.3** (Duals and pairings). The *dual* of  $V$  is  $V^* = \text{Hom}_k(V, k)$ . A *pairing* of  $V$  and  $W$  is a bilinear map  $V \times W \rightarrow k$ . It is *nondegenerate* if each nonzero argument pairs nontrivially with some argument in the other space.

**Definition 1.4** (Tensor products). A *tensor product* of  $V$  and  $W$  is a vector space  $V \otimes W$  with a bilinear map  $(v, w) \mapsto v \otimes w$  such that every bilinear map  $b : V \times W \rightarrow X$  factors uniquely through a linear map  $\tilde{b} : V \otimes W \rightarrow X$ . Tensor powers and multilinear maps are defined by iteration. Such a space exists: let  $k^{(V \times W)}$  be the free vector space on symbols  $[v, w]$ , and quotient by the span of

$$\begin{aligned} [v + v', w] - [v, w] - [v', w], & \quad [v, w + w'] - [v, w] - [v, w'], \\ [av, w] - a[v, w], & \quad [v, aw] - a[v, w] \end{aligned}$$

for  $a \in k$ . The class of  $[v, w]$  is  $v \otimes w$ , and sending  $[v, w]$  to  $b(v, w)$  proves the universal property. It also shows that any two tensor products are uniquely isomorphic in a way preserving pure tensors.

**Definition 1.5** (Finite-dimensional duality, trace, and dimension). The *evaluation* map is

$$\text{ev}_V : V^* \otimes V \longrightarrow k, \quad \varphi \otimes v \longmapsto \varphi(v).$$

The vector space  $V$  is *finite-dimensional* if there is a coevaluation map  $\text{coev}_V : k \rightarrow V \otimes V^*$  satisfying the snake identities

$$\begin{aligned} (\text{id}_V \otimes \text{ev}_V) \circ (\text{coev}_V \otimes \text{id}_V) &= \text{id}_V, \\ (\text{ev}_V \otimes \text{id}_{V^*}) \circ (\text{id}_{V^*} \otimes \text{coev}_V) &= \text{id}_{V^*}. \end{aligned}$$

Let  $s_{V, V^*} : V \otimes V^* \rightarrow V^* \otimes V$  exchange the tensor factors. For  $T \in \text{End}(V)$ , define

$$\text{tr}(T) = \text{ev}_V \circ s_{V, V^*} \circ (T \otimes \text{id}_{V^*}) \circ \text{coev}_V \in k, \quad \dim V = \text{tr}(\text{id}_V).$$

**Definition 1.6** (Exterior powers, determinant, and characteristic polynomial). The *tensor algebra* and *exterior algebra* of  $V$  are

$$T(V) = \bigoplus_{r \geq 0} V^{\otimes r}, \quad \Lambda V = T(V) / \langle v \otimes v : v \in V \rangle;$$

write  $\Lambda^r V$  for the degree- $r$  component. For finite-dimensional  $V$ , define its *rank* to be the unique integer  $n \geq 0$  such that  $\Lambda^n V \neq 0$  and  $\Lambda^{n+1} V = 0$ ; the exercise below proves that this integer exists and that  $\Lambda^n V$  is one-dimensional. For  $T \in \text{End}(V)$ , the induced map  $\Lambda^n T$  is multiplication by a scalar, the *determinant*  $\det(T)$ . The *characteristic polynomial* is

$$\chi_T(t) = \det(t \text{id}_V - T).$$

Its root multiplicity at  $\lambda$  is the *algebraic multiplicity* of  $\lambda$ ; the dimension of  $\ker(T - \lambda \text{id}_V)$  is its *geometric multiplicity*.

**Exercise 1.1** (Top exterior powers and determinant identities). Using only the universal property of alternating maps and the duality maps of  $V$ , prove that the rank  $n$  exists, that  $\Lambda^n V$  is one-dimensional, and that  $\Lambda^r V = 0$  for  $r > n$ . Prove

$$\det(ST) = \det(S) \det(T), \quad \det(T \oplus R) = \det(T) \det(R).$$

Relate the scalar dimension to rank by  $\dim V = (\text{rank } V)1_k$  and explain why the scalar dimension alone need not distinguish ranks in positive characteristic. For  $V = \mathbb{R}^n$ , identify  $|\det T|$  with the volume-scaling factor and the sign with orientation. Deduce the change-of-variables rule for parallelepipeds and Cramer's rule.

**Definition 1.7** (Inner products and orthogonality). An *inner product* on a real vector space is a positive-definite symmetric bilinear form. On a complex vector space it is a positive-definite sesquilinear form, conjugate-linear in the first variable. Write  $\|v\| = \sqrt{\langle v, v \rangle}$ . Vectors are *orthogonal* when their inner product is zero, and  $U^\perp = \{v : \langle u, v \rangle = 0 \text{ for every } u \in U\}$ .

**Definition 1.8** (Projections and adjoints). A *projection* is an endomorphism  $P$  with  $P^2 = P$ . An *orthogonal projection* is a projection with  $\ker P = (\text{im } P)^\perp$ . The *adjoint* of  $T : V \rightarrow W$  is a map  $T^* : W \rightarrow V$  satisfying  $\langle Tv, w \rangle = \langle v, T^*w \rangle$ .

**Exercise 1.2** (Orthogonal decomposition and best approximation). Let  $V$  be finite-dimensional and  $U \subseteq V$ . Prove  $V = U \oplus U^\perp$ , construct the orthogonal projection  $P_U$ , and prove that  $P_U v$  is the unique element of  $U$  minimizing  $\|v - u\|$ . Prove that  $P$  is an orthogonal projection if and only if  $P = P^* = P^2$ .

**Definition 1.9** (Isometries and spectral classes). A linear map is *orthogonal* over  $\mathbb{R}$ , or *unitary* over  $\mathbb{C}$ , if it preserves the inner product. An endomorphism  $A$  is *self-adjoint* if  $A = A^*$ , *normal* if  $AA^* = A^*A$ , and *positive* if  $A = A^*$  and  $\langle v, Av \rangle \geq 0$  for every  $v$ . A scalar  $\lambda$  is an *eigenvalue* if  $\ker(A - \lambda \text{id}) \neq 0$ ; a subspace  $U$  is *invariant* if  $A(U) \subseteq U$ . In finite dimensions,  $\text{spec}(A)$  denotes the set of eigenvalues of  $A$  in the scalar field.

**Definition 1.10** (Complete commuting spectral projections). A family  $(P_\lambda)$  of commuting orthogonal projections is *complete and mutually orthogonal* if

$$P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda, \quad \sum_\lambda P_\lambda = \text{id}.$$

A *spectral decomposition* of  $A$  is such a family satisfying

$$A = \sum_{\lambda \in \text{spec}(A)} \lambda P_\lambda.$$

**Exercise 1.3** (Finite-dimensional spectral theorem). Prove that adjoints exist uniquely. Prove that a complex endomorphism is normal if and only if it admits a complete family of mutually orthogonal commuting spectral projections. Deduce the reality of self-adjoint spectra, the unit modulus of unitary spectra, and nonnegative spectra for positive operators. Prove that every positive operator  $A$  has a unique positive square root  $A^{1/2}$  by applying the square-root function to its spectral projections. For a map  $T : V \rightarrow W$  between finite-dimensional inner-product spaces, let  $(Q_s)$  be the complete spectral projections of  $|T| = (T^*T)^{1/2}$ . For each  $s > 0$ , construct an isometric isomorphism  $U_s : Q_s V \rightarrow T(Q_s V)$  satisfying  $TQ_s = sU_s Q_s$ , and obtain the coordinate-free singular-value decomposition

$$T = \sum_{s>0} sU_s Q_s.$$

## 2 Fourier Series

**Definition 2.1** (Linear oscillator). A *linear oscillator* on a finite-dimensional real inner-product space  $V$  is an equation

$$M\ddot{q} + Kq = 0,$$

where  $M$  is positive definite and  $K$  is self-adjoint and positive semidefinite. A *normal mode* is a nonzero  $v \in V$  for which  $q(t) = a(t)v$  solves the equation for some nonconstant  $a$ .

**Exercise 2.1** (Coupled oscillators and normal modes). Set  $x = M^{1/2}q$  and  $L = M^{-1/2}KM^{-1/2}$ . Use the spectral projections of  $L$  to derive the normal-mode decomposition. For a fixed-end chain of  $N$  equal masses joined by equal springs, compute the mode shapes and frequencies and relate them to the modes of a taut string.

**Definition 2.2** (Vibrating string). For length  $L$ , tension  $\tau$ , and linear density  $\rho$ , a *vibrating string* is a function  $u : [0, L] \times \mathbb{R} \rightarrow \mathbb{R}$  satisfying fixed-end conditions  $u(0, t) = u(L, t) = 0$  and

$$\rho \partial_t^2 u = \tau \partial_x^2 u.$$

**Exercise 2.2** (String modes and measured pitch). Derive the vibrating-string equation from transverse force balance in the small-slope limit. Determine all separated normal modes and their frequencies. Prove that the fundamental frequency is  $f_1 = (2L)^{-1} \sqrt{\tau/\rho}$  and compare the predicted ratios  $f_n/f_1 = n$  with the harmonics of a stretched string.

**Definition 2.3** (Orthogonal expansions). Let  $H$  be an inner-product space. For an orthogonal family  $(e_n)$ , the *orthogonal coefficients* of  $f$  are  $\hat{f}(n) = \langle e_n, f \rangle / \|e_n\|^2$ . A *Fourier series* on  $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$  is the formal expansion  $f \sim \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{inx}$ , where  $\hat{f}(n) = (2\pi)^{-1} \int_0^{2\pi} f(x) e^{-inx} dx$ .

**Exercise 2.3** (Completeness of Fourier modes). Construct the Fejér kernels, prove that they form an approximate identity, and use convolution with them to prove completeness of  $(e^{inx})_{n \in \mathbb{Z}}$  in  $L^2(\mathbb{T})$ . Deduce Parseval's identity and  $L^2$  convergence of Fourier series.

## 3 Hilbert Space

### 3.1 Measure, integration, and Hilbert space

**Definition 3.1** (Measure spaces and simple functions). A *measure space* is a triple  $(X, \Sigma, \mu)$  in which  $\Sigma$  is a  $\sigma$ -algebra and  $\mu : \Sigma \rightarrow [0, \infty]$  is countably additive on disjoint families, with  $\mu(\emptyset) = 0$ .

A function is *measurable* when inverse images of Borel sets are in  $\Sigma$ . A nonnegative *simple function* is a finite sum  $\sum_j a_j 1_{E_j}$  with  $a_j \geq 0$  and  $E_j \in \Sigma$ .

**Definition 3.2** ( $L^p$  spaces). For  $1 \leq p < \infty$ , the space  $L^p(X, \mu)$  consists of measurable functions modulo equality almost everywhere with

$$\|f\|_p = \left( \int_X |f|^p d\mu \right)^{1/p} < \infty;$$

$L^\infty$  uses the essential supremum. For Lebesgue measure on  $\mathbb{R}^n$  and normalized arc measure on  $\mathbb{T}$ , write  $L^p(\mathbb{R}^n)$  and  $L^p(\mathbb{T})$ . The inner product on  $L^2(\mathbb{T})$  is  $\langle f, g \rangle = (2\pi)^{-1} \int_0^{2\pi} \overline{f(x)} g(x) dx$ . The exercises below establish the analytic properties used in this section.

**Exercise 3.1** (Integration and monotone convergence). Define the integral of a nonnegative simple function and then of an arbitrary nonnegative measurable function by monotone approximation. Prove monotone convergence and Fatou's lemma.

**Exercise 3.2** (Dominated convergence and product measures). Construct the integral of an integrable complex function from its positive and negative real and imaginary parts. Prove dominated convergence. Construct product measure for  $\sigma$ -finite spaces and prove Tonelli's and Fubini's theorems.

**Definition 3.3** (Hilbert spaces and bounded operators). From this point onward, Hilbert spaces are complex. A *Hilbert space* is a complete inner-product space. A linear map  $T : H \rightarrow K$  is *bounded* if  $\|T\| = \sup_{\|x\| \leq 1} \|Tx\| < \infty$ . Write  $B(H, K)$  for the space of bounded maps and  $B(H) = B(H, H)$ . For  $T \in B(H)$ , the *resolvent set* consists of  $\lambda \in \mathbb{C}$  for which  $T - \lambda \text{id}$  has a bounded inverse; its complement is the *spectrum*  $\text{spec}(T)$ .

**Exercise 3.3** (Hilbert-space foundations). Prove the projection theorem for closed subspaces and the Riesz representation theorem. Deduce  $H = M \oplus M^\perp$  for closed  $M$  and prove existence, uniqueness, and  $\|T^*\| = \|T\|$  for bounded adjoints.

### 3.2 Spectral measures and quantum evolution

**Definition 3.4** (Projection-valued measures). A *projection-valued measure* on a measurable space  $X$  assigns an orthogonal projection  $E(B)$  to each measurable set, with  $E(X) = \text{id}$  and countable additivity in the strong operator topology. For bounded measurable  $f$ , the operator  $\int_X f dE$  is defined by uniform approximation from simple functions.

**Exercise 3.4** (Bounded spectral theorem). Prove that every bounded self-adjoint operator  $A$  has a unique projection-valued measure on  $\text{spec}(A) \subset \mathbb{R}$  such that  $A = \int \lambda dE(\lambda)$ . Extend the statement to bounded normal operators, derive continuous functional calculus, and characterize unitary and projection spectra.

**Definition 3.5** (Unbounded operators and domains). A *densely defined operator*  $T$  has a dense linear domain  $\text{Dom}(T) \subset H$ . It is *closed* when its graph is closed in  $H \oplus H$ ; its graph norm is  $\|x\|_T^2 = \|x\|^2 + \|Tx\|^2$ . The domain of  $T^*$  consists of those  $y$  for which  $x \mapsto \langle Tx, y \rangle$  is bounded in the Hilbert norm on  $\text{Dom}(T)$ ; for such  $y$ , define  $T^*y$  by  $\langle Tx, y \rangle = \langle x, T^*y \rangle$  for all  $x \in \text{Dom}(T)$ . Density of the domain makes  $T^*y$  unique. The operator is *symmetric* when  $\langle Tx, y \rangle = \langle x, Ty \rangle$  on its domain and *self-adjoint* when  $T = T^*$ , including equality of domains.

**Definition 3.6** (Weak derivatives and first Sobolev spaces). A function  $g \in L^1_{\text{loc}}(0, L)$  is the *weak derivative* of  $f \in L^1_{\text{loc}}(0, L)$  when

$$\int_0^L f \phi' dx = - \int_0^L g \phi dx \quad (\phi \in C_c^\infty(0, L)).$$

The space  $H^1(0, L)$  consists of the  $f \in L^2(0, L)$  having weak derivative  $f' \in L^2(0, L)$ , with norm  $\|f\|_{H^1}^2 = \|f\|_2^2 + \|f'\|_2^2$ .

**Exercise 3.5** (Self-adjoint momentum and boundary holonomy). On  $L^2([0, L])$ , let  $P_\theta = -i\hbar d/dx$  on the absolutely continuous functions with derivative in  $L^2$  and boundary condition  $\psi(L) = e^{i\theta}\psi(0)$ . Use integration by parts and the adjoint domain to prove that  $P_\theta$  is self-adjoint. Determine its mutually orthogonal spectral projections  $(E_n)_{n \in \mathbb{Z}}$ , prove that they sum strongly to the identity, and show that their spectral values are

$$p_n = \frac{\hbar}{L}(2\pi n + \theta), \quad n \in \mathbb{Z}.$$

Interpret  $\theta$  as a boundary holonomy, later realized by magnetic flux through a ring, and explain why changing the domain changes the measured spectrum.

**Exercise 3.6** (Cayley transform and spectral measure). Use the Cayley transform and the bounded spectral theorem to prove that every self-adjoint  $H$  has a unique projection-valued measure with

$$H = \int_{\mathbb{R}} \lambda dE_H(\lambda), \quad \text{Dom}(H) = \left\{ \psi : \int_{\mathbb{R}} \lambda^2 d\langle \psi, E_H(\lambda) \psi \rangle < \infty \right\}.$$

**Exercise 3.7** (Stone's theorem). Prove that  $U_t = e^{-itH/\hbar}$  is a strongly continuous unitary group for every self-adjoint  $H$ . Conversely, recover a unique self-adjoint generator  $H$  from every strongly continuous unitary group and identify its domain by the strong derivative at  $t = 0$ .

**Exercise 3.8** (Fourier spectra and free quantum evolution). Use the Fourier transform to realize  $-i\hbar\partial_j$  and  $-\Delta$  as multiplication by  $\hbar\xi_j$  and  $|\xi|^2$  on their natural domains. Deduce self-adjointness, identify their spectral projections, and derive the free Schrödinger evolution for  $H = -(\hbar^2/2m)\Delta$ . Prove preservation of the domain, norm, and expected energy, and use this concrete multiplication-operator model to verify the preceding spectral and Stone theorems, including their domain statements. Identify the continuous energy spectrum and relate it to the continuum of outgoing-electron energies above an ionization threshold.

**Exercise 3.9** (Spectral lines from unitary evolution). If  $H|n\rangle = E_n|n\rangle$ , use  $U_t = e^{-itH/\hbar}$  to show that  $\langle A(t) \rangle$  contains frequencies  $(E_m - E_n)/\hbar$ . Explain why spectroscopic peaks measure energy differences and why line strengths depend on matrix elements between the corresponding spectral subspaces. Convert the Balmer wavelengths 656.3, 486.1, 434.0, and 410.2 nm to energy differences using  $E = hc/\lambda$ . Identify the levels  $E_n$ , rather than their differences, with atoms of the spectral measure of  $H$ , and identify the transition frequencies with spectral values of the Liouvillian  $A \mapsto [H, A]/\hbar$ .

## 4 Harmonic Analysis

### 4.1 Euclidean Fourier analysis

**Definition 4.1** (Schwartz functions, Fourier transform, and convolution). A *Schwartz function* on  $\mathbb{R}^n$  is a smooth function  $f$  such that  $\sup_x |x^\alpha \partial^\beta f(x)| < \infty$  for every pair of multi-indices  $\alpha, \beta$ . The



Schwartz space is denoted  $\mathcal{S}(\mathbb{R}^n)$ . For  $f, g \in C_c(\mathbb{R}^n)$ , define the *Fourier transform* and *convolution* by

$$\mathcal{F}f(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx, \quad (f * g)(x) = \int_{\mathbb{R}^n} f(y) g(x - y) dy.$$

The estimates  $\|\mathcal{F}f\|_\infty \leq (2\pi)^{-n/2} \|f\|_1$  and  $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$  extend these operations to the  $L^1$  completion.

**Exercise 4.1** (Plancherel theorem). Prove on  $\mathcal{S}(\mathbb{R}^n)$  that  $\mathcal{F}$  preserves the  $L^2$  inner product. Prove that  $\mathcal{S}(\mathbb{R}^n)$  is dense in  $L^2(\mathbb{R}^n)$  and extend  $\mathcal{F}$  uniquely to a unitary operator and deduce Plancherel's identity.

**Exercise 4.2** (Wave and heat propagators). Use the Fourier transform to solve the Cauchy problems for

$$\partial_t^2 u = c^2 \Delta u, \quad \partial_t u = \kappa \Delta u$$

on  $\mathbb{R}^n$ . Identify their spectral-mode evolution, derive the Gaussian heat kernel and its convolution law, and prove finite propagation speed for the one-dimensional wave equation.

## 4.2 Lattices, Poisson summation, and diffraction

**Definition 4.2** (Lattices and reciprocal lattices). A *lattice* in  $\mathbb{R}^n$  is a subgroup  $\Gamma = A\mathbb{Z}^n$  for some invertible linear map  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ . Its covolume is  $\text{vol}(\Gamma) = |\det A|$ , and a fundamental cell is  $F = A[0, 1)^n$ . Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi \cdot \gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

**Definition 4.3** (Periodic summation and structure factors). For  $f \in \mathcal{S}(\mathbb{R}^n)$ , its  $\Gamma$ -*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma \in \Gamma} f(x + \gamma).$$

For a finite motif  $B \subset \mathbb{R}^n$  with scattering weights  $c_b$ , its *structure factor* is

$$S_B(q) = \sum_{b \in B} c_b e^{-iq \cdot b}.$$

**Exercise 4.3** (Poisson summation). Prove that  $P_\Gamma f$  converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\text{vol}(\Gamma)} \sum_{\xi \in \Gamma^*} \mathcal{F}f(\xi) e^{i\xi \cdot x}.$$

Set  $x = 0$  to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

**Exercise 4.4** (Crystal diffraction and Bragg peaks). Let  $\Gamma_N$  be a finite rectangular portion of  $\Gamma$ . In the single-scattering approximation, prove that the amplitude at momentum transfer  $q$  factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma \in \Gamma_N} e^{-iq \cdot \gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at  $q \in \Gamma^*$  as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude  $2\pi/\lambda$ , derive the Laue condition and, for planes spaced by  $d$ , the Bragg relation

$$2d \sin \theta = m\lambda.$$

Show how zeros of  $S_B$  produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

### 4.3 Groups, representations, and abelian duality

**Definition 4.4** (Groups, homomorphisms, and quotients). A *group* is a set  $G$  with an associative multiplication, an identity, and inverses. A *homomorphism*  $\phi : G \rightarrow H$  preserves multiplication. A subgroup  $N \leq G$  is *normal* if  $gNg^{-1} = N$  for every  $g \in G$ ; the *quotient group*  $G/N$  consists of cosets with induced multiplication.

**Definition 4.5** (Actions, orbits, and stabilizers). A *left action* of  $G$  on  $X$  is a homomorphism  $G \rightarrow \text{Bij}(X)$ . The *orbit* and *stabilizer* of  $x \in X$  are  $Gx = \{gx : g \in G\}$  and  $G_x = \{g : gx = x\}$ .

**Definition 4.6** (Representations and intertwiners). A *representation* of  $G$  on  $V$  is a homomorphism  $\rho : G \rightarrow \text{GL}(V)$ . An *intertwiner*  $T : (V, \rho) \rightarrow (W, \sigma)$  satisfies  $T\rho(g) = \sigma(g)T$ . A subspace is *invariant* if it is preserved by every  $\rho(g)$ ; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions  $\rho \oplus \sigma$  and  $g \mapsto \rho(g) \otimes \sigma(g)$ .

**Exercise 4.5** (Maschke's theorem). Let  $G$  be finite and let the characteristic of  $k$  not divide  $|G|$ . Average an arbitrary projection over  $G$  to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

**Exercise 4.6** (Schur's lemma). Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

**Definition 4.7** (Characters). The *character* of a finite-dimensional representation  $V$  is the class function  $\chi_V(g) = \text{Tr}(\rho_V(g))$ . For class functions on a finite group, define

$$\langle f, h \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} h(g).$$

**Exercise 4.7** (Character orthogonality and decomposition). Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that  $\langle \chi_U, \chi_V \rangle_G = \dim \text{Hom}_G(U, V)$ . Deduce the multiplicity formula and the character table constraints.

**Exercise 4.8** (Molecular symmetry). For the equilibrium geometry of  $\text{NH}_3$ , determine the  $C_{3v}$ -action on nuclear displacements, remove translations and rotations, and prove

$$V_{\text{vib}} \cong 2A_1 \oplus 2E.$$

Deduce four fundamental frequencies, two of them doubly degenerate. For  $\text{NH}_2\text{D}$ , restrict to  $C_s$  and prove  $E|_{C_s} \cong A' \oplus A''$ . Deduce that symmetry no longer enforces either degeneracy and compare with the observed band splitting.

**Definition 4.8** (Lie groups and Lie algebras). A *Lie group* is a smooth manifold  $G$  whose multiplication and inversion are smooth. Its *Lie algebra* is  $\mathfrak{g} = T_e G$  with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism  $\mathbb{R} \rightarrow G$ . The *exponential map*  $\exp : \mathfrak{g} \rightarrow G$  sends  $X$  to the value at 1 of the one-parameter subgroup with initial velocity  $X$ .

**Exercise 4.9** (One-parameter subgroups and the exponential map). Prove existence and uniqueness of the one-parameter subgroup generated by  $X \in \mathfrak{g}$ . For a closed matrix group, prove that it is  $t \mapsto e^{tX}$ , identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

**Definition 4.9** (Infinitesimal representations). For a finite-dimensional smooth representation  $\rho : G \rightarrow \mathrm{GL}(V)$ , its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \rightarrow \mathrm{End}(V), \quad d\rho(X) = \left. \frac{d}{dt} \right|_0 \rho(\exp tX).$$

**Definition 4.10** (Haar measure and convolution). Let  $G$  be a locally compact Hausdorff group. A *left Haar measure* is a nonzero regular Borel measure  $\mu$  satisfying  $\mu(gE) = \mu(E)$ . On a compact group, a Haar measure may be normalized by  $\mu(G) = 1$ . For  $f, g \in C_c(G)$ , define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) d\mu(y).$$

The *left regular representation* on  $L^2(G)$  is  $(\lambda_g f)(x) = f(g^{-1}x)$ .

**Exercise 4.10** (Haar measure and the regular representation). Identify Haar measure on  $\mathbb{R}^n$ ,  $\mathbb{Z}^n$ , and  $\mathbb{T}^n$ . Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the  $L^1$  convolution inequality, and  $\|f * g\|_2 \leq \|f\|_1 \|g\|_2$ . Prove that  $g \mapsto \lambda_g$  is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over  $G$  to obtain an invariant one.

**Definition 4.11** (Pontryagin dual). For a locally compact abelian group  $G$ , its *Pontryagin dual*  $\widehat{G}$  is the group of continuous characters  $\chi : G \rightarrow U(1)$ , with pointwise multiplication and the compact-open topology. For  $f \in L^1(G)$ , define

$$\widehat{f}(\chi) = \int_G f(x) \overline{\chi(x)} d\mu(x).$$

The *double-dual map* sends  $x \in G$  to the character  $\chi \mapsto \chi(x)$  of  $\widehat{G}$ .

**Exercise 4.11** (Abelian Fourier duality). Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \quad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \quad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. Show that the group Fourier transform recovers Fourier transforms on  $\mathbb{R}^n$  and Fourier series on  $\mathbb{T}^n$ . For a closed subgroup  $H \leq G$ , define its annihilator  $H^\perp \subseteq \widehat{G}$  and prove  $\widehat{\widehat{G}/H} \cong H^\perp$ .

**Exercise 4.12** (Interference and Fourier optics). For coherent scalar waves with complex amplitudes  $a_1, a_2$ , derive  $|a_1 + a_2|^2$  and the fringe spacing in Young’s double-slit geometry. In the Fraunhofer regime, derive that the far-field amplitude is the Fourier transform of the aperture. Compute the single-slit diffraction intensity, locate its zeros, and compare the predicted angular spacing with measured optical diffraction.

## 4.4 Compact groups, rotations, and spin

**Definition 4.12** (Rotations and spin). Define

$$\begin{aligned} SO(3) &= \{R \in \text{End}(\mathbb{R}^3) : R^*R = \text{id}, \det R = 1\}, \\ SU(2) &= \{U \in \text{End}(\mathbb{C}^2) : U^*U = \text{id}, \det U = 1\}. \end{aligned}$$

An *angular-momentum representation* is a unitary representation of  $SU(2)$ ; with Planck constant  $\hbar$ , its infinitesimal generators  $J_1, J_2, J_3$  are normalized by  $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$ .

**Exercise 4.13** ( $SU(2)$  double-covers  $SO(3)$ ). Let  $SU(2)$  act by conjugation on the real space of traceless Hermitian  $2 \times 2$  operators. Prove that the resulting homomorphism  $SU(2) \rightarrow SO(3)$  is surjective with kernel  $\{\pm \text{id}\}$ . Deduce the effect of a  $2\pi$  rotation in integer- and half-integer-spin representations and the  $4\pi$  restoration observed by neutron interferometry.

**Exercise 4.14** (Irreducible representations of  $SU(2)$ ). Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by  $j \in \{0, \frac{1}{2}, 1, \dots\}$ , has dimension  $2j + 1$ , and has  $J_3$ -eigenvalues  $m\hbar$  for  $m = -j, -j + 1, \dots, j$ . Compute the spectrum of  $J^2 = J_1^2 + J_2^2 + J_3^2$ . For  $j = \frac{1}{2}$ , relate the two  $J_3$  outcomes to the two Stern–Gerlach beams.

**Exercise 4.15** (Clebsch–Gordan decomposition). Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of  $V_{1/2} \otimes V_{1/2}$ .

**Definition 4.13** (Quantum symmetries). Let a compact group  $G$  act continuously and unitarily by  $U : G \rightarrow U(H)$  on a Hilbert space with self-adjoint Hamiltonian  $\mathcal{H}$ . It is a *symmetry group* when  $U_g \text{Dom}(\mathcal{H}) = \text{Dom}(\mathcal{H})$  and  $\mathcal{H}U_g\psi = U_g\mathcal{H}\psi$  for every  $g \in G$ . A *quantum number* is an eigenvalue of a self-adjoint operator whose spectral projections commute with those of  $\mathcal{H}$ . A bounded operator multiplet  $A$  has *symmetry type*  $W$  when its components define an intertwiner from  $W$  to  $B(H)$  under conjugation by  $G$ .

**Exercise 4.16** (Symmetry, degeneracy, and selection rules). Prove that every finite-dimensional eigenspace of  $\mathcal{H}$  is symmetry-invariant and decomposes into irreducible  $G$ -representations. Deduce symmetry-forced degeneracies. Prove that a transition matrix element  $\langle f, Ai \rangle$  can be nonzero only if the trivial representation occurs in  $V_f^* \otimes W \otimes V_i$ , and derive the angular-momentum rules for an electric-dipole operator.

**Exercise 4.17** (Atomic spectra). For the Coulomb Hamiltonian, separate the Schrödinger equation in spherical coordinates, obtain the quantum numbers  $n, \ell, m$ , and derive

$$E_n = -\frac{m_e e^4}{2(4\pi\varepsilon_0)^2 \hbar^2 n^2}.$$

Recover the Balmer series and compare its predicted energy differences with the empirical values computed in Exercise 3.9. Apply the electric-dipole rule to identify forbidden transitions.

**Definition 4.14** (Compact-group Fourier transform). For a compact group  $G$ , let  $\widehat{G}$  denote the equivalence classes of irreducible finite-dimensional unitary representations. For  $f \in L^1(G)$  and  $\pi \in \widehat{G}$ , define

$$\widehat{f}(\pi) = \int_G f(g) \pi(g)^* dg \in \text{End}(V_\pi).$$

Thus the Fourier transform is scalar-valued for abelian  $G$  and matrix-valued in general.

**Exercise 4.18** (Peter–Weyl theorem). Use Haar averaging and Schur’s lemma to prove orthogonality of matrix coefficients of irreducible unitary representations of a compact group. Prove that their span is dense in  $C(G)$  and  $L^2(G)$  and obtain

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}} V_\pi \otimes V_\pi^*}.$$

Derive Fourier inversion, the Plancherel formula

$$\|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim V_\pi) \|\widehat{f}(\pi)\|_{\text{HS}}^2,$$

and  $\widehat{f * g}(\pi) = \widehat{g}(\pi) \widehat{f}(\pi)$ . Recover ordinary Fourier series for  $G = \mathbb{T}$ . For  $G = SO(3)$ , identify the matrix coefficients with the Wigner functions  $D_{mn}^\ell$ .

**Exercise 4.19** (Spherical harmonics and atomic orbitals). Identify  $S^2$  with  $SO(3)/SO(2)$  and  $L^2(S^2)$  with the right  $SO(2)$ -invariant subspace of  $L^2(SO(3))$ . Use Peter–Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty} V_\ell}$$

and identify a basis of  $V_\ell$  with  $Y_{\ell m} \propto D_{m0}^\ell$ . Derive  $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1) Y_{\ell m}$  and completeness of the spherical harmonics. Using the Coulomb separation from Exercise 4.17, identify its angular factor and recover  $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$ . Determine the angular nodes of the  $s$ ,  $p$ , and  $d$  orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare with Matsui et al., *Physical Review B* **72** (2005).

**Exercise 4.20** (Molecular rotational spectra). Model a rigid molecule with orientation  $R \in SO(3)$  and principal moments  $I_1, I_2, I_3$ . On  $L^2(SO(3))$ , derive

$$H_{\text{rot}} = \frac{J_1^2}{2I_1} + \frac{J_2^2}{2I_2} + \frac{J_3^2}{2I_3}.$$

Use the Peter–Weyl basis to reduce  $H_{\text{rot}}$  to finite-dimensional blocks. For a symmetric top, derive

$$E_{JK} = \frac{\hbar^2 J(J+1)}{2I_\perp} + \frac{\hbar^2 K^2}{2} \left( \frac{1}{I_\parallel} - \frac{1}{I_\perp} \right).$$

Derive the electric-dipole selection rules and the splitting in a static electric field. Compare the predicted components with microwave measurements of the symmetric-top molecule s-trioxane reported in NBS Special Publication 343 (1971).

## 4.5 Semisimple and noncommutative Plancherel theory

**Definition 4.15** (Cartan and Iwasawa decompositions). Let  $G$  be a connected real semisimple Lie group with finite center and let  $\mathfrak{g}$  be its Lie algebra. A *Cartan involution* gives a vector space decomposition

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$$

into its  $+1$  and  $-1$  eigenspaces; the subgroup  $K$  with Lie algebra  $\mathfrak{k}$  is maximal compact, and  $G/K$  is a noncompact Riemannian symmetric space. Choose a maximal abelian subspace  $\mathfrak{a} \subseteq \mathfrak{p}$ , put  $A = \exp \mathfrak{a}$ , choose positive restricted roots, and let  $N$  be the subgroup generated by their root spaces. The *Iwasawa decomposition* is the diffeomorphism

$$K \times A \times N \longrightarrow G, \quad (k, a, n) \longmapsto kan.$$

If  $\mathfrak{a}^+$  is a positive Weyl chamber, the *Cartan decomposition* is  $G = K\overline{A^+}K$ . Write  $M = Z_K(A)$  and  $W = N_K(A)/M$ , and let  $\rho$  be half the sum of the positive restricted roots, counted with multiplicity.

**Definition 4.16** (Spherical principal series and spherical transform). For  $\lambda \in \mathfrak{a}^*$ , the normalized *spherical principal series* is

$$\pi_\lambda = \text{Ind}_{MAN}^G(1_M \otimes e^{i\lambda} \otimes 1_N),$$

where normalized induction incorporates the modular factor determined by  $\rho$ . It contains a unit  $K$ -fixed vector  $v_\lambda$ , and its *zonal spherical function* is

$$\varphi_\lambda(g) = \langle v_\lambda, \pi_\lambda(g)v_\lambda \rangle.$$

It is  $K$ -bi-invariant, satisfies  $\varphi_\lambda(e) = 1$ , and is a joint eigenfunction of the  $G$ -invariant differential operators on  $G/K$ .

For  $f \in C_c^\infty(K \backslash G/K)$ , its *spherical transform* is

$$\tilde{f}(\lambda) = \int_G f(g) \varphi_{-\lambda}(g) dg.$$

Harish-Chandra's  $c$ -function is the coefficient governing the leading asymptotics of  $\varphi_\lambda(\exp H)$  as  $H \rightarrow \infty$  in a positive chamber.

**Exercise 4.21** (Harish-Chandra transform on  $\mathbb{H}^2$ ). For the sign convention in which the Laplace–Beltrami operator is

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\text{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta \varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \quad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

For a smooth compactly supported radial function, define

$$\tilde{f}(\lambda) = 2\pi \int_0^\infty f(r) \varphi_\lambda(r) \sinh r \, dr.$$

Prove directly from the differential equation that distinct real values of  $\lambda^2$  give orthogonal generalized eigenfunctions after truncation to  $[0, R]$  with a common self-adjoint boundary condition. Compare the large- $r$  equation with a constant-coefficient wave equation and show that its oscillation parameter is continuous. Contrast this with the discrete Dirichlet spectrum in the finite hyperbolic-drum exercise below.

**Exercise 4.22** (A finite hyperbolic drum). Discretize the Laplace–Beltrami operator by a real symmetric weighted graph Laplacian on a finite regular hyperbolic tiling with Dirichlet boundary. Prove that the finite operator has a discrete orthogonal eigenbasis, and explain why those eigenvalues approximate a bounded domain with a boundary rather than the continuous spectrum of all  $\mathbb{H}^2$ . Compare the predicted mode ordering and propagation along discrete geodesics with the circuit measurements in Lenggenger et al., *Nature Communications* **13** (2022). State which observations support the effective hyperbolic metric and which do not by themselves establish the infinite-space Plancherel theorem.

**Definition 4.17** (Harish-Chandra characters and discrete series). Let  $\pi$  be an irreducible admissible unitary representation of a connected real reductive group. Although the infinite-dimensional operator  $\pi(g)$  usually has no trace, for  $f \in C_c^\infty(G)$  the integrated operator

$$\pi(f) = \int_G f(g) \pi(g) dg$$

is trace class. The *Harish-Chandra character* is the conjugation-invariant distribution

$$\Theta_\pi(f) = \text{Tr}(\pi(f)).$$

An irreducible unitary representation belongs to the *discrete series* if it has a nonzero matrix coefficient square-integrable modulo the center.

**Exercise 4.23** (From compact characters to the semisimple Plancherel formula). For a compact group, prove

$$\Theta_\pi(f) = \int_G f(g) \text{Tr}(\pi(g)) dg$$

and recover the ordinary character used in Peter–Weyl theory. Show that a unitary operator on an infinite-dimensional Hilbert space is not trace class, which explains the distributional definition above. Prove directly that  $\Theta_\pi$  is invariant under conjugation.

For  $SL_2(\mathbb{R})$ , identify the  $K$ -types of the spherical principal series and show that they contain a  $K$ -fixed vector. Contrast them with the holomorphic and antiholomorphic discrete series, whose nonzero one-sided  $K$ -types imply that they have no  $K$ -fixed vector. Deduce that spherical analysis on  $G/K$  sees the class-one principal series but not the discrete series. Compare the two families tabulated in Harish-Chandra, *Proceedings of the National Academy of Sciences* **43** (1957) for spherical functions and Harish-Chandra, *Transactions of the American Mathematical Society* **119** (1965) for invariant eigendistributions.

**Definition 4.18** (Noncommutative Plancherel transform). A locally compact group is *unimodular* when its left and right Haar measures agree. It is *type I* when its unitary representations have essentially unique direct-integral decompositions into irreducibles. Let  $G$  be second-countable, unimodular, and type I. For  $f \in L^1(G)$  and an irreducible unitary representation  $\pi : G \rightarrow U(H_\pi)$ , define the operator-valued Fourier transform

$$\widehat{f}(\pi) = \int_G f(g) \pi(g)^* dg.$$

A *Plancherel measure* for  $G$  is a measure  $\nu_G$  on the unitary dual  $\widehat{G}$  such that, initially for  $f \in L^1(G) \cap L^2(G)$ ,

$$\|f\|_2^2 = \int_{\widehat{G}} \|\widehat{f}(\pi)\|_{\text{HS}}^2 d\nu_G(\pi), \quad \widehat{f * g}(\pi) = \widehat{g}(\pi)\widehat{f}(\pi).$$

For abelian groups the representation spaces are one-dimensional; for compact groups the integral becomes the dimension-weighted Peter–Weyl sum.

**Definition 4.19** (Heisenberg group and Schrödinger representations). On the standard symplectic space  $\mathbb{R}^{2n} = \mathbb{R}_q^n \oplus \mathbb{R}_p^n$ , set

$$\sigma((q, p), (q', p')) = p \cdot q' - q \cdot p'.$$

The *Heisenberg group* is  $\mathbb{H}_n = \mathbb{R}^{2n} \times \mathbb{R}$  with multiplication

$$(z, s)(z', s') = (z + z', s + s' + \tfrac{1}{2}\sigma(z, z')).$$

For  $\lambda \neq 0$ , its *Schrödinger representation* on  $L^2(\mathbb{R}^n)$  is

$$(\pi_\lambda(q, p, s)\psi)(x) = e^{i\lambda(s + p \cdot (x - q/2))}\psi(x - q).$$

Here  $\sigma = -\omega_0$  for the standard symplectic-form convention used earlier; this sign is paired with the translation convention in the displayed representation. The physical central character is  $\lambda = 1/\hbar$ .

**Exercise 4.24** (Harmonic analysis of the Heisenberg group). Verify the group law, identify the center, and prove that Lebesgue measure is Haar measure on  $\mathbb{H}_n$ . Prove that each  $\pi_\lambda$  is an irreducible unitary representation and derive its Weyl commutation relations. At  $\lambda = 1/\hbar$ , differentiate the position and momentum subgroups and recover  $Q_j$ ,  $P_j$ , and  $[Q_j, P_k] = i\hbar\delta_{jk}$ .

Compute the operator-valued group Fourier transform in these representations and derive the Plancherel decomposition with measure  $c_n|\lambda|^n d\lambda$ , determining  $c_n$  from the Haar and Fourier normalizations above. Show that the one-dimensional representations with trivial central character have zero Plancherel measure. Derive the distributional orthogonality relation for Weyl operators and the resulting Hilbert–Schmidt Plancherel and inversion formulas. Prove that

$$S \longmapsto \chi_S(u, v) = \text{Tr}\left(Se^{i(u \cdot X + v \cdot Y)}\right)$$

is injective on trace-class operators, where  $[X_j, Y_k] = i\delta_{jk}$ . Interpret this as noncommutative Fourier analysis turning functions on a group into operators rather than scalar frequency amplitudes.

## Part II

# Geometry

## 5 Symplectic Geometry

### 5.1 Manifolds and flows

**Definition 5.1** (Topological spaces). A *topology* on  $X$  is a family of subsets containing  $\emptyset, X$  and closed under arbitrary unions and finite intersections. A map is *continuous* when inverse images of open sets are open. A space is *Hausdorff* when distinct points have disjoint neighborhoods, *second countable* when its topology is generated by a countable family of open sets, and *compact* when every open cover has a finite subcover.



**Definition 5.2** (Topological and smooth manifolds). A *topological manifold* of dimension  $n$  is a Hausdorff, second-countable space locally homeomorphic to  $\mathbb{R}^n$ . A *smooth manifold* is a topological manifold with a maximal atlas whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth.

**Definition 5.3** (Tangent and cotangent spaces). The *tangent space*  $T_p M$  is the vector space of derivations  $v : C^\infty(M) \rightarrow \mathbb{R}$  at  $p$ , satisfying  $v(fg) = f(p)v(g) + g(p)v(f)$ . The *cotangent space* is  $T_p^* M = (T_p M)^*$ . The differential of  $F : M \rightarrow N$  is  $dF_p(v)(f) = v(f \circ F)$ . The disjoint union  $TM = \coprod_{p \in M} T_p M$  has a canonical smooth vector-bundle structure: if  $x = (x^1, \dots, x^n)$  is a coordinate chart on  $U$ , its bundle chart is

$$TM|_U \longrightarrow x(U) \times \mathbb{R}^n, \quad v \in T_p M \longmapsto (x(p), v(x^1), \dots, v(x^n)).$$

The chain rule makes the transition maps smooth and fiberwise linear. The dual bundle charts similarly define the cotangent bundle  $T^*M$ .

**Definition 5.4** (Vector fields and flows). A *vector field* is a smooth section  $X : M \rightarrow TM$ . A *flow* of  $X$  is a smooth local action  $\Phi$  of  $\mathbb{R}$  on  $M$  satisfying  $\left. \frac{d}{dt} \right|_0 \Phi_t(p) = X_p$ . The *Lie bracket* is the vector field  $[X, Y](f) = X(Yf) - Y(Xf)$ .

**Exercise 5.1** (Flows and brackets). Prove local existence and uniqueness of flows. Prove that  $[X, Y] = 0$  if and only if their local flows commute, and that the Lie derivative  $\mathcal{L}_X Y = [X, Y]$  is natural under diffeomorphisms.

## 5.2 Symplectic linear algebra and differential forms

**Definition 5.5** (Symplectic vector spaces and groups). A *symplectic vector space* is a finite-dimensional real vector space  $V$  equipped with a nondegenerate alternating bilinear form  $\omega$ . Its *symplectic group* is

$$\mathrm{Sp}(V, \omega) = \{A \in \mathrm{GL}(V) : \omega(Au, Av) = \omega(u, v)\}.$$

For  $V = \mathbb{R}_q^n \oplus \mathbb{R}_p^n$ , the standard form is  $\omega_0((q, p), (q', p')) = q \cdot p' - p \cdot q'$ .

**Exercise 5.2** (Symplectic bases and transformations). Prove that every symplectic vector space has even dimension and admits a basis in which the form has matrix  $\begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ . Show that  $A$  is symplectic exactly when  $A^* J A = J$ , and compute the Lie algebra of  $\mathrm{Sp}(2n, \mathbb{R})$ . Identify translations and linear symplectic maps as the transformations preserving the affine symplectic geometry of  $\mathbb{R}^{2n}$ .

**Definition 5.6** (Differential forms and pullback). A *differential  $k$ -form* on  $M$  is a smooth section of  $\Lambda^k T^*M$ ; write  $\Omega^k(M) = \Gamma(\Lambda^k T^*M)$ . The *pullback* of  $\omega \in \Omega^k(N)$  by  $F : M \rightarrow N$  is

$$(F^* \omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(dF_p v_1, \dots, dF_p v_k).$$

**Definition 5.7** (Exterior derivative). The *exterior derivative* is the unique family of maps  $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$  satisfying  $df(X) = Xf$ , the graded Leibniz rule,  $d^2 = 0$ , and naturality  $F^* d = dF^*$ .

**Exercise 5.3** (Cartan calculus). Construct  $d$  locally and prove its uniqueness. For contraction  $\iota_X$ , prove Cartan's identity  $\mathcal{L}_X = d\iota_X + \iota_X d$  and the naturality of  $d$ , wedge products, and pullbacks.

### 5.3 Hamiltonian dynamics and phase space

**Definition 5.8** (Symplectic manifolds and dynamics). A *symplectic form* on a manifold  $M$  is a closed, nondegenerate two-form  $\omega$ . The *Hamiltonian vector field* of  $H \in C^\infty(M)$  is defined by  $\iota_{X_H}\omega = dH$ . Its flow is the *Hamiltonian flow*, and the *Poisson bracket* is  $\{f, g\} = \omega(X_f, X_g)$ .

**Definition 5.9** (Configuration and phase spaces). A *configuration space* is a smooth manifold  $Q$ . Its *phase space* is the cotangent bundle  $T^*Q$ . The tautological one-form on  $T^*Q$  is  $\theta_\alpha(v) = \alpha(d\pi(v))$ , and its canonical symplectic form is  $\omega = -d\theta$ .

**Definition 5.10** (Classical states and observables). On a phase space  $M$ , a *pure classical state* is a point  $x \in M$ , and a *classical observable* is a smooth real-valued function  $f \in C^\infty(M, \mathbb{R})$ . A *statistical state* is a Borel probability measure  $\mu$  on  $M$ , with expectation

$$\mathbb{E}_\mu[f] = \int_M f d\mu$$

whenever the integral exists. If  $\phi_t$  is a Hamiltonian flow, Schrödinger evolution pushes states forward by  $\mu_t = (\phi_t)_*\mu_0$ , while Heisenberg evolution pulls observables back by  $f_t = f \circ \phi_t$ ; hence  $\int_M f d\mu_t = \int_M f_t d\mu_0$ .

**Exercise 5.4** (Hamilton's equations and Poisson algebra). Prove that the canonical form on  $T^*Q$  is symplectic. In canonical coordinates, derive Hamilton's equations. Prove antisymmetry, the Leibniz rule, and the Jacobi identity for the Poisson bracket, and prove  $\dot{f} = \{f, H\} + \partial_t f$  along Hamiltonian flow.

### 5.4 Momentum maps, reduction, and conservation laws

**Definition 5.11** (Symplectic actions and momentum maps). An action of a Lie group  $G$  on  $(M, \omega)$  is *symplectic* if it preserves  $\omega$ . A *momentum map* is an equivariant map  $J : M \rightarrow \mathfrak{g}^*$  such that  $d\langle J, \xi \rangle = \iota_{\xi_M}\omega$  for every  $\xi \in \mathfrak{g}$ .

**Definition 5.12** (Symplectic reduction). Let  $J : M \rightarrow \mathfrak{g}^*$  be a momentum map, let  $\xi$  be a regular value, and let  $G_\xi = \{g \in G : \text{Ad}_g^*\xi = \xi\}$ . If  $G_\xi$  acts freely and properly on  $J^{-1}(\xi)$ , the *reduced phase space* is

$$M//_\xi G = J^{-1}(\xi)/G_\xi.$$

Writing  $i : J^{-1}(\xi) \hookrightarrow M$  and  $q : J^{-1}(\xi) \rightarrow M//_\xi G$ , its *reduced symplectic form* is the form  $\omega_\xi$  characterized by  $q^*\omega_\xi = i^*\omega$ .

**Exercise 5.5** (Noether's theorem). Prove that if a Hamiltonian is invariant under a symplectic  $G$ -action with momentum map  $J$ , then every component of  $J$  is conserved. Compute the momentum maps for translations and rotations of a particle and recover linear and angular momentum conservation.

**Exercise 5.6** (Reduction and central-force dynamics). Under the hypotheses of symplectic reduction, prove that  $i^*\omega$  is basic for the  $G_\xi$ -action and that  $\omega_\xi$  exists uniquely and is symplectic. For a  $G$ -invariant Hamiltonian  $H$ , prove that  $X_H$  is tangent to  $J^{-1}(\xi)$ , that  $H$  descends to a Hamiltonian  $H_\xi$ , and that the flow of  $H_\xi$  is the quotient of the flow of  $H$ .

Apply this to the cotangent-lifted rotation action on  $T^*(\mathbb{R}^3 \setminus \{0\})$ , whose momentum map is  $J(q, p) = q \times p$ . For a nonzero angular momentum  $\ell$ , identify  $J^{-1}(\ell)/SO(3)_\ell$  with the radial phase

space having coordinates  $(r, p_r)$ , prove that its symplectic form is  $dr \wedge dp_r$ , and reduce

$$H(q, p) = \frac{\|p\|^2}{2m} + V(\|q\|)$$

to

$$H_\ell(r, p_r) = \frac{p_r^2}{2m} + V(r) + \frac{\|\ell\|^2}{2mr^2}.$$

For  $V(r) = -\kappa/r$ , derive the elliptical bound orbits and the equal-area law.

**Exercise 5.7** (Gauss constraints and transverse modes). Let  $d : V \rightarrow W$  be a linear map between finite-dimensional real inner-product spaces. On  $M = W \oplus W$  set

$$\omega((a, e), (a', e')) = \langle a, e' \rangle - \langle a', e \rangle.$$

Let the additive group  $(\ker d)^\perp$  act by  $\chi \cdot (A, E) = (A + d\chi, E)$ . Prove that this action is free and Hamiltonian with momentum map

$$\langle J(A, E), \chi \rangle = \langle d\chi, E \rangle = \langle \chi, d^*E \rangle.$$

Show that reduction at zero imposes  $d^*E = 0$  and identifies

$$M/\!/_0(\ker d)^\perp \cong (W/\operatorname{im} d) \oplus \ker d^* \cong \ker d^* \oplus \ker d^*.$$

Interpret  $d$  as a finite-dimensional approximation to the exterior derivative:  $d^*E = 0$  is the vacuum Gauss constraint,  $A \sim A + d\chi$  is gauge equivalence, and the reduced variables are the transverse electromagnetic modes.

**Exercise 5.8** (Liouville's theorem). On a  $2n$ -dimensional symplectic manifold, prove that Hamiltonian flow preserves  $\omega$ , the phase-space volume  $\omega^n/n!$ , and every density depending only on  $H$ . Deduce that Hamiltonian evolution cannot compress phase volume.

## 6 Differential Geometry

### 6.1 Bundles, sections, and geometric structures

**Definition 6.1** (Smooth bundles and sections). A *smooth fiber bundle* with fiber  $F$  is a smooth map  $\pi : E \rightarrow M$  locally isomorphic over  $M$  to  $U \times F \rightarrow U$ . A *smooth vector bundle* has vector-space fibers and linear transition maps. A *smooth principal  $G$ -bundle* has a free right  $G$ -action, transitive on each fiber, and local equivariant trivializations. A *section* is a smooth map  $s : M \rightarrow E$  with  $\pi s = \operatorname{id}_M$ . For a representation  $G \rightarrow \operatorname{GL}(V)$ , the *associated vector bundle* is  $P \times_G V = (P \times V)/((pg, v) \sim (p, gv))$ .

**Definition 6.2** (Frame bundles and  $G$ -structures). The *frame bundle* of an  $n$ -manifold is the principal  $\operatorname{GL}(n, \mathbb{R})$ -bundle

$$\operatorname{Fr}(M)_p = \operatorname{Iso}(\mathbb{R}^n, T_p M).$$

For a closed subgroup  $G \leq \operatorname{GL}(n, \mathbb{R})$ , a  $G$ -*structure* is a principal  $G$ -subbundle of  $\operatorname{Fr}(M)$ . It is a choice of the frames related by the transformations in  $G$ . Riemannian, Lorentzian, and almost symplectic structures correspond respectively to reductions to  $O(n)$ ,  $O(1, n-1)$ , and  $\operatorname{Sp}(2n, \mathbb{R})$ ; an oriented, time-oriented spacetime has structure group  $SO^+(1, 3)$ .

**Definition 6.3** (Tensor fields and metrics). A  $(r, s)$ -*tensor field* is a smooth section of  $TM^{\otimes r} \otimes T^*M^{\otimes s}$ . Equivalently, an  $O(p, q)$ -structure is a smooth nondegenerate symmetric tensor  $g$  of signature  $(p, q)$ . The cases  $(n, 0)$  and  $(1, n - 1)$  give Riemannian and Lorentzian metrics. The length of a Riemannian curve is  $\int \sqrt{g(\dot{\gamma}, \dot{\gamma})} dt$ ; the proper time of a timelike curve is  $c^{-1} \int \sqrt{-g(\dot{\gamma}, \dot{\gamma})} dt$ . A metric and an orientation define a volume form  $\text{vol}_g$ .

**Exercise 6.1** (Proper time). Prove invariance of length and proper time under reparametrization. In Minkowski spacetime, prove that the inertial path maximizes proper time among piecewise smooth future-directed timelike paths with fixed timelike-separated endpoints, and derive special-relativistic time dilation. Using the muon rest lifetime  $\tau_\mu \simeq 2.2 \mu\text{s}$ , derive the mean laboratory travel distance  $\gamma v \tau_\mu$  and compare it with the distance  $v \tau_\mu$  predicted without time dilation.

## 6.2 Differential forms, cohomology, and Hodge theory

**Definition 6.4** (Integration of forms). An *orientation* of an  $n$ -manifold is a reduction of its frame bundle to  $\text{GL}^+(n, \mathbb{R})$ , equivalently a continuous choice of component of nonzero elements in  $\Lambda^n T_p^*M$ . The integral  $\int_M \omega$  of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

**Exercise 6.2** (Stokes' theorem). Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form  $B$  and electric one-form  $E$ , prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt} \int_S B \quad \text{and} \quad dE = -\partial_t B,$$

recovering the observed flux–emf law.

**Definition 6.5** (de Rham cohomology). A differential form  $\omega$  is *closed* if  $d\omega = 0$  and *exact* if  $\omega = d\eta$  for some form  $\eta$ . Since  $d^2 = 0$ , the *de Rham cohomology* of  $M$  is

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M))}{\text{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes  $H_{\text{dR}}^*(M)$  a graded-commutative algebra.

**Exercise 6.3** (de Rham classes and periods). On a star-shaped open subset of  $\mathbb{R}^n$ , construct a homotopy operator and prove the Poincaré lemma: every closed form of positive degree is exact. Use Stokes' theorem to prove that an exact  $k$ -form has zero integral over every closed oriented  $k$ -manifold mapped smoothly into  $M$ . Prove  $H_{\text{dR}}^1(S^1) \cong \mathbb{R}$ , with the isomorphism given by integration.

On  $\mathbb{R}^2 \setminus \{0\}$ , show that

$$\alpha = \frac{-y dx + x dy}{x^2 + y^2}$$

is closed and has integral  $2\pi$  around the unit circle, hence is not exact. Compute  $H_{\text{dR}}^1(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$ . Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential  $F = dA$  exactly when  $[F] = 0$  in de Rham cohomology.

**Definition 6.6** (Hodge star). On an oriented pseudo-Riemannian  $n$ -manifold, the metric induces a nondegenerate pairing  $\langle \cdot, \cdot \rangle_g$  on differential forms. The *Hodge star* is the bundle map

$$* : \Omega^p(M) \longrightarrow \Omega^{n-p}(M)$$

characterized by

$$\alpha \wedge * \beta = \langle \alpha, \beta \rangle_g \text{ vol}_g$$

for  $p$ -forms  $\alpha, \beta$ .

**Exercise 6.4** (Metric volume and Hodge star). Construct  $\text{vol}_g$  and the Hodge star intrinsically and derive their coordinate expressions. For a metric with  $q$  negative directions, prove on  $p$ -forms that

$$*^2 = (-1)^{p(n-p)+q} \text{id}.$$

Compute  $*$  on the standard coordinate forms of Euclidean three-space and Minkowski four-space, stating the orientation and signature conventions.

### 6.3 Connections, curvature, and gauge fields

**Definition 6.7** (Connections and parallel transport). A *connection* on a vector bundle  $E \rightarrow M$  is an  $\mathbb{R}$ -bilinear map  $\nabla : \Gamma(TM) \times \Gamma(E) \rightarrow \Gamma(E)$ , linear over smooth functions in the first variable and satisfying  $\nabla_X(fs) = X(f)s + f\nabla_X s$ . Along a curve  $\gamma$  it induces a covariant derivative  $D/dt$ ; its solutions  $Ds/dt = 0$  define *parallel transport* between endpoint fibers. On  $TM$ , the torsion is  $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$ , and a *geodesic* satisfies  $D\dot{\gamma}/dt = 0$ .

**Definition 6.8** (Curvature and holonomy). The *curvature* of  $\nabla$  is the  $\text{End}(E)$ -valued two-form

$$R^\nabla(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

The *holonomy group* at  $p$  consists of the automorphisms of  $E_p$  obtained by parallel transport around loops based at  $p$ .

**Exercise 6.5** (Parallel transport and curvature). Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

**Definition 6.9** (Principal connections and gauge fields). Let  $P \rightarrow M$  be a principal  $G$ -bundle. A *principal connection* is an equivariant  $\mathfrak{g}$ -valued one-form  $A$  on  $P$  reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is  $F_A = dA + \frac{1}{2}[A \wedge A]$ . A *gauge transformation* is a  $G$ -equivariant bundle automorphism covering  $\text{id}_M$ . A reduction of the frame bundle describes spacetime geometry; a separate principal bundle describes an internal gauge symmetry.

**Exercise 6.6** (Gauge covariance and Bianchi identity). Choose local sections  $s_i$  and derive transition maps  $g_{ij} : U_i \cap U_j \rightarrow G$  and the transformation laws for the local potentials  $A_i = s_i^* A$  and curvatures  $F_i = s_i^* F_A$ . Prove  $d_A F_A = 0$  and gauge invariance of holonomy up to conjugacy. For  $G = U(1)$ , recover  $A \mapsto A + d\chi$  and  $F = dA$ . Show that a flat potential is locally pure gauge, and explain why removing it globally can fail on a nonsimply connected base. The global obstruction, holonomy, and measured phase are analyzed in Exercise 12.2.

**Definition 6.10** (Electromagnetic field). An *electromagnetic gauge field* is a connection on an internal principal  $U(1)$ -bundle over an oriented Lorentzian four-manifold. Its field strength is the curvature two-form  $F$ ; locally  $F = dA$ . Given a current one-form  $J$ , Maxwell's equations are

$$dF = 0, \quad d*F = \mu_0 *J.$$

For the SI vector decomposition on Minkowski spacetime, fix

$$(x^0, x^1, x^2, x^3) = (ct, x, y, z), \quad g = -(dx^0)^2 + dx^2 + dy^2 + dz^2,$$

with orientation  $dx^0 \wedge dx \wedge dy \wedge dz$ , and set

$$\begin{aligned} F &= \frac{1}{c} dx^0 \wedge (E_x dx + E_y dy + E_z dz) \\ &\quad - (B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy), \\ J &= c\rho dx^0 - j_x dx - j_y dy - j_z dz. \end{aligned}$$

With these sign conventions, the force law for a particle of charge  $q$  and four-velocity  $u$  is  $m\nabla_u u = q(\iota_u F)^\sharp$ .

**Exercise 6.7** (Maxwell theory and electromagnetic waves). Derive charge conservation from Maxwell's equations. Using the stated Minkowski conventions, recover the electric and magnetic vector equations and verify that the covariant force law reduces to  $q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ . Define the experimentally measured vacuum permittivity  $\varepsilon_0$  by the electrostatic Gauss law  $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ ; comparison with the covariant equation should give  $\mu_0 \varepsilon_0 c^2 = 1$ . In vacuum, impose Lorenz gauge and derive wave equations with speed  $c$ . Prove transverse polarization and compare  $c = (\mu_0 \varepsilon_0)^{-1/2}$  with the measured speed of light.

**Exercise 6.8** (Yang–Mills equations and instantons). For  $G = SU(r)$ , use  $\langle X, Y \rangle = -2 \operatorname{Tr}(XY)$  and define

$$S_{\text{YM}}(A) = \frac{1}{2g^2} \int_M \langle F_A \wedge *F_A \rangle$$

on a closed oriented Riemannian four-manifold. Prove gauge invariance and derive  $d_A^* F_A = 0$ , where  $d_A^* = -*d_A*$ . Decompose  $F_A = F_A^+ + F_A^-$  and, for

$$k(A) = -\frac{1}{8\pi^2} \int_M \operatorname{Tr}(F_A \wedge F_A),$$

prove  $S_{\text{YM}}(A) \geq 8\pi^2 |k(A)|/g^2$ , with equality precisely for  $F_A = \pm *F_A$ . Use the Bianchi identity to show that these instantons solve the Yang–Mills equation.

## 6.4 Riemannian geometry and gravitation

**Definition 6.11** (Levi–Civita geometry). An  $O(p, q)$ -structure determines a unique torsion-free connection preserving the metric, the *Levi–Civita connection*. On the orthonormal frame bundle it is a principal  $O(p, q)$ -connection; the solder form identifies its associated vector bundle with  $TM$ . Its curvature gives the *Riemann tensor*  $\operatorname{Riem}(X, Y, Z, W) = g(R^\nabla(X, Y)Z, W)$ , whose contractions are the *Ricci tensor*  $\operatorname{Ric}$  and scalar curvature  $S$ .

**Exercise 6.9** (Levi–Civita connection and holonomy). Prove existence and uniqueness of the Levi–Civita connection and derive the Koszul formula. Prove that its geodesics are precisely the critical curves of the energy functional with fixed endpoints and that its parallel transport preserves the metric. Compute the rotation obtained by transporting a tangent vector around a latitude on the unit sphere and relate it to the enclosed curvature.

**Exercise 6.10** (Curvature identities and geodesic deviation). For the Levi–Civita connection, prove the algebraic symmetries of the Riemann tensor and the first and second Bianchi identities. For a variation through geodesics with separation field  $J$ , prove

$$\frac{D^2 J}{dt^2} + R^\nabla(J, \dot{\gamma})\dot{\gamma} = 0.$$

Deduce the leading relative acceleration of nearby freely falling bodies.

**Definition 6.12** (Einstein equation). The *Einstein tensor* of a Lorentzian metric is  $G = \text{Ric} - \frac{1}{2}Sg$ . A *stress–energy tensor* is a symmetric two-tensor  $T$  whose contraction with an observer’s velocity gives the observer’s energy–momentum current. The *Einstein equation* with cosmological constant  $\Lambda$  is

$$G + \Lambda g = \frac{8\pi G_N}{c^4} T.$$

**Exercise 6.11** (Gravitational-wave detection). For a quasi-circular binary, derive the leading quadrupole waveform and the chirp relation between frequency, frequency derivative, and chirp mass. For GW150914, use the reported detector-frame component-mass medians  $36M_\odot$  and  $29M_\odot$  to infer the chirp mass. Compare with the model-independent reconstruction  $\mathcal{M} \simeq 30M_\odot$  and with the reported peak strain  $h \simeq 10^{-21}$  near 150 Hz in Abbott et al., *Physical Review D* **93** (2016) and the LIGO GW150914 fact sheet. Limit the conclusion to agreement of the observed waveform with general relativity in that regime.

## Part III

# Probability

## 7 Ergodic Theory

### 7.1 Probability, conditioning, and entropy

**Definition 7.1** (Probability and conditioning). A *probability space* is a measure space  $(\Omega, \mathcal{F}, \mathbb{P})$  with  $\mathbb{P}(\Omega) = 1$ . A *random variable* is a measurable map, and its *expectation* is its integral. For  $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$  and a sub- $\sigma$ -algebra  $\mathcal{G} \subseteq \mathcal{F}$ , a *conditional expectation*  $\mathbb{E}[X \mid \mathcal{G}]$  is a  $\mathcal{G}$ -measurable integrable random variable satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P} \quad (G \in \mathcal{G}).$$

**Definition 7.2** (Shannon entropy). For a discrete probability vector  $p$ , its *Shannon entropy* is

$$H(p) = - \sum_i p_i \log p_i.$$

The corresponding physical entropy is  $S(p) = k_B H(p)$ . For probability vectors  $p, q$  with  $p_i = 0$  whenever  $q_i = 0$ , their *relative entropy* is

$$D(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

For a joint distribution  $p_{XY}$ , the *mutual information* is  $I(X : Y) = D(p_{XY} \| p_X p_Y)$ . For a phase-space probability density  $\rho$  relative to a fixed dimensionless normalization of Liouville measure  $\mu_L$  (equivalently, to a chosen phase-space cell), its *statistical entropy* is  $S(\rho) = -k_B \int \rho \log \rho d\mu_L$ ; for densities  $\rho, \sigma$ , set  $D(\rho \| \sigma) = \int \rho \log(\rho/\sigma) d\mu_L$  when defined. Rescaling the reference measure changes  $S(\rho)$  by an additive constant, whereas  $D(\rho \| \sigma)$  is independent of that normalization.

## 7.2 Time–bandwidth concentration and Gaussian channels

**Definition 7.3** (Time and band limiting). On  $L^2(\mathbb{R})$ , define the time-limiting and band-limiting projections by

$$(Q_T f)(x) = \mathbf{1}_{[-T, T]}(x) f(x), \quad B_W = \mathcal{F}^{-1} \mathbf{1}_{[-W, W]} \mathcal{F},$$

where  $T, W > 0$ . The *time–band concentration operator* is

$$K_{T, W} = Q_T B_W Q_T.$$

After identifying  $Q_T L^2(\mathbb{R})$  with  $L^2([-T, T])$ , its normalized eigenfunctions are the time-limited representatives of the *prolate spheroidal wave functions*, also called *Slepian functions*. The dimensionless parameter  $c = TW$  is the time–bandwidth product.

**Exercise 7.1** (Optimal simultaneous concentration). Use Plancherel’s theorem to prove that  $Q_T$  and  $B_W$  are orthogonal projections. Starting from the Fourier-transform convention in Spectral Theory, derive

$$(K_{T, W} f)(x) = \int_{-T}^T \frac{\sin W(x-y)}{\pi(x-y)} f(y) dy, \quad |x| \leq T,$$

where the diagonal value of the kernel is  $W/\pi$ . Prove that  $K_{T, W}$  is a positive compact self-adjoint contraction. If its positive eigenvalues are  $\lambda_0 \geq \lambda_1 \geq \dots > 0$ , use the Rayleigh–Ritz principle to show

$$\lambda_0 = \sup_{\substack{f \neq 0 \\ \text{supp } f \subseteq [-T, T]}} \frac{\int_{-W}^W |\widehat{f}(\xi)|^2 d\xi}{\int_{\mathbb{R}} |f(x)|^2 dx}.$$

Show likewise that  $B_W Q_T B_W$  has the same nonzero eigenvalues and that, for a band-limited eigenfunction  $g_n$ ,  $\lambda_n$  is the fraction of its energy lying in  $[-T, T]$ . Thus successive eigenfunctions solve the optimal concentration problem subject to orthogonality to the preceding solutions.

**Exercise 7.2** (Slepian modes and Gaussian channel capacity). Let

$$A_{T, W} = B_W Q_T : Q_T L^2(\mathbb{R}) \longrightarrow B_W L^2(\mathbb{R}).$$

Show that  $A_{T, W}^* A_{T, W} = K_{T, W}$ . If  $f_n$  is a normalized  $\lambda_n$ -eigenfunction of  $K_{T, W}$ , prove that  $g_n = \lambda_n^{-1/2} A_{T, W} f_n$  is normalized, that the  $g_n$  are mutually orthogonal, and that

$$A_{T, W} f_n = \sqrt{\lambda_n} g_n.$$

Thus the numbers  $\sqrt{\lambda_n}$  are the singular gains of the time–bandwidth channel.

Restrict to the first  $N$  real modes and add independent Gaussian noise:

$$Y_n = \sqrt{\lambda_n} X_n + Z_n, \quad Z_n \sim \mathcal{N}(0, N_0), \quad 0 \leq n < N.$$



For a real random vector with density  $p$ , write  $h(p) = -\int p \log p$ . Prove nonnegativity of relative entropy for such densities from  $\log u \leq u - 1$ , and use it to prove that a Gaussian maximizes  $h$  among densities with fixed covariance. Taking channel capacity to be the supremum of  $I(X : Y)$ , show that under the power constraint  $\sum_n \mathbb{E}[X_n^2] \leq P$  it is, in nats per channel use,

$$C_N(P) = \max_{\substack{p_n \geq 0 \\ \sum_n p_n \leq P}} \frac{1}{2} \sum_{n=0}^{N-1} \log \left( 1 + \frac{\lambda_n p_n}{N_0} \right).$$

Show that independent centered Gaussian inputs attain the maximum and use Lagrange multipliers to obtain the water-filling allocation

$$p_n = \left( \nu - \frac{N_0}{\lambda_n} \right)_+, \quad \sum_n p_n = P.$$

Explain why  $2TW/\pi = \sum_n \lambda_n$  estimates the number of useful modal channels, while capacity depends on the full eigenvalue distribution, the noise level, and the power constraint.

### 7.3 Ergodic dynamics and equilibrium states

**Definition 7.4** (Ergodic dynamics and dynamical entropy). A measurable transformation  $T$  of a probability space is *measure preserving* if  $\mu(T^{-1}A) = \mu(A)$ . Its *Koopman operator* is the isometry  $U_T f = f \circ T$  on  $L^2(\mu)$ . The system is *ergodic* if every invariant measurable set has measure zero or one, and *mixing* if

$$\mu(T^{-n}A \cap B) \longrightarrow \mu(A)\mu(B)$$

for all measurable  $A, B$ . An *equilibrium state* is an invariant probability measure for the dynamics.

For a finite measurable partition  $\mathcal{P}$ , set

$$H_\mu(\mathcal{P}) = - \sum_{A \in \mathcal{P}} \mu(A) \log \mu(A), \quad h_\mu(T, \mathcal{P}) = \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu \left( \bigvee_{j=0}^{n-1} T^{-j} \mathcal{P} \right).$$

The *Kolmogorov–Sinai entropy* is  $h_\mu(T) = \sup_{\mathcal{P}} h_\mu(T, \mathcal{P})$ . It measures the asymptotic information per time step revealed by successively refined observations.

**Exercise 7.3** (Ergodic averages, mixing, and entropy rate). Prove that the fixed vectors of  $U_T$  are precisely the invariant  $L^2$ -observables, and that ergodicity is equivalent to all such observables being constant almost everywhere. Prove the von Neumann mean ergodic theorem

$$\frac{1}{N} \sum_{n=0}^{N-1} U_T^n f \longrightarrow P_{\text{inv}} f \quad \text{in } L^2(\mu),$$

where  $P_{\text{inv}}$  is orthogonal projection onto the invariant observables. Prove that mixing implies ergodicity and that it gives decay of correlations  $\int (f \circ T^n) g d\mu - \int f d\mu \int g d\mu \rightarrow 0$  for bounded observables.

For every finite partition  $\mathcal{P}$ , prove by subadditivity that the limit defining  $h_\mu(T, \mathcal{P})$  exists. For the two-sided Bernoulli shift on  $\{1, \dots, r\}^{\mathbb{Z}}$  with product probabilities  $p_1, \dots, p_r$ , prove mixing first for cylinder sets and then for all measurable sets. Show that the time-zero coordinate partition satisfies

$$h_\mu(T, \mathcal{P}) = - \sum_{i=1}^r p_i \log p_i = H(p).$$

Interpret this equality as the information produced per observed symbol.

**Definition 7.5** (Gibbs states). For a classical Hamiltonian  $H$  on a phase space with Liouville measure  $\mu_L$ , the *canonical Gibbs state* at inverse temperature  $\beta$  is

$$d\mathbb{P}_\beta = Z(\beta)^{-1} e^{-\beta H} d\mu_L, \quad Z(\beta) = \int e^{-\beta H} d\mu_L,$$

provided  $0 < Z(\beta) < \infty$ . It is stationary under the Hamiltonian dynamics generated by  $H$ .

**Exercise 7.4** (Entropy and equilibrium). For every density  $\rho$  with finite mean energy, prove

$$D(\rho \| \rho_\beta) = -\frac{S(\rho)}{k_B} + \beta \langle H \rangle_\rho + \log Z(\beta).$$

Deduce that the Gibbs ensemble uniquely maximizes entropy at fixed mean energy, equivalently minimizes free energy at fixed temperature. Prove  $\langle H \rangle = -\partial_\beta \log Z$  and  $\text{Var}(H) = \partial_\beta^2 \log Z$ . Prove invariance of Gibbs measure under Hamiltonian flow. For a finite measure-preserving system, apply the  $L^2$  mean ergodic result proved above to an observable and identify the long-time mean-square limit in the ergodic case.

**Exercise 7.5** (Ideal-gas thermodynamics). Compute the classical canonical partition function of  $N$  indistinguishable noninteracting particles in a box, including the phase-space normalization  $h^{3N} N!$ . Derive the ideal-gas law, internal energy, and entropy, and the one-particle Maxwell–Boltzmann speed distribution. Derive its temperature scaling and compare both predictions with dilute-gas measurements in the classical regime.

For a wall of area  $A$ , express the pressure measured during time  $\tau$  as the time-averaged normal momentum transfer

$$P_\tau = \frac{1}{A\tau} \sum_{\substack{\text{collisions with the wall} \\ 0 < t_j \leq \tau}} 2m|v_{j,\perp}|.$$

Average the collision flux using the Maxwell–Boltzmann distribution and recover the same pressure  $P = Nk_B T/V$ . Identify the stationarity and ergodic assumptions that relate the long-time reading of one gas sample to the ensemble calculation. Explain why agreement of pressure, velocity statistics, and relaxation rates for selected observables and finite times tests consequences of those assumptions but does not prove that the full microscopic dynamics is mathematically ergodic.

## 8 Operator Algebras

### 8.1 $C^*$ -algebras, states, and GNS representations

**Definition 8.1** ( $C^*$ -algebras and states). A *unital Banach algebra* is a complete normed algebra with identity and  $\|ab\| \leq \|a\|\|b\|$ . An *involution* satisfies  $(ab)^* = b^*a^*$  and  $(a^*)^* = a$ . A  *$C^*$ -algebra* is an involutive Banach algebra satisfying  $\|a^*a\| = \|a\|^2$ . An element is *positive* if it is  $b^*b$  for some  $b$ . A *state* is a positive linear functional  $\varphi$  with  $\varphi(1) = 1$ ; positivity then implies  $\|\varphi\| = 1$ .

**Definition 8.2** (Noncommutative probability). A *noncommutative probability space* is a unital  $*$ -algebra  $A$  together with a linear functional  $\varphi : A \rightarrow \mathbb{C}$  satisfying  $\varphi(a^*a) \geq 0$  and  $\varphi(1) = 1$ . Its random variables are elements of  $A$ , and the joint  $*$ -distribution of  $X_1, \dots, X_r$  is the functional

$$p \mapsto \varphi(p(X_1, \dots, X_r, X_1^*, \dots, X_r^*))$$

on noncommutative  $*$ -polynomials. For a self-adjoint variable  $X$ , its moments are  $\varphi(X^n)$ . A *tracial probability space* also satisfies  $\varphi(ab) = \varphi(ba)$ . Random Hermitian matrices form such spaces after combining expectation with the normalized trace.

**Definition 8.3** (Representations and GNS construction). A *representation* of a  $C^*$ -algebra  $A$  is a  $*$ -homomorphism  $\pi : A \rightarrow B(H)$ . For a state  $\varphi$ , define  $N_\varphi = \{a : \varphi(a^*a) = 0\}$  and  $\langle [a], [b] \rangle = \varphi(a^*b)$  on  $A/N_\varphi$ .

**Exercise 8.1** (GNS theorem). Complete  $A/N_\varphi$  and prove that left multiplication induces a cyclic representation  $(\pi_\varphi, H_\varphi, \Omega_\varphi)$  satisfying  $\varphi(a) = \langle \Omega_\varphi, \pi_\varphi(a)\Omega_\varphi \rangle$ . Prove uniqueness up to a cyclic unitary equivalence.

## 8.2 Quantum mechanics and canonical quantization

**Definition 8.4** (Quantum states and observables). A positive operator  $S$  has the coordinate-free trace

$$\mathrm{Tr} S = \sup_P \mathrm{tr} ((PSP)|_{PH}),$$

where  $P$  ranges over orthogonal projections with finite-dimensional image. A bounded operator  $T$  is *trace class* when  $\mathrm{Tr} |T| < \infty$ , where  $|T| = (T^*T)^{1/2}$ ; its trace is obtained by linearity from the positive and negative parts of its real and imaginary parts. A *density operator* is positive and trace class with  $\mathrm{Tr} \rho = 1$ . An *observable* is a self-adjoint operator  $A$  with spectral measure  $E_A$ . The *Born probability* of an event  $B$  is  $\mathrm{Tr}(\rho E_A(B))$ . A closed-system *unitary evolution* is  $\rho(t) = U_t \rho(0) U_t^*$ , where  $(U_t)_{t \in \mathbb{R}}$  is a strongly continuous one-parameter unitary group. For unbounded observables and generators, spectral measures, expectations, and exponentials use the earlier unbounded spectral calculus and its natural domains.

**Exercise 8.2** (Canonical quantization and Weyl relations). On  $\mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$ , set  $Q_j \psi = x_j \psi$  and  $P_j \psi = -i\hbar \partial_j \psi$ . Prove  $[Q_j, P_k] = i\hbar \delta_{jk}$  and derive the Weyl relations for their unitary groups. Using symmetric ordering, verify the Poisson-bracket-commutator rule for affine and quadratic observables. Compare different orderings of  $q^2 p^2$  and explain the Groenewold–van Hove obstruction to quantizing all observables by this rule.

**Exercise 8.3** (Quantized harmonic oscillator). For  $H = P^2/(2m) + m\omega^2 Q^2/2$ , define creation and annihilation operators on the Schwartz core  $\mathcal{S}(\mathbb{R})$  and prove there that  $[a, a^*] = 1$  and  $H = \hbar\omega(a^*a + 1/2)$ . Construct the eigenstates with  $E_n = \hbar\omega(n + 1/2)$  and use the spectral-lines exercise to obtain the frequency spacing  $\omega$ . Prove that their span is a core for the self-adjoint closure of  $H$ . For a two-level internal transition of frequency  $\Omega_0$  coupled linearly to  $Q$ , compute the oscillator matrix elements, obtain motional sidebands at  $\Omega_0 \pm \omega$ , and compare their spacing with trapped-ion spectra.

**Exercise 8.4** (Liouville and von Neumann dynamics). Prove that the Born rule defines a probability measure and that its expected value is  $\mathrm{Tr}(\rho A)$  when defined. Derive the following classical and quantum evolution equations.

Let  $\mu_t = (\phi_t)_* \mu_0$  have smooth density  $p_t$  relative to the Liouville measure  $\mu_L$  of a time-independent Hamiltonian system. Using Exercise 5.8, prove

$$\partial_t p_t + \{p_t, H\} = 0, \quad \frac{d}{dt} \int_M f d\mu_t = \int_M \{f, H\} d\mu_t.$$

For  $U_t = e^{-itH/\hbar}$ , set  $\rho_t = U_t \rho_0 U_t^*$  and  $A_t = U_t^* A_0 U_t$ . Prove

$$\dot{\rho}_t = -\frac{i}{\hbar} [H, \rho_t], \quad \dot{A}_t = \frac{i}{\hbar} [H, A_t].$$

Prove

$$\int_M f d\mu_t = \int_M (f \circ \phi_t) d\mu_0, \quad \text{Tr}(\rho_t A_0) = \text{Tr}(\rho_0 A_t),$$

and identify these as the equivalence of Schrödinger and Heisenberg pictures. Compare the derivations  $f \mapsto \{f, H\}$  and  $A \mapsto (i\hbar)^{-1}[A, H]$ , and prove conservation of expected energy in both theories.

**Exercise 8.5** (Blackbody radiation from quantized Fourier modes). For electromagnetic radiation in a large conducting cavity, use its Fourier modes and two polarizations to derive the mode density  $g(\omega) = \omega^2/(\pi^2 c^3)$  per unit volume. Show that classical Gibbs equipartition assigns mean energy  $k_B T$  to each mode and produces the ultraviolet divergence. Using the oscillator levels derived in Exercise 8.3, compute the quantum Gibbs partition function and the thermal mean energy  $\hbar\omega/(e^{\beta\hbar\omega} - 1)$ . Deduce Planck's law

$$u(\omega, T) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/(k_B T)} - 1},$$

the Stefan–Boltzmann  $T^4$  law, and Wien scaling  $\omega_{\max} \propto T$ . Compare the low- and high-frequency limits with the cavity spectra discussed by Planck, *Annalen der Physik* **309** (1901), and identify precisely where quantization removes the classical divergence.

### 8.3 Fermionic statistics and the CAR algebra

**Definition 8.5** (CAR algebra and fermionic Fock space). For a complex Hilbert space  $K$ , the *canonical anticommutation relations algebra*  $\text{CAR}(K)$  is the unital  $C^*$ -algebra generated by conjugate-linear symbols  $a(f)$  satisfying

$$\{a(f), a(g)\} = 0, \quad \{a(f), a(g)^*\} = \langle f, g \rangle 1.$$

Let  $\Lambda_2^n K$  be the Hilbert-space completion of the  $n$ th antisymmetric tensor power of  $K$ , with  $\Lambda_2^0 K = \mathbb{C}$ . The *antisymmetric Fock space* is the Hilbert direct sum

$$\mathcal{F}_-(K) = \bigoplus_{n \geq 0} \Lambda_2^n K.$$

Its vacuum is  $\Omega = 1 \in \Lambda_2^0 K$ ; creation by  $f$  is exterior multiplication, and annihilation by  $f$  is its Hilbert-space adjoint.

**Exercise 8.6** (CAR and GNS predict fermionic antibunching). Prove that creation and annihilation on  $\mathcal{F}_-(K)$  satisfy the CAR and that  $\omega_0(x) = \langle \Omega, x\Omega \rangle$  is a state. Construct the cyclic map  $[x] \mapsto x\Omega$  from its GNS pre-Hilbert space and prove that its completion is the Fock representation. Let  $\omega_C$  be a gauge-invariant quasifree state with one-particle density  $0 \leq C \leq 1$ , and use its determinantal Wick rule to prove, for orthonormal detector modes  $f, g$ , that

$$\omega_C(N_f N_g) = \omega_C(N_f) \omega_C(N_g) - |\langle f, Cg \rangle|^2, \quad N_f = a(f)^* a(f).$$

Deduce  $N_f^2 = N_f$  and fermionic antibunching. Compare the sign and separation dependence of the correlation deficit with the neutral-atom measurements in Fölling et al., *Nature* **444** (2006).

## 8.4 Von Neumann algebras and noncommutative conditioning

**Definition 8.6** (Operator topologies and von Neumann algebras). The *weak operator topology* is generated by  $T \mapsto \langle x, Ty \rangle$ ; the *strong operator topology* is generated by  $T \mapsto \|Tx\|$ . For  $S \subseteq B(H)$ , its *commutant* is  $S' = \{T : TS = ST \text{ for all } S \in S\}$ . A *von Neumann algebra* is a unital  $*$ -subalgebra  $M \subseteq B(H)$  with  $M = M''$ .

**Exercise 8.7** (Bicommutant theorem). Prove that for a unital  $*$ -subalgebra  $A \subseteq B(H)$ , the strong and weak operator closures of  $A$  equal  $A''$ . Deduce that von Neumann algebras have sufficient projections to approximate their self-adjoint elements spectrally.

**Exercise 8.8** (Macroscopic magnetization and von Neumann closure). Let  $\mathcal{A}_N = M_2(\mathbb{C})^{\otimes N}$ , include  $\mathcal{A}_N$  in  $\mathcal{A}_{N+1}$  by  $a \mapsto a \otimes 1$ , and form the quasilocal spin algebra

$$\mathcal{A} = \overline{\bigcup_{N \geq 1} \mathcal{A}_N}^{\|\cdot\|}.$$

For  $Z = \text{diag}(1, -1)$ , let  $Z_j$  act as  $Z$  at site  $j$  and set

$$M_N = \frac{1}{N} \sum_{j=1}^N Z_j.$$

Prove  $\|M_{2N} - M_N\| = 1$ , so  $(M_N)$  has no norm limit in  $\mathcal{A}$ .

Let  $|+\rangle, |-\rangle$  be the  $Z$ -eigenvectors and define product states on local elementary tensors by

$$\omega_{\pm}(a_1 \otimes \cdots \otimes a_N) = \prod_{j=1}^N \langle \pm | a_j | \pm \rangle.$$

Prove that these extend to states on  $\mathcal{A}$ . In their GNS triples  $(H_{\pm}, \pi_{\pm}, \Omega_{\pm})$ , prove

$$\pi_+(M_N) \longrightarrow I_{H_+}, \quad \pi_-(M_N) \longrightarrow -I_{H_-}$$

in the strong operator topology. First prove the convergence on vectors  $\pi_{\pm}(a)\Omega_{\pm}$  with  $a \in \mathcal{A}_k$ , using

$$\|[M_N, a]\| \leq \frac{2k}{N} \|a\|,$$

and then extend it to all vectors using  $\|M_N\| \leq 1$ .

On  $H_+ \oplus H_-$ , let  $\pi = \pi_+ \oplus \pi_-$  and  $\Omega = 2^{-1/2}(\Omega_+ \oplus \Omega_-)$ . Prove that  $\Omega$  represents the mixed state  $\omega = (\omega_+ + \omega_-)/2$  and is cyclic. Show that

$$\pi(M_N) \longrightarrow C = I_{H_+} \oplus (-I_{H_-})$$

strongly, that  $C \in \pi(\mathcal{A})'' \cap \pi(\mathcal{A})'$ , and that the spectral projections  $(I \pm C)/2$  select the two magnetized phases. Thus the  $C^*$ -algebra describes the common local experiments, while its von Neumann closure in a chosen state also contains the macroscopic phase observable. Interpret this as an idealized zero-temperature ferromagnet; in an extremal low-temperature phase of an interacting ferromagnet, the corresponding limit is  $\pm m_{\beta} I$  under the appropriate ergodic or clustering hypotheses.

**Definition 8.7** (Normality). A positive functional  $\varphi$  on a von Neumann algebra is *normal* if  $a_\lambda \uparrow a$  implies  $\varphi(a_\lambda) \uparrow \varphi(a)$  for every bounded increasing net of positive elements. A representation is *normal* if it preserves such suprema.

**Definition 8.8** (Spatial tensor products). For concrete von Neumann algebras  $M \subseteq B(H)$  and  $N \subseteq B(K)$ , their *spatial tensor product* is

$$M \overline{\otimes} N = (M \odot N)'' \subseteq B(H \otimes K).$$

**Definition 8.9** (Noncommutative conditional expectation). For a von Neumann subalgebra  $N \subseteq M$ , a *conditional expectation* is a normal positive unital projection  $E : M \rightarrow N$  satisfying

$$E(n_1 a n_2) = n_1 E(a) n_2 \quad (a \in M, n_1, n_2 \in N).$$

It is  $\varphi$ -*preserving* for a normal state  $\varphi$  when  $\varphi \circ E = \varphi$ . When  $M = L^\infty(\Omega, \mathcal{F}, \mathbb{P})$  and  $N = L^\infty(\Omega, \mathcal{G}, \mathbb{P})$ , this recovers classical conditional expectation onto  $\mathcal{G}$ .

**Exercise 8.9** (Conditioning on subalgebras). Show that classical conditional expectation is positive, unital, contractive, and satisfies the bimodule identity above. For a normal state  $\psi$  on  $N$ , prove that

$$\text{id}_M \overline{\otimes} \psi : M \overline{\otimes} N \longrightarrow M$$

is a normal conditional expectation onto  $M \otimes 1$  after the canonical identification, and determine which product states it preserves. If  $M$  is a finite von Neumann algebra with normalized trace  $\tau$  and  $N \subseteq M$ , show that the orthogonal projection  $L^2(M, \tau) \rightarrow L^2(N, \tau)$  restricts to the unique  $\tau$ -preserving conditional expectation  $M \rightarrow N$ . Derive the tower property for nested von Neumann subalgebras.

## 8.5 Factor types, modular theory, and relative entropy

**Definition 8.10** (Traces and factor types). A *finite trace* on a von Neumann algebra is a normal positive functional  $\tau$  satisfying  $\tau(ab) = \tau(ba)$ . A possibly unbounded *normal tracial weight* is an additive, positively homogeneous map  $\tau : M_+ \rightarrow [0, \infty]$  satisfying  $\tau(x^*x) = \tau(xx^*)$  and  $\tau(x_\lambda) \uparrow \tau(x)$  whenever  $x_\lambda \uparrow x$ . It is *semifinite* if every nonzero  $x \in M_+$  majorizes a nonzero  $y \in M_+$  with  $\tau(y) < \infty$ . Thus the standard traces on infinite type-I and type-II $_\infty$  factors are normal semifinite weights, not finite functionals.

Projections  $p, q \in M$  are *Murray-von Neumann equivalent*, written  $p \sim q$ , if some  $v \in M$  satisfies  $v^*v = p$  and  $vv^* = q$ . A projection  $p$  is *finite* if  $q \leq p$  and  $q \sim p$  imply  $q = p$ , and it is *minimal* if its only subprojections are 0 and  $p$ .

A *factor* is a von Neumann algebra with center  $\mathbb{C}1$ . A factor is *type I* if it has a nonzero minimal projection, equivalently if it is isomorphic to some  $B(K)$ ; it is *type II* if it has nonzero finite projections but no minimal projections; and it is *type III* if it has no nonzero finite projection. Thus a type-III factor has no normal tracial state and no intrinsic trace-density representation of its normal states. A factor is *hyperfinite* if it is the weak closure of an increasing union of finite-dimensional \*-subalgebras.

**Exercise 8.10** (Projection dimension and factors). For a finite factor with normalized trace  $\tau$ , prove additivity and unitary invariance of  $\tau$  on projections. Relate the possible projection values to matrix factors and to diffuse finite factors. Prove that  $B(K)$  is type I and that a factor with a faithful normal tracial state cannot be type III. Explain why finite-dimensional quantum systems see only type-I factors, whereas the classification permits genuinely different infinite systems.

**Definition 8.11** (Cyclic and separating vectors). For  $M \subseteq B(H)$ , a vector  $\Omega$  is *cyclic* if  $\overline{M\Omega} = H$  and *separating* if  $a\Omega = 0$  implies  $a = 0$ . For such a pair  $(M, \Omega)$ , define the densely defined antilinear operator  $S_0(a\Omega) = a^*\Omega$ .

**Definition 8.12** (KMS states). With the present sign convention, a state  $\varphi$  is  $\beta$ -KMS for a flow  $\alpha_t$  if, for analytic  $a, b$ ,

$$\varphi(a\alpha_{t+i\beta}(b)) = \varphi(\alpha_t(b)a).$$

**Exercise 8.11** (Finite modular flow and Gibbs equilibrium). Let  $H = H^*$  on a finite-dimensional Hilbert space and  $\rho = Z^{-1}e^{-\beta H}$ . On  $B(H)$  define  $\alpha_t(a) = e^{itH}ae^{-itH}$ . Prove directly, first on matrix units in an energy eigenbasis and then by linearity, that

$$\text{Tr}(\rho a \alpha_{t+i\beta}(b)) = \text{Tr}(\rho \alpha_t(b)a).$$

Show that the modular flow of  $\rho$  is  $a \mapsto \rho^{it}a\rho^{-it} = \alpha_{-\beta t}(a)$  and that  $-\log \rho = \beta H + \log Z$ . Deduce that the same spectral data determine energy gaps, equilibrium weights, and modular frequencies.

**Definition 8.13** (Relative entropy of operator-algebraic states). Let  $\varphi$  and  $\psi$  be faithful normal states on  $M$ , represented by vectors  $\Omega_\varphi$  and  $\Omega_\psi$  in standard form. Their relative Tomita operator is the closure of

$$S_{\varphi|\psi,0}(a\Omega_\psi) = a^*\Omega_\varphi,$$

and their *relative modular operator* is

$$\Delta_{\varphi|\psi} = S_{\varphi|\psi}^* S_{\varphi|\psi}.$$

The *Araki relative entropy* is the extended nonnegative number

$$D_M(\varphi\|\psi) = \langle \Omega_\varphi, \log \Delta_{\varphi|\psi} \Omega_\varphi \rangle.$$

For  $M = B(H)$  and faithful density operators  $\rho, \sigma$ , the standard Hilbert–Schmidt representation gives  $\Delta_{\rho|\sigma}(X) = \rho X \sigma^{-1}$  and hence

$$D(\rho\|\sigma) = \text{Tr}(\rho(\log \rho - \log \sigma)).$$

Unlike the entropy of an individual density matrix, relative entropy and modular flow are intrinsic to an algebra and its states and remain meaningful for type-III factors.

## 9 Free Probability

### 9.1 Free independence and additive convolution

**Definition 9.1** (Free independence and free cumulants). Unital subalgebras  $(A_i)_i$  of  $(A, \varphi)$  are *freely independent* if

$$\varphi(a_1 \cdots a_m) = 0$$

whenever  $a_j \in A_{i_j}$ ,  $\varphi(a_j) = 0$ , and neighboring indices are distinct. Variables are free when their generated unital subalgebras are.

Let  $NC(n)$  be the lattice of noncrossing partitions of  $\{1, \dots, n\}$ . The *free cumulants*  $\kappa_n$  are the multilinear functionals determined recursively by

$$\varphi(a_1 \cdots a_n) = \sum_{\pi \in NC(n)} \kappa_\pi(a_1, \dots, a_n),$$

where  $\kappa_\pi$  is the product of the cumulants associated to the blocks of  $\pi$ . Freeness is equivalently the vanishing of every mixed free cumulant.

**Definition 9.2** (Semicircular variables). A self-adjoint variable  $S$  is *semicircular* with variance  $\sigma^2$  if its distribution is

$$d\mu_{\text{sc},\sigma}(x) = \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - x^2} \mathbf{1}_{[-2\sigma, 2\sigma]}(x) dx.$$

Equivalently, its only nonzero free cumulant is  $\kappa_2(S, S) = \sigma^2$ .

**Exercise 9.1** (Free central limit theorem). Use noncrossing pairings to prove

$$\varphi(S^{2m+1}) = 0, \quad \varphi(S^{2m}) = C_m \sigma^{2m}, \quad C_m = \frac{1}{m+1} \binom{2m}{m}.$$

Let  $X_1, X_2, \dots$  be freely independent, identically distributed, centered self-adjoint variables with  $\varphi(X_i^2) = \sigma^2$  and moments of every order. Use multilinearity and vanishing mixed cumulants to prove that the moments of  $n^{-1/2}(X_1 + \dots + X_n)$  converge to these semicircular moments. Compare this free central limit theorem with the Gaussian limit for classically independent variables.

**Definition 9.3** (Cauchy and  $R$ -transforms). For a compactly supported probability measure  $\mu$  on  $\mathbb{R}$ , its *Cauchy transform* is

$$G_\mu(z) = \int_{\mathbb{R}} \frac{d\mu(t)}{z - t}, \quad z \in \mathbb{C} \setminus \text{supp } \mu.$$

Near infinity it has a local compositional inverse  $K_\mu(w) = G_\mu^{(-1)}(w)$  near  $w = 0$ . The  $R$ -transform is defined by

$$K_\mu(w) = \frac{1}{w} + R_\mu(w), \quad R_\mu(w) = \sum_{n \geq 1} \kappa_n(\mu) w^{n-1}.$$

If freely independent self-adjoint variables  $X, Y$  have laws  $\mu, \nu$ , then the law of  $X + Y$  is their *free additive convolution*  $\mu \boxplus \nu$ , characterized by

$$R_{\mu \boxplus \nu} = R_\mu + R_\nu.$$

**Exercise 9.2** (Free convolution and the semicircle equation). Derive the additivity of the  $R$ -transform from the vanishing of mixed free cumulants. Show that a centered semicircular law of variance  $\sigma^2$  has  $R(w) = \sigma^2 w$  and hence

$$\sigma^2 G(z)^2 - zG(z) + 1 = 0.$$

Select the solution satisfying  $zG(z) \rightarrow 1$  at infinity and recover the semicircle density from the Stieltjes inversion formula. Deduce that the free sum of semicircular variables of variances  $\sigma_1^2$  and  $\sigma_2^2$  is semicircular of variance  $\sigma_1^2 + \sigma_2^2$ , and rederive the free central limit theorem at the level of transforms.

## 9.2 Random matrices and spectral statistics

**Definition 9.4** (Wigner and Gaussian matrix ensembles). A *Wigner matrix*  $H_N$  is an  $N \times N$  Hermitian random matrix whose upper-triangular entries are independent and centered, with  $\mathbb{E}|H_{ij}|^2 = \sigma^2/N$  off the diagonal and, for each fixed  $m$ ,

$$\sup_{N,i,j} \mathbb{E} |\sqrt{N} H_{ij}|^m < \infty.$$



Its *empirical spectral measure* is

$$L_{H_N} = \frac{1}{N} \sum_{j=1}^N \delta_{\lambda_j(H_N)}, \quad \int x^k dL_{H_N}(x) = \frac{1}{N} \text{Tr}(H_N^k).$$

The *Gaussian unitary ensemble* is the unitarily invariant Wigner ensemble with density proportional to  $\exp[-N \text{Tr}(H^2)/(2\sigma^2)]$  on Hermitian matrices. Sequences of random matrices are *asymptotically free* when their normalized joint trace moments converge to those of freely independent variables.

**Exercise 9.3** (Wigner semicircle law and asymptotic freeness). Expand  $N^{-1} \mathbb{E} \text{Tr}(H_N^k)$  as a sum over closed index walks. Show that only pair-matched walks survive as  $N \rightarrow \infty$ , that crossing pairings are lower order, and that the limiting moments are the semicircular moments. Prove that the variance tends to zero and hence that  $L_{H_N}$  converges in probability, in moments, to  $\mu_{\text{sc}, \sigma}$ .

For independent GUE matrices  $H_N^{(1)}, \dots, H_N^{(r)}$ , apply Wick's rule to normalized mixed traces of centered polynomials and prove that the family converges in joint moments to freely independent semicircular variables. Explain why ordinary independence of matrix entries becomes free independence of the limiting noncommuting matrices.

**Definition 9.5** (Dyson ensembles and unfolded spectra). The Gaussian ensembles with Dyson index  $\beta = 1, 2, 4$  consist respectively of real symmetric, complex Hermitian, and quaternionic self-dual matrices. They are the Gaussian orthogonal (GOE), unitary (GUE), and symplectic (GSE) ensembles. They model an antiunitary symmetry squaring to  $+1$ , no antiunitary symmetry, and an antiunitary symmetry squaring to  $-1$  after Kramers pairs are removed. Their ordered eigenvalues have joint density

$$p_{N, \beta}(\lambda_1, \dots, \lambda_N) = \frac{1}{Z_{N, \beta}} \exp \left[ -\frac{\beta N}{4\sigma^2} \sum_{j=1}^N \lambda_j^2 \right] \prod_{i < j} |\lambda_i - \lambda_j|^\beta.$$

The Vandermonde factor suppresses coincident eigenvalues and produces *level repulsion*.

To compare spectra with nonconstant mean density, let  $\overline{N}(E)$  be a smooth approximation to the cumulative level count. The *unfolded levels* and nearest-neighbor spacings are

$$x_j = \overline{N}(E_j), \quad s_j = x_{j+1} - x_j,$$

so their local mean spacing is one. Exact symmetry sectors must be separated before unfolding, since levels from independent sectors do not repel.

**Exercise 9.4** (Level repulsion and resonance spectra). Use the eigenvalue–eigenvector change of variables to derive the Vandermonde factor in the Gaussian eigenvalue density. For  $2 \times 2$  GOE and GUE matrices, separate the trace from the traceless coordinates and compute the radial distribution of the eigenvalue gap. After rescaling to unit mean spacing, obtain the Wigner surmises

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \quad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Explain from the joint density why the full ensembles satisfy  $P_\beta(s) \sim c_\beta s^\beta$  as  $s \downarrow 0$ , in contrast with the Poisson spacing law  $e^{-s}$  for uncorrelated levels.

Finally, describe how one must separate nuclear levels by spin and parity, or microwave resonances by exact cavity symmetry, fit  $\overline{N}(E)$ , unfold, and then compare spacing and correlation statistics

with the appropriate Dyson class. Time-reversal-invariant compound nuclei and chaotic microwave billiards without spin are predicted by the GOE. Compare with Haq–Pandey–Bohigas, *Physical Review Letters* **48** (1982) and Stöckmann–Stein, *Physical Review Letters* **64** (1990). Identify missing or misassigned levels and an inaccurate unfolding function as experimental effects that can imitate deviations from universality.

### 9.3 Bulk and edge universality

**Definition 9.6** (Global, bulk, and edge scales). Order the eigenvalues of a Wigner matrix as  $\lambda_1 \leq \dots \leq \lambda_N$ , and let

$$\rho_{\text{sc}}(x) = \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - x^2} \mathbf{1}_{[-2\sigma, 2\sigma]}(x).$$

The *classical location*  $\gamma_j$  is defined by

$$\frac{j}{N} = \int_{-2\sigma}^{\gamma_j} \rho_{\text{sc}}(x) dx.$$

The semicircle law is a *global law*: it concerns intervals containing order  $N$  eigenvalues. At a bulk energy  $E \in (-2\sigma, 2\sigma)$ , the microscopic coordinate is

$$u = N\rho_{\text{sc}}(E)(\lambda - E),$$

so the mean spacing is normalized to one. At the upper edge the density vanishes as a square root and the microscopic coordinate is

$$\xi = \frac{N^{2/3}}{\sigma}(\lambda - 2\sigma).$$

Bulk and edge universality concern the limiting point processes in these coordinates, not merely convergence of the empirical measure.

**Exercise 9.5** (Deriving the microscopic scales). At a bulk point  $E$ , use the expected count  $N\rho_{\text{sc}}(E) dE$  to derive the spacing scale  $[N\rho_{\text{sc}}(E)]^{-1}$ . Next put  $x = 2\sigma - s$  and show

$$\rho_{\text{sc}}(2\sigma - s) \sim \frac{1}{\pi\sigma^{3/2}} \sqrt{s} \quad (s \downarrow 0).$$

Estimate the expected number of eigenvalues in  $[2\sigma - s, 2\sigma]$  and show that it becomes order one when  $s$  is of order  $\sigma N^{-2/3}$ . Explain why a single rescaling cannot describe both the bulk and the edge.

**Definition 9.7** (Airy kernel and the Tracy–Widom law). Let  $\text{Ai}$  be the solution of  $y'' = xy$  that decays as  $x \rightarrow +\infty$ . The *Airy kernel* is

$$K_{\text{Ai}}(x, y) = \frac{\text{Ai}(x) \text{Ai}'(y) - \text{Ai}'(x) \text{Ai}(y)}{x - y},$$

with the diagonal defined by continuity. It determines the Airy determinantal point process at the soft edge. The GUE Tracy–Widom distribution is

$$F_2(s) = \det(I - K_{\text{Ai}})_{L^2(s, \infty)},$$

where the right side is the Fredholm determinant of the Airy integral operator. This formula is taken as a definition here; its analysis requires more operator theory than the present section develops.

## 10 Quantum Information Theory

### 10.1 Measurements, entropy, and tomography

**Definition 10.1** (Measurements). A *positive operator-valued measure* is a map  $E$  from measurable events to positive operators with  $E(X) = 1$  that is countably additive in the weak operator topology. In state  $\rho$ , its outcome probability is  $\text{Tr}(\rho E(B))$ .

**Exercise 10.1** (POVMs and dilation). Prove that projection-valued measures are POVMs. Prove Naimark's dilation theorem in finite dimensions, realizing every POVM's outcome probabilities by a projective measurement on a larger Hilbert space. Show that state discrimination can require a nonprojective POVM.

**Definition 10.2** (Von Neumann entropy). For a density operator with discrete spectral resolution  $\rho = \sum_{\lambda>0} \lambda P_\lambda$ , its *von Neumann entropy* is

$$S(\rho) = -k_B \text{Tr}(\rho \log \rho) = -k_B \sum_{\lambda>0} (\text{rank } P_\lambda) \lambda \log \lambda,$$

with  $0 \log 0 = 0$ ; in infinite dimension the value may be  $+\infty$ . It is  $k_B$  times the Shannon entropy of the outcomes of a projective measurement in an eigenbasis of  $\rho$ , and it vanishes exactly for pure states.

**Definition 10.3** (Quadratures, Wigner distributions, and homodyne data). For one canonical degree of freedom, choose  $q_0, p_0 > 0$  with  $q_0 p_0 = \hbar$  and set

$$X = Q/q_0, \quad Y = P/p_0, \quad X_\theta = X \cos \theta + Y \sin \theta,$$

so that  $[X, Y] = i$ . For a density operator  $\rho$ , its *Weyl characteristic function* is

$$\chi_\rho(u, v) = \text{Tr}(\rho e^{i(uX+vY)}),$$

and its *Wigner distribution* is the inverse Fourier transform

$$W_\rho(x, y) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \chi_\rho(u, v) e^{-i(ux+vy)} du dv,$$

interpreted as a tempered distribution when necessary. If  $E_\theta$  is the spectral measure of  $X_\theta$ , the *homodyne quadrature distribution* is

$$p_\rho(r \mid \theta) dr = \text{Tr}(\rho E_\theta(dr)).$$

In balanced optical homodyne detection, the local-oscillator phase selects  $\theta$  and repeated measurements estimate this distribution.

**Exercise 10.2** (Optical homodyne tomography). Use Exercise 4.24 and the spectral theorem to prove the Fourier-slice identity

$$\tilde{p}_\rho(k \mid \theta) = \int_{\mathbb{R}} p_\rho(r \mid \theta) e^{ikr} dr = \chi_\rho(k \cos \theta, k \sin \theta).$$

Deduce, in the distributional sense, that the measured quadrature density is the Radon transform of the Wigner distribution:

$$p_\rho(r \mid \theta) = \int_{\mathbb{R}^2} W_\rho(x, y) \delta(r - x \cos \theta - y \sin \theta) dx dy.$$

Prove that the data for  $0 \leq \theta < \pi$  determine  $\chi_\rho$ , hence  $W_\rho$  and  $\rho$ , and derive the filtered-backprojection formula

$$W_\rho(x, y) = \frac{1}{(2\pi)^2} \int_0^\pi \int_{\mathbb{R}} |k| \tilde{p}_\rho(k | \theta) e^{-ik(x \cos \theta + y \sin \theta)} dk d\theta.$$

Show that every  $p_\rho(\cdot | \theta)$  is a genuine probability density although  $W_\rho$  may be negative. Compute the data and reconstruction for vacuum, coherent, squeezed, and one-photon states. Explain why the factor  $|k|$  amplifies high-frequency noise and why finite-angle experimental data require regularization or constrained density-matrix estimation. Compare the predicted reconstruction of vacuum and squeezed states with Smithey et al., *Physical Review Letters* **70** (1993).

**Definition 10.4** (Quantum relative entropy). For finite-dimensional density operators with  $\text{supp } \rho \leq \text{supp } \sigma$ , their *quantum relative entropy* is

$$D(\rho \| \sigma) = \text{Tr}(\rho(\log \rho - \log \sigma)),$$

and it is  $+\infty$  otherwise. It reduces to classical relative entropy when  $\rho$  and  $\sigma$  commute.

**Exercise 10.3** (Entropy and quantum Gibbs states). Let  $K$  be  $n$ -dimensional, let  $P_1, \dots, P_n$  be mutually orthogonal rank-one projections with sum  $1_K$ , and let  $\mathcal{D} = \text{span}\{P_1, \dots, P_n\} \cong \mathbb{C}^n$ . Identify a state on  $\mathcal{D}$  with a probability vector  $p = (p_i)$  and the density operator  $\rho_p = \sum_i p_i P_i$ . Prove

$$S(\rho_p) = -k_B \sum_i p_i \log p_i,$$

so von Neumann entropy restricts to classical entropy on a commutative observable algebra. Prove more generally that  $S(U\rho U^*) = S(\rho)$  for every unitary  $U$ .

Let  $H = H^* \in B(K)$  have at least two distinct eigenvalues, and let the prescribed energy  $E$  lie strictly between its smallest and largest eigenvalues. Set

$$\rho_\beta = \frac{e^{-\beta H}}{Z(\beta)}, \quad Z(\beta) = \text{Tr}(e^{-\beta H}).$$

Use the spectral resolution of  $H$  to reduce the partition function to a finite classical Gibbs distribution. Referring to Exercise 7.4 and its variance identity, prove that there is a unique  $\beta \in \mathbb{R}$  for which  $\text{Tr}(\rho_\beta H) = E$ . For an arbitrary density operator  $\rho$ , now prove the genuinely noncommutative identity

$$D(\rho \| \rho_\beta) = -\frac{S(\rho)}{k_B} + \beta \text{Tr}(\rho H) + \log Z(\beta).$$

Use nonnegativity to obtain the quantum variational principle from the classical one:  $\rho_\beta$  uniquely maximizes entropy at fixed mean energy, equivalently minimizes free energy at fixed temperature. Finally, restrict  $\rho_\beta$  to the commutative algebra generated by the spectral projections of  $H$  and recover the classical Gibbs probabilities, including their energy-level degeneracies. If  $g_i$  is the degeneracy of  $E_i$  and  $p_i$  is the total population of that energy level, derive

$$\frac{p_i}{p_j} = \frac{g_i}{g_j} e^{-\beta(E_i - E_j)}.$$

Explain why the degeneracy factor is absent for the ratio of populations of individual eigenstates, and compare both formulas with measured level-population ratios.

## 10.2 Composite systems and entanglement

**Definition 10.5** (Composite and reduced states). The Hilbert space of a composite system is  $H_A \otimes H_B$ , and its normal states are the density operators on this tensor product. A density operator is *separable* if it lies in the trace-norm closed convex hull of product density operators and *entangled* otherwise. The *reduced state*  $\rho_A = \text{Tr}_B \rho$  is determined by  $\text{Tr}(\rho_A A) = \text{Tr}(\rho(A \otimes \text{id}))$  for all bounded  $A$ .

**Definition 10.6** (Classical and tensor independence). Families of  $\sigma$ -algebras are *independent* when the probability of every finite intersection of events, one from each family, is the product of their probabilities. Equivalently, bounded random variables measurable with respect to distinct families have factorizing mixed expectations.

Commuting unital subalgebras  $M_1, M_2 \subseteq M$  are independent in the state  $\varphi$  when  $\varphi(ab) = \varphi(a)\varphi(b)$  for  $a \in M_1$  and  $b \in M_2$ . They are *tensor independent* when the algebra they generate, together with the state, is a spatial tensor product with a product state.

**Exercise 10.4** (Wishart matrices and typical entanglement). Let  $G$  be a  $d_A \times d_B$  matrix of independent standard complex Gaussian entries and set

$$W = GG^*, \quad \rho_A = \frac{W}{\text{Tr } W}.$$

Show that  $\rho_A$  has the same distribution as the reduced state of a Haar-random unit vector in  $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ . Using Gaussian or Haar integration, prove

$$\mathbb{E} \text{Tr}(\rho_A^2) = \frac{d_A + d_B}{d_A d_B + 1}.$$

Deduce that when  $d_B \gg d_A$ , a typical reduced state is close to maximally mixed in Hilbert–Schmidt norm. Interpret this as a random-matrix explanation of why a typical pure state of a large composite system is highly entangled.

**Definition 10.7** (Entanglement entropy and mutual information). For a pure state  $\rho = |\psi\rangle\langle\psi|$  on  $H_A \otimes H_B$ , its *entanglement entropy* is  $S_A(\psi) = S(\rho_A)$ ; the Schmidt spectral data will show that this also equals  $S(\rho_B)$ . For an arbitrary bipartite state, its *quantum mutual information* is

$$I(A : B)_\rho = D(\rho \| \rho_A \otimes \rho_B) = \frac{S(\rho_A) + S(\rho_B) - S(\rho)}{k_B}.$$

It measures total correlation and vanishes exactly for product states.

**Exercise 10.5** (Schmidt projections and entanglement). Let  $H_A$  and  $H_B$  be finite-dimensional. For a pure vector  $\psi \in H_A \otimes H_B$ , apply the projection form of the singular-value decomposition to the contraction map  $C_\psi : H_A^* \rightarrow H_B$ . If  $Q_s$  and  $R_s$  are its domain and range projections, decompose  $\psi$  into mutually orthogonal spectral blocks supported on  $(Q_s H_A^*)^* \otimes R_s H_B$ . Explain why a rank-one refinement within a block of multiplicity greater than one is not canonical. Prove that a pure state is separable exactly when its reduced state has rank one, and that the two reduced states have the same nonzero spectral values with multiplicities  $\text{rank } Q_s = \text{rank } R_s$ . Derive

$$S_A(\psi) = -k_B \sum_{s>0} (\text{rank } Q_s) s^2 \log s^2$$

and prove that it vanishes exactly for separable pure states. Compute the joint and reduced states, local outcome frequencies, and entanglement entropy of a Bell state, and compare them with entangled-pair tomography.

**Exercise 10.6** (Bell correlations and CHSH). For a polarization-entangled two-photon state, derive the CHSH bound 2 for local hidden-variable models. Choose four polarization angles for which the quantum correlations attain  $2\sqrt{2}$ , and derive the predicted coincidence frequencies. As experimental input, use the adjusted local-realistic  $p$ -value  $2.3 \times 10^{-7}$  reported by Shalm et al., *Physical Review Letters* **115** (2015). Determine at the 5%, 1%, and five-standard-deviation significance levels, using the one-sided Gaussian five-sigma threshold  $2.87 \times 10^{-7}$ , whether the local-realistic null is excluded. State the tested freedom-of-choice and spacelike-separation assumptions and explain why the result does not exclude arbitrary superdeterministic or signaling models.

### 10.3 Quantum channels and data processing

**Definition 10.8** (Positive and completely positive maps). A linear map  $\Phi : A \rightarrow B$  between  $C^*$ -algebras is *positive* if it preserves positive elements and *completely positive* if every  $\text{id}_{M_n} \otimes \Phi$  is positive. A *quantum channel* is a completely positive trace-preserving map on trace-class operators, equivalently a normal unital completely positive map on observables in the adjoint picture.

**Exercise 10.7** (Stinespring and Kraus representations). Prove that a unital completely positive map  $\Phi : A \rightarrow B(H)$  has a representation  $\Phi(a) = V^* \pi(a) V$ . In finite dimensions, when  $\Phi = \mathcal{N}^*$  is the Heisenberg adjoint of a Schrödinger-picture channel  $\mathcal{N}$ , derive the paired Kraus formulas

$$\mathcal{N}(\rho) = \sum_i K_i \rho K_i^*, \quad \Phi(a) = \sum_i K_i^* a K_i, \quad \sum_i K_i^* K_i = 1.$$

Prove that every channel is unitary evolution on a larger system followed by discarding an environment. Use the bimodule identity to prove that every conditional expectation introduced earlier is completely positive. For a finite POVM  $(E_x)_x$ , prove that

$$\mathcal{M}(\rho) = \sum_x \text{Tr}(\rho E_x) |x\rangle\langle x|$$

is a quantum-to-classical channel, compute its adjoint, and identify its Stinespring dilation with the Naimark dilation constructed earlier.

**Exercise 10.8** (Relative entropy and data processing). Prove the log-sum inequality and deduce  $D(p\|q) \geq 0$ , with equality exactly when  $p = q$ . For a stochastic map  $K(y|x)$ , prove

$$D(Kp\|Kq) \leq D(p\|q).$$

Interpret a random variable, finite-resolution detector, or forgotten label as a stochastic map and explain why none can increase distinguishability. Prove  $I(X : Y) = S(p_X)/k_B + S(p_Y)/k_B - S(p_{XY})/k_B$ , and show that it vanishes exactly when  $X$  and  $Y$  are independent.

**Exercise 10.9** (Amplitude damping reproduces spontaneous emission). For a qubit with ground state  $|0\rangle$ , excited state  $|1\rangle$ , and environment vacuum  $|0\rangle_E$ , let

$$\begin{aligned} V_t|0\rangle &= |0\rangle|0\rangle_E, \\ V_t|1\rangle &= \sqrt{\eta_t}|1\rangle|0\rangle_E + \sqrt{1-\eta_t}|0\rangle|1\rangle_E, \quad \eta_t = e^{-t/T_1}. \end{aligned}$$

In the Schrödinger picture, trace out the environment and derive

$$K_0 = |0\rangle\langle 0| + \sqrt{\eta_t}|1\rangle\langle 1|, \quad K_1 = \sqrt{1-\eta_t}|0\rangle\langle 1|, \quad \mathcal{N}_t(\rho) = K_0 \rho K_0^* + K_1 \rho K_1^*.$$

Prove complete positivity, trace preservation, and the semigroup law. Derive  $\rho_{11}(t) = e^{-t/T_1} \rho_{11}(0)$  and  $\rho_{01}(t) = e^{-t/(2T_1)} \rho_{01}(0)$ . Compare the predicted decay and the environmental dependence of  $T_1$  with the transmon measurements in Houck et al., *Physical Review Letters* **101** (2008).

## 10.4 Quantum error correction and operator systems

**Definition 10.9** (Quantum codes and correctable errors). Let  $V : K \rightarrow H$  be an isometry between finite-dimensional Hilbert spaces. Its image  $C = \text{im } V$ , with projection  $P = VV^*$ , is a *quantum code*;  $K$  is the logical system and  $H$  the physical system. Write  $\mathcal{T}(H)$  for the trace-class operators on  $H$ . A channel  $\mathcal{N} : \mathcal{T}(H) \rightarrow \mathcal{T}(H')$  is *correctable on  $C$*  if there is a recovery channel  $\mathcal{R}$  such that

$$(\mathcal{R} \circ \mathcal{N})(V\rho V^*) = V\rho V^*$$

for every state  $\rho$  on  $K$ . A *stabilizer code* is the common  $+1$  spectral subspace of commuting self-adjoint unitaries. When  $H = \bigotimes_{j=1}^n H_j$ , the *distance* of  $C$  is the least number of tensor factors supporting an operator  $E$  for which  $PEP$  is not a scalar multiple of  $P$ , or  $+\infty$  if there is no such operator.

**Exercise 10.10** (Knill–Laflamme correction and decoupling). Let  $\mathcal{N}(\rho) = \sum_a E_a \rho E_a^*$  and let  $P$  project onto a code  $C$ . Prove that  $\mathcal{N}$  is correctable on  $C$  exactly when there is a positive semidefinite matrix  $(\lambda_{ab})$  such that

$$PE_a^* E_b P = \lambda_{ab} P \quad \text{for every } a, b.$$

Show that this condition is independent of the chosen Kraus representation, and construct a recovery from the polar decomposition of the combined error map

$$W : C \otimes \ell^2(\{a\}) \longrightarrow H', \quad W(\psi \otimes \delta_a) = E_a \psi,$$

together with the spectral projections of  $(\lambda_{ab})$ . Deduce that a code of distance  $d$  corrects every error supported on at most  $\lfloor (d-1)/2 \rfloor$  tensor factors.

For an encoding  $V : K \rightarrow H_A \otimes H_B$ , prove that erasure of  $B$  is correctable exactly when the complementary channel  $\rho \mapsto \text{Tr}_A(V\rho V^*)$  is constant on logical states. Interpret this as the equivalence between recovery from  $A$  and absence of encoded information in  $B$ ; see Knill–Laflamme, *Physical Review A* **55** (1997).

**Definition 10.10** (Operator systems). An *operator system* is a closed self-adjoint linear subspace  $S \subseteq B(H)$  containing the identity. Its matrix order is the family of cones

$$M_n(S)_+ = M_n(S) \cap B(H^{\oplus n})_+.$$

Completely positive maps are precisely the maps that preserve these cones at every matrix level.

**Definition 10.11** (The confusability operator system of a channel). Let  $H, H'$  be finite-dimensional and let  $\mathcal{N} : \mathcal{T}(H) \rightarrow \mathcal{T}(H')$  be a channel with Kraus operators  $E_a : H \rightarrow H'$ . Its *confusability operator system* is

$$S_{\mathcal{N}} = \text{span}\{E_a^* E_b : a, b\} \subseteq B(H).$$

It is self-adjoint, and it contains the identity because  $\sum_a E_a^* E_a = 1_H$ .

**Definition 10.12** (Protected observable algebras). Let  $P$  project onto a code  $C \subseteq H$ . A unital  $*$ -subalgebra  $\mathcal{A} \subseteq PB(H)P$ , whose identity is  $P$ , is *protected* or *correctable* for  $\mathcal{N}$  if there is a recovery channel  $\mathcal{R} : \mathcal{T}(H') \rightarrow \mathcal{T}(H)$  such that

$$P(\mathcal{R} \circ \mathcal{N})^*(A)P = A \quad (A \in \mathcal{A}).$$

Thus every observable in  $\mathcal{A}$ , rather than necessarily every encoded state, can be recovered. Write  $\mathcal{A}' = \{X \in PB(H)P : [X, A] = 0 \text{ for all } A \in \mathcal{A}\}$  for the commutant in the code corner.

**Exercise 10.11** (Exact operator-algebra quantum error correction). Prove that  $\mathcal{A}$  is correctable for  $\mathcal{N}$  if and only if

$$[PE_a^*E_bP, A] = 0 \quad (a, b, A \in \mathcal{A}),$$

or, equivalently,

$$PS_{\mathcal{N}}P \subseteq \mathcal{A}'.$$

Derive necessity by applying a Stinespring dilation to the recovery identity, and prove sufficiency by decomposing the finite-dimensional algebra and constructing a recovery on each summand. For  $\mathcal{A} = PB(H)P \cong B(C)$ , use  $\mathcal{A}' = \mathbb{C}P$  to recover the ordinary Knill–Laflamme equations  $PE_a^*E_bP = \lambda_{ab}P$ . Show that a commutative algebra protects a classical label, while a decomposition

$$\mathcal{A} \cong \bigoplus_{\alpha} (B(H_{\alpha}^A) \otimes 1_{H_{\alpha}^B})$$

protects hybrid information: the central label  $\alpha$  is classical, the  $H_{\alpha}^A$  factor is quantum, and  $H_{\alpha}^B$  is an unprotected gauge subsystem. Compare with Bény–Kempf–Kribs, *Quantum Error Correction of Observables*, Physical Review A **76** (2007).

## 11 Operator Space Theory

### 11.1 Matrix norms and channel discrimination

**Definition 11.1** (Operator spaces). An *operator space* is a closed subspace  $E \subseteq B(H)$  with matrix norms inherited from  $M_n(E) \subseteq B(H^{\oplus n})$ . A map  $u : E \rightarrow F$  is *completely bounded* if

$$\|u\|_{cb} = \sup_n \|\text{id}_{M_n} \otimes u\| < \infty.$$

Thus an operator space records not only the norm of an observable but all norms obtained after coupling it to matrix-valued ancillary systems.

**Definition 11.2** (Ancilla-assisted channel discrimination). A *one-use discrimination strategy* for channels  $\Phi_1, \dots, \Phi_m$  prepares a state on  $H \otimes K$ , applies  $\Phi_j \otimes \text{id}_K$  to the unknown channel input, and measures the output with a POVM. The strategy is *ancilla assisted* when  $\dim K > 1$ . It is *unambiguous* when every conclusive outcome identifies the channel without error. Its *inconclusive probability* is denoted  $P_I$ .

**Exercise 11.1** (An ancilla improves measurement-channel discrimination). For  $0 < \theta < \pi/4$ , define

$$|\phi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle, \quad |\psi\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle,$$

and let  $\mathcal{M}, \mathcal{N}$  be the two projective measurement channels with rank-one projection families  $\{P_{\phi}, P_{\phi^{\perp}}\}$  and  $\{P_{\psi}, P_{\psi^{\perp}}\}$ . Use a singlet probe, retain the second qubit as an ancilla, and condition on the classical measurement outcome. Reduce the task to unambiguous discrimination of  $|\phi\rangle$  and  $|\psi\rangle$  and prove

$$P_I^{\text{anc}} = \cos(2\theta), \quad P_I^{\text{sep}} = \frac{1 + \cos^2(2\theta)}{2}$$

for the optimal ancilla-assisted and single-qubit strategies. Deduce the strict advantage for  $0 < \theta < \pi/4$  and express it as a distinction between the first and amplified matrix levels of the measurement channels. Compare the predicted success and inconclusive probabilities with Miková et al., *Physical Review A* **90** (2014).



**Exercise 11.2** (Ruan’s representation theorem). Prove conversely that every vector space with matrix norms satisfying these axioms admits a complete isometry into  $B(H)$ . Apply matrix Hahn–Banach separation to a norming family of matrices and take the direct sum of the resulting maps.

## 11.2 Fourier algebras and completely bounded multipliers

**Definition 11.3** (Group von Neumann and Fourier algebras). For a locally compact group  $G$  with left regular representation  $\lambda$  on  $L^2(G)$ , its *group von Neumann algebra* is

$$VN(G) = \{\lambda_s : s \in G\}''.$$

The *Fourier algebra*  $A(G)$  consists of the coefficient functions

$$u(s) = \langle \xi, \lambda_s \eta \rangle, \quad \xi, \eta \in L^2(G),$$

with norm

$$\|u\|_{A(G)} = \inf \{ \|\xi\| \|\eta\| : u(s) = \langle \xi, \lambda_s \eta \rangle \}.$$

The pairing  $\langle u, \lambda_s \rangle = u(s)$  identifies  $A(G) = VN(G)_*$  and equips  $A(G)$  with its canonical predual operator-space structure. The normal unital  $*$ -homomorphism

$$\Delta : VN(G) \longrightarrow VN(G) \overline{\otimes} VN(G), \quad \Delta(\lambda_s) = \lambda_s \otimes \lambda_s,$$

is the *group coproduct*; its preadjoint gives pointwise multiplication on  $A(G)$ .

**Exercise 11.3** (Fourier algebras as operator-space preduals). For a discrete group  $G$ , prove that the canonical trace on  $VN(G)$  satisfies  $\tau(\lambda_s) = \delta_{s,e}$  and that the coefficient functions above separate the generators  $\lambda_s$ . Verify the coproduct identities

$$(\Delta \overline{\otimes} \text{id})\Delta = (\text{id} \overline{\otimes} \Delta)\Delta, \quad \Delta(\lambda_s \lambda_t) = \Delta(\lambda_s) \Delta(\lambda_t).$$

Show from the predual matrix norms that pointwise multiplication on  $A(G)$  is completely contractive at every matrix level. For abelian  $G$ , use Pontryagin duality to identify  $A(G)$  with  $L^1(\widehat{G})$  under the inverse Fourier transform and show that this pointwise product corresponds to convolution on  $L^1(\widehat{G})$ . Compare this coefficient-space viewpoint with the operator-valued Plancherel transform developed in Harmonic Analysis.

**Definition 11.4** (Factorizable Fourier multipliers). A function  $\phi : G \rightarrow \mathbb{C}$  is a *Fourier multiplier* when  $m_\phi(u) = \phi u$  maps  $A(G)$  to itself. It is *completely bounded* when  $m_\phi$  is completely bounded for the predual operator-space structure. Its adjoint is the normal map

$$M_\phi : VN(G) \rightarrow VN(G), \quad M_\phi(\lambda_s) = \phi(s) \lambda_s.$$

A Fourier multiplier is *factorizable* when the invariant kernel

$$k_\phi(x, y) = \phi(xy^{-1})$$

has the form

$$\phi(xy^{-1}) = \langle \xi(x), \eta(y) \rangle$$

Its *factorization norm* is the infimum of  $\|\xi\|_\infty \|\eta\|_\infty$  over all such factorizations.

**Exercise 11.4** (Fourier–Schur transference and quantum channels). Let  $G$  be discrete. On the matrix units of  $B(\ell^2(G))$ , define

$$S_\phi(E_{x,y}) = \phi(xy^{-1})E_{x,y}.$$

Verify directly that  $S_\phi(\lambda_s) = \phi(s)\lambda_s$ , so its restriction to  $VN(G)$  is  $M_\phi$ . Use the factorization in the preceding definition to prove that  $S_\phi$  and  $M_\phi$  are completely bounded and obtain the bound  $\|M_\phi\|_{cb} \leq \|\xi\|_\infty \|\eta\|_\infty$ .

A function  $\phi$  is *positive definite* if  $[\phi(s_i^{-1}s_j)]_{i,j} \geq 0$  for every finite family  $(s_i)$ . Prove that if  $\phi$  is positive definite and  $\phi(e) = 1$ , then  $S_\phi$  and  $M_\phi$  are normal unital completely positive maps. For finite  $G$ , prove that  $M_\phi$  preserves the canonical trace and hence is a quantum channel in the Schrödinger adjoint picture. Show

$$M_\phi M_\psi = M_{\phi\psi}.$$

Let  $(\phi_t)_{t \geq 0}$  be normalized positive-definite functions satisfying  $\phi_0 \equiv 1$  and  $\phi_{s+t} = \phi_s \phi_t$ , and suppose  $t \mapsto \phi_t(r)$  is continuous for every  $r \in G$ . Prove that  $(M_{\phi_t})_{t \geq 0}$  is a weak-\* continuous completely positive Markov semigroup. When  $\phi_t(r) = e^{-t\ell(r)}$ , interpret it as symmetry-diagonal decoherence: modes with larger  $\ell(r)$  decay faster while the identity mode is preserved.

### 11.3 Compact quantum groups and quantum Peter–Weyl theory

**Definition 11.5** (Compact quantum groups and their Haar states). A *compact quantum group*  $\mathbb{G}$  consists of a unital  $C^*$ -algebra  $C(\mathbb{G})$  and a unital  $*$ -homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where  $\otimes_{\min}$  is the spatial  $C^*$ -tensor product obtained from faithful Hilbert-space representations, such that

$$(\Delta \otimes \text{id})\Delta = (\text{id} \otimes \Delta)\Delta$$

and the linear spans of  $\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))$  and  $\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)$  are dense in the minimal tensor product. The density conditions are the cancellation laws without points.

A *Haar state* is a state  $h$  satisfying

$$(h \otimes \text{id})\Delta(a) = h(a)1 = (\text{id} \otimes h)\Delta(a) \quad (a \in C(\mathbb{G})).$$

In the GNS representation of the reduced compact quantum group, set

$$L^\infty(\mathbb{G}) = \pi_h(C(\mathbb{G}))'', \quad L^1(\mathbb{G}) = L^\infty(\mathbb{G})_*.$$

When the coproduct has a normal extension to  $L^\infty(\mathbb{G})$ , its preadjoint convolution is

$$(\omega * \eta)(a) = (\omega \otimes \eta)(\Delta a) \quad (\omega, \eta \in L^1(\mathbb{G})).$$

**Definition 11.6** (Unitary corepresentations and quantum dimension). A finite-dimensional *unitary corepresentation* of  $\mathbb{G}$  on  $H_U$  is a unitary

$$U \in B(H_U) \otimes C(\mathbb{G})$$

satisfying

$$(\text{id} \otimes \Delta)(U) = U_{12}U_{13}.$$

If  $U = (u_{ij})$  in an orthonormal basis, this says  $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$ . Tensor products are formed using the coproduct; their order can matter because  $\Delta$  need not equal its opposite.

A *coefficient algebra* is the  $*$ -algebra generated by the matrix coefficients  $u_{ij}$ . The coefficients of all finite-dimensional unitary corepresentations span a dense Hopf  $*$ -algebra  $\text{Pol}(\mathbb{G})$ , whose antipode  $S$  is determined by  $(\text{id} \otimes S)(U) = U^*$ . For every irreducible corepresentation  $U^\alpha$  there is a positive invertible matrix  $Q_\alpha$ , unique after the normalization below, that implements the square of the antipode:

$$(\text{id} \otimes S^2)(U^\alpha) = (Q_\alpha \otimes 1)U^\alpha(Q_\alpha^{-1} \otimes 1).$$

Normalize it by

$$\text{Tr}(Q_\alpha) = \text{Tr}(Q_\alpha^{-1}),$$

this common value is its *quantum dimension*  $d_q(\alpha)$ . In the tracial, classical compact-group case,  $Q_\alpha = 1$  and  $d_q(\alpha) = \dim H_\alpha$ .

**Exercise 11.5** ( $SU_q(2)$  and arithmetic dimensions). Fix  $0 < q < 1$ . Let  $C(SU_q(2))$  be generated by  $\alpha, \gamma$  so that

$$u = \begin{pmatrix} \alpha & -q\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary, equivalently

$$\begin{aligned} \alpha^* \alpha + \gamma^* \gamma &= 1, & \alpha \alpha^* + q^2 \gamma \gamma^* &= 1, & \gamma^* \gamma &= \gamma \gamma^*, \\ \alpha \gamma &= q \gamma \alpha, & \alpha \gamma^* &= q \gamma^* \alpha. \end{aligned}$$

Define  $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$ . Check on generators that  $\Delta$  preserves the relations and is coassociative. Explain why the noncommuting coordinate functions describe a noncommutative compact space, while the matrix  $u$  still transforms like a fundamental representation.

Let symbols  $U^\ell$ ,  $\ell = 0, \frac{1}{2}, 1, \dots$ , obey the fusion recurrence

$$U^{1/2} \boxtimes U^\ell \cong U^{\ell+1/2} \oplus U^{\ell-1/2}.$$

Set  $d_q(0) = 1$ ,  $d_q(1/2) = q + q^{-1}$ , and require dimension to be additive on  $\oplus$  and multiplicative on  $\boxtimes$ . Derive

$$d_q(\ell) = [2\ell + 1]_q = \frac{q^{2\ell+1} - q^{-(2\ell+1)}}{q - q^{-1}}.$$

Recover the ordinary dimensions as  $q \rightarrow 1$ . Distinguish the Hilbert-space dimension  $2\ell + 1$  from the arithmetic Laurent polynomial  $[2\ell + 1]_q$ .

## 11.4 Equivariant compression and Toeplitz operator systems

**Definition 11.7** (Equivariant compression). Let a compact group  $G$  act on a Hilbert space  $H$  by a unitary representation  $U$ , and let a unital  $C^*$ -algebra  $A \subseteq B(H)$  be invariant under conjugation by  $U_g$ . If a finite-rank orthogonal projection  $P$  commutes with every  $U_g$ , define

$$S_P = PAP = \{PaP : a \in A\} \subseteq B(PH).$$

The restricted representation  $U_g^P = U_g|_{PH}$  acts on  $S_P$  by conjugation. We call  $S_P$  an *equivariant compression* of  $A$ .

**Exercise 11.6** (Rotation-covariant Toeplitz compression). Let  $A = C(S^1)$  act by multiplication on  $H = L^2(S^1)$ , with normalized Fourier basis  $e_k(\theta) = (2\pi)^{-1/2}e^{ik\theta}$ . Let  $P_N$  project onto  $H_N = \text{span}\{e_k : -N \leq k \leq N\}$  and set  $S_N = P_N A P_N$ . Prove

$$\langle e_j, P_N M_f P_N e_k \rangle = \hat{f}(j - k), \quad -N \leq j, k \leq N,$$

and deduce that  $S_N$  is exactly the operator system of  $(2N + 1)$ -dimensional Toeplitz matrices.

For  $z(\theta) = e^{i\theta}$ , put  $T_N = P_N M_z P_N$ . Show that  $T_N$  is a truncated shift and that  $T_N T_N^*$  has nonconstant main diagonal. Conclude that  $S_N$  is not an algebra, although the  $C^*$ -algebra generated by  $S_N$  is the full matrix algebra  $B(H_N)$ .

Finally, let rotations act by  $(U_\phi \xi)(\theta) = \xi(\theta - \phi)$ . Prove that  $P_N$  commutes with  $U_\phi$  and that conjugation by  $U_\phi|_{H_N}$  preserves  $S_N$ . Verify directly that the cutoff retains rotation covariance although its compressed observables do not form an algebra.

## 11.5 Extension, tensor norms, and bilinear forms

**Exercise 11.7** (Finite-dimensional Arveson–Wittstock extension). Prove in finite dimensions that a completely positive map from an operator system extends completely positively to the containing  $C^*$ -algebra. Deduce, by the  $2 \times 2$  operator-system construction, that completely bounded maps between finite-dimensional operator spaces admit norm-preserving extensions.

**Definition 11.8** (Operator-space tensor products). The *minimal operator-space tensor norm* on  $E \otimes F$  is inherited from concrete inclusions in  $B(H \otimes K)$ . The *Haagerup norm* is

$$\|z\|_h = \inf\{\|[x_1 \ \cdots \ x_r]\| \|[y_1 \ \cdots \ y_r]^T\| : z = \sum_i x_i \otimes y_i\}.$$

A bilinear form  $u : E \times F \rightarrow \mathbb{C}$  is *jointly completely bounded* when the maps

$$u_n([x_{ij}], [y_{kl}]) = [u(x_{ij}, y_{kl})]_{(i,k),(j,l)}$$

from  $M_n(E) \times M_n(F)$  to  $M_{n^2}$  are uniformly bounded; the supremum of their norms is  $\|u\|_{jcb}$ .

**Exercise 11.8** (Tensor norms linearize bilinear maps). Prove the universal property of the projective Banach-space tensor norm. Prove that the Haagerup tensor product linearizes completely bounded factorizations and compare the minimal, Haagerup, and projective norms on finite-dimensional matrix spaces.

## 11.6 Nonlocal games as operator-space tensors

**Definition 11.9** (Bell functionals and nonlocal games). A *Bell functional* is an array  $M = (M_{xy}^{ab})$  evaluated on conditional distributions by  $\langle M, p \rangle = \sum_{a,b,x,y} M_{xy}^{ab} p(a, b \mid x, y)$ . A two-player *nonlocal game*  $G = (\pi, V)$  consists of a question distribution  $\pi(x, y)$  and a payoff  $0 \leq V(a, b \mid x, y) \leq 1$ . Its classical and entangled values  $\omega(G)$  and  $\omega^*(G)$  are the optimal expected payoffs using, respectively, shared randomness and local POVMs  $\{E_a^x\}_a, \{F_b^y\}_b$  on a shared finite-dimensional state  $\rho$ , with  $p(a, b \mid x, y) = \text{Tr}(\rho(E_a^x \otimes F_b^y))$ . An *XOR game* has binary answers and scores only their parity, equivalently a correlation  $\sum_{x,y} M_{xy} \mathbb{E}[a_x b_y]$  for signs  $a_x, b_y$ .

**Exercise 11.9** (Magic-square pseudotelepathy). The referee chooses  $(x, y) \in \{1, 2, 3\}^2$  uniformly. Alice returns a row  $a \in \{\pm 1\}^3$  with product +1, Bob returns a column  $b \in \{\pm 1\}^3$  with product

$-1$ , and they win when  $a_y = b_x$ . Prove by multiplying all row and column constraints that no deterministic classical strategy wins every question, construct one losing on only one question, and hence show  $\omega(G_{\text{MS}}) = 8/9$ . Using two Bell pairs and a Mermin–Peres square of commuting two-qubit Pauli observables, construct a strategy with  $\omega^*(G_{\text{MS}}) = 1$ .

Compare the ideal query-by-query predictions with the hyperentangled-photon experiment of Xu et al., *Physical Review Letters* **129** (2022). Explain separately the locality and fair-sampling assumptions in that experiment. Show why a score above  $8/9$  is a device-independent witness under the stated assumptions.

**Definition 11.10** (The tensor of a game). Set

$$E_{X,A} = \ell_1^X(\ell_\infty^A), \quad \|z\|_{E_{X,A}} = \sum_x \max_a |z_{x,a}|,$$

with its standard dual operator-space structure. The tensor associated to  $G = (\pi, V)$  is

$$\hat{G} = \sum_{x,y,a,b} \pi(x,y) V(a,b | x,y) (e_x \otimes e_a) \otimes (e_y \otimes e_b) \in E_{X,A} \otimes E_{Y,B}.$$

**Exercise 11.10** (Game values as Banach and operator-space norms). Identify deterministic strategies with extreme points of the relevant unit balls and prove

$$\omega(G) = \|\hat{G}\|_{E_{X,A} \otimes_\varepsilon E_{Y,B}}, \quad \omega^*(G) = \|\hat{G}\|_{E_{X,A} \otimes_{\min} E_{Y,B}}.$$

Equivalently, show that these are the bounded and completely bounded norms of the bilinear payoff form on the dual mixed-norm spaces. Locate in the proof the use of positivity of a game tensor, and explain why the same simple formula need not hold for an arbitrary signed Bell functional. Use Palazuelos–Vidick, *Survey on Nonlocal Games and Operator Space Theory*, *Journal of Mathematical Physics* **57** (2016) as a guide.

## Part IV

# Topology

## 12 Index Theory

### 12.1 Homotopy, homology, and generalized cohomology

**Definition 12.1** (Homotopy and fundamental group). A *homotopy* from  $f : X \rightarrow Y$  to  $g : X \rightarrow Y$  is a continuous map  $H : X \times [0, 1] \rightarrow Y$  with endpoints  $f$  and  $g$ . The *fundamental group*  $\pi_1(X, x_0)$  is the group of endpoint-fixed homotopy classes of loops based at  $x_0$ , with concatenation.

**Definition 12.2** (Homology and cohomology). Fix a commutative ring  $R$  with identity. The *singular chain group*  $C_n(X; R)$  is the free  $R$ -module on continuous maps  $\Delta^n \rightarrow X$ , with alternating face boundary  $\partial$ . The *homology* is  $H_n(X; R) = \ker \partial_n / \text{im } \partial_{n+1}$ . The *cochain complex* is  $C^n(X; R) = \text{Hom}_R(C_n(X; R), R)$  with coboundary  $\delta\phi = \phi\partial$ , and its cohomology is  $H^n(X; R)$ .

**Definition 12.3** (Cofibers, finite CW complexes, and cohomology theories). A based inclusion  $A \hookrightarrow X$  is a *cofibration* if every homotopy on  $A$  that extends at its initial time to  $X$  extends over

the whole homotopy. Its *cofiber* is  $X/A$ . More generally, the cofiber of a based map  $f : A \rightarrow X$  is the pushout  $X \cup_f CA$ , where  $CA$  is the reduced cone; the reduced suspension is  $\Sigma A = CA/A$ .

A *finite CW complex* is obtained from the empty space by finitely many pushouts

$$\begin{array}{ccc} \coprod_{\alpha} S^{n-1} & \longrightarrow & X^{n-1} \\ \downarrow & & \downarrow \\ \coprod_{\alpha} D^n & \longrightarrow & X^n, \end{array}$$

where  $X^n$  is the  $n$ th skeleton. A *reduced cohomology theory* on based finite CW complexes is a graded contravariant functor  $\tilde{E}^*$  that is homotopy invariant, vanishes on a point, takes finite wedges to direct sums, has natural suspension isomorphisms  $\tilde{E}^{q+1}(\Sigma X) \cong \tilde{E}^q(X)$ , and makes

$$\tilde{E}^q(X/A) \longrightarrow \tilde{E}^q(X) \longrightarrow \tilde{E}^q(A)$$

exact for every cofibration  $A \hookrightarrow X$ .

**Definition 12.4** (Products and pairings). The *Kronecker pairing* is  $H^n(X; R) \times H_n(X; R) \rightarrow R$ ,  $([\phi], [c]) \mapsto \phi(c)$ . The *cup product* is the product on cohomology induced by the front-back face formula on cochains.

**Exercise 12.1** (Homology, cohomology, and degree). Prove  $\partial^2 = 0$ ,  $\delta^2 = 0$ , and homotopy invariance. For a CW pair  $(X, A)$ , define  $C_*(X, A; R) = C_*(X; R)/C_*(A; R)$ , construct barycentric subdivision, and use it to prove excision and  $H^q(X, A; R) \cong \tilde{H}^q(X/A; R)$ . Construct the Puppe sequence of a cofibration and derive the long exact sequence of a pair and the Mayer–Vietoris sequence. Deduce that reduced singular cohomology is a reduced cohomology theory.

Prove that  $X^{n-1} \hookrightarrow X^n$  is a cofibration and  $X^n/X^{n-1}$  is a finite wedge of  $n$ -spheres. By induction over the skeleta, prove that a natural transformation between reduced cohomology theories that is an isomorphism on  $S^0$  in every degree is an isomorphism on every finite CW complex; prove the required five-term diagram chase. Compute the homology and cohomology rings of spheres and tori. For a continuous map  $f : S^n \rightarrow S^n$ , define its degree by  $f_*[S^n] = \deg(f)[S^n]$  and prove multiplicativity and agreement with winding number for  $n = 1$ .

## 12.2 Bundles, holonomy, and characteristic classes

**Exercise 12.2** (Aharonov–Bohm holonomy). Using the connection, curvature, and holonomy of Definitions 6.7 and 6.8, consider on  $X = \mathbb{R}^2 \setminus \{0\}$  the  $U(1)$  gauge potential  $A = (\Phi/2\pi)d\theta$ . Verify that  $F = dA = 0$  on  $X$  but  $\oint_{\gamma} A = \Phi$  for a loop of winding number one. Show that a particle of charge  $q$  acquires the gauge-invariant phase  $\exp(iq\Phi/\hbar)$ , derive the relative phase of two paths enclosing the excluded flux, and obtain the flux period  $\hbar/|q|$ . Prove that  $A$  cannot be removed by a single-valued gauge transformation unless its holonomy is trivial. Compare the predicted fringe displacement with the shielded-flux electron-holography experiment of Tonomura et al., *Physical Review Letters* **56** (1986). Explain why the result establishes the physical significance of holonomy, not of a gauge-dependent potential by itself, even though curvature vanishes along the electron paths.

**Definition 12.5** (Grassmannians and universal bundles). For a finite-dimensional Hermitian space  $V$ , the *Grassmannian*  $\text{Gr}_n(V)$  is the space of  $n$ -dimensional subspaces of  $V$ . Its *tautological bundle* has fiber  $W$  over  $W \in \text{Gr}_n(V)$ . Isometric inclusions of Hermitian spaces induce compatible inclusions of their Grassmannians and tautological bundles, and

$$BU(n) = \varinjlim_N \text{Gr}_n(\mathbb{C}^N), \quad \gamma_n = \varinjlim_N \gamma_{n,N}.$$

**Exercise 12.3** (Universal Chern classes). Identify  $BU(1)$  with  $\mathbb{P}(\mathbb{C}^\infty)$  and use its even-dimensional cell decomposition to compute  $H^*(BU(1); \mathbb{Z}) = \mathbb{Z}[x]$ , where  $x$  evaluates as  $-1$  on the projective line carrying the tautological line bundle. Let

$$s : BU(1)^n \longrightarrow BU(n)$$

classify the direct sum of the  $n$  tautological lines, and set  $x_i = \text{pr}_i^* x$ . Construct the Schubert cell decompositions of the finite Grassmannians and their flag bundles. Use the cellular filtration and the splitting map  $s$  to prove that  $s^*$  is injective with image

$$\mathbb{Z}[x_1, \dots, x_n]^{\mathfrak{S}_n} = \mathbb{Z}[e_1(x), \dots, e_n(x)].$$

Define  $c_j(\gamma_n) \in H^{2j}(BU(n); \mathbb{Z})$  by  $s^* c_j(\gamma_n) = e_j(x_1, \dots, x_n)$ . For the block-sum map  $BU(n) \times BU(m) \rightarrow BU(n+m)$ , prove

$$c(\gamma_n \oplus \gamma_m) = c(\gamma_n)c(\gamma_m), \quad c = 1 + c_1 + c_2 + \dots$$

**Definition 12.6** (Chern classes and character). If a rank- $n$  complex vector bundle  $E \rightarrow X$  is classified by  $f : X \rightarrow BU(n)$ , its *Chern classes* are  $c_j(E) = f^* c_j(\gamma_n) \in H^{2j}(X; \mathbb{Z})$ . If the formal roots of  $1 + c_1(E) + \dots + c_n(E)$  are  $y_1, \dots, y_n$ , the *Chern character* is the degreewise class

$$\text{ch}(E) = \sum_{i=1}^n e^{y_i} \in \prod_{k \geq 0} H^{2k}(X; \mathbb{Q}),$$

where each symmetric polynomial in the  $y_i$  is expressed in the Chern classes by the Newton identities and the definition extends additively to  $K^0(X)$ .

**Exercise 12.4** (Curvature representatives). For a Hermitian bundle with unitary connection, prove that

$$\det \left( 1 + \frac{i}{2\pi} F_\nabla \right) \quad \text{and} \quad \text{Tr} \exp \left( \frac{i}{2\pi} F_\nabla \right)$$

are closed forms whose cohomology classes are independent of the connection. Identify their de Rham classes with the images of the Chern classes and Chern character, and prove their Whitney-sum formulas.

### 12.3 Berry geometry and topological K-theory

**Definition 12.7** (Berry geometry). For a smooth family of one-dimensional spectral projections  $P_x$  on a Hilbert space, the ranges form a line bundle. The *Berry connection* is  $\nabla = P \circ d$  and its curvature is the *Berry curvature*. Its holonomy is the *Berry phase*. The integral  $\langle c_1(L), [\Sigma] \rangle$  is the *Chern number* over an oriented closed surface  $\Sigma$ .

**Exercise 12.5** (Berry phase and quantized response). Compute the Berry curvature of a two-level Hamiltonian  $H(n) = n \cdot \sigma$  over  $S^2$  for both the lower and upper eigenline bundles. With the standard orientation on  $S^2$ , compute their opposite Chern numbers and state the sign convention used. Identify the two eigenlines with classifying maps  $f_\pm : S^2 \rightarrow \mathbb{P}(\mathbb{C}^2)$  and, for the Fubini–Study normalization  $\omega_{\text{FS}}([u], [v]) = -\text{Im} \langle u, v \rangle$ , prove

$$F_{\text{Berry}}^\pm = -2i f_\pm^* \omega_{\text{FS}}.$$

For a gapped two-dimensional periodic Hamiltonian, derive the Kubo formula expressing Hall conductance as  $e^2/h$  times the occupied-band Chern number. Deduce integer quantization under gap-preserving perturbations and compare with integer quantum Hall plateaus. For an adiabatic closed parameter loop, compute the Berry holonomy, show that it is independent of traversal rate, and identify the corresponding interference-fringe shift.

**Definition 12.8** (Grothendieck completion and stable equivalence). The *Grothendieck group* of a commutative monoid  $M$  is the universal abelian group receiving a monoid map from  $M$ . Vector bundles  $E, F$  are *stably equivalent* if  $E \oplus \mathbb{C}^m \cong F \oplus \mathbb{C}^n$  for some  $m, n$ ; this relation forgets rank and therefore describes reduced stable classes. The group  $K^0(X)$  is the Grothendieck group of complex vector bundles on compact  $X$ . For based  $X$ , let  $\widetilde{K}^0(X)$  be the kernel of restriction to the basepoint and, for every  $j \geq 0$ , define

$$\widetilde{K}^j(X) = \widetilde{K}^0(\Sigma^j X).$$

For unbased  $X$ , write  $X_+ = X \sqcup \{*\}$  and set  $K^j(X) = \widetilde{K}^j(X_+)$  for  $j \geq 0$ .

**Definition 12.9** (Relative and compactly supported K-theory). For a compact pair  $(X, A)$  and a locally compact space  $Y$ , define

$$K^j(X, A) = \widetilde{K}^j(X/A), \quad K_c^j(Y) = \widetilde{K}^j(Y^+),$$

where  $Y^+$  is the one-point compactification. A triple  $(F^0, F^1, \sigma)$  of vector bundles on  $Y$ , with  $\sigma : F^0 \rightarrow F^1$  invertible off a compact set, represents a class in  $K_c^0(Y)$  up to homotopy and stabilization.

**Definition 12.10** (The K-theory Thom class). Let  $\pi : E \rightarrow X$  be a rank- $r$  Hermitian complex vector bundle. Exterior multiplication by  $v$  and its adjoint give the odd endomorphism

$$c(v) = \varepsilon(v) - \iota(v) : \Lambda^* E_{\pi(v)} \longrightarrow \Lambda^* E_{\pi(v)}, \quad c(v)^2 = -\|v\|^2.$$

Writing  $c^+(v)$  for its even-to-odd part, the *K-theory Thom class* is

$$u_E = [\pi^* \Lambda^{\text{even}} E, \pi^* \Lambda^{\text{odd}} E, c^+] \in K_c^0(E).$$

**Exercise 12.6** (Thom isomorphism and Bott periodicity). Apply K-theory exactness and cellular induction to the preceding point calculation and prove the trivial-line result over finite CW complexes. Iterate it for trivial bundles and then use Mayer–Vietoris over bundle trivializations to prove the complex Thom isomorphism

$$K^j(X) \longrightarrow K_c^j(E), \quad a \longmapsto \pi^* a u_E, \quad j = 0, 1,$$

for finite CW complexes  $X$ . For the trivial line over a compact based finite CW complex, identify its relative Thom space with  $S^2 \wedge X$  and deduce complex Bott periodicity:

$$\widetilde{K}^j(X) \xrightarrow{\boxtimes \beta} \widetilde{K}^j(S^2 \wedge X)$$

is an isomorphism.

**Exercise 12.7** (*K*-theory of tori and class-A phases). Derive from the cofibration for  $X \times S^1$  and Bott periodicity the split formula  $K^j(X \times S^1) \cong K^j(X) \oplus K^{j-1}(X)$ , with indices taken modulo two, and compute the complex *K*-groups of tori. Show that a gapped family of finite-rank Hamiltonians defines the stable class of its occupied bundle and identify the strong and weak class-A invariants on Brillouin tori in dimensions one through three.

**Definition 12.11** (Operator K-theory). For a unital  $C^*$ -algebra  $A$ ,  $K_0(A)$  is the Grothendieck group of stable homotopy classes of projections in matrix algebras over  $A$ , where orthogonal sum gives the underlying monoid, and  $K_1(A)$  is the stable homotopy group of unitaries in matrix algebras over  $A$ .



## 12.4 Fredholm operators and elliptic index theory

**Definition 12.12** (Compact and Fredholm operators). An operator on a Hilbert space is *compact* if it is a norm limit of finite-rank operators. A bounded operator  $T : H_0 \rightarrow H_1$  is *Fredholm* if its kernel and cokernel are finite-dimensional and its range is closed; its *index* is  $\text{ind } T = \dim \ker T - \dim \text{coker } T$ . The *Calkin algebra* is  $\mathcal{Q}(H) = B(H)/\mathcal{K}(H)$ .

**Exercise 12.8** (Fredholm parametrices and the Calkin algebra). For  $T : H_0 \rightarrow H_1$ , prove that  $T$  is Fredholm exactly when there is a bounded  $S : H_1 \rightarrow H_0$  such that  $ST - \text{id}_{H_0}$  and  $TS - \text{id}_{H_1}$  are compact. When  $H_0 = H_1 = H$ , identify this condition with invertibility of the image of  $T$  in  $\mathcal{Q}(H)$ . Prove homotopy invariance, compact-perturbation invariance, and additivity of the index.

**Definition 12.13** (Elliptic symbols). For vector bundles  $E, F$  over a compact smooth manifold, the *principal symbol* of an order- $m$  differential operator  $D : \Gamma(E) \rightarrow \Gamma(F)$  is the homogeneous bundle map  $\sigma_D : \pi^*E \rightarrow \pi^*F$  on  $T^*M$ . The operator is *elliptic* if  $\sigma_D(x, \xi)$  is invertible for every  $\xi \neq 0$ . When the Sobolev extension of  $D$  is Fredholm, its Fredholm index is the *analytic index*. The symbol defines a compactly supported K-class on  $T^*M$ .

**Exercise 12.9** (Symbol winding and Toeplitz index). Let  $H^2(S^1)$  be the closed span of  $1, z, z^2, \dots$  in  $L^2(S^1)$  and let  $P_+$  be the orthogonal projection onto it. For  $m \in \mathbb{Z}$ , compute the kernel and cokernel of  $T_{z^m} = P_+ M_{z^m} P_+$  and prove

$$\text{ind}(T_{z^m}) = -m.$$

Identify  $m$  with the winding number of the symbol  $z^m$ . Use compact-perturbation invariance and multiplication of monomials to show that this index is stable under compact perturbations of the infinite-dimensional Toeplitz operator. If  $P_N$  projects onto  $\text{span}\{1, z, \dots, z^N\}$ , compute the kernel and cokernel of  $P_N T_{z^m} P_N$  for  $N \geq |m|$  and show that its finite-dimensional index is zero: the artificial upper cutoff creates a defect that cancels the one-sided Toeplitz defect. Explain why square finite-rank spectral truncations therefore do not preserve the Fredholm index. Compare this compression with Exercise 11.6.

## 12.5 Dirac operators, anomalies, and topological phases

**Definition 12.14** (Spin structures and Dirac operators). A *spin structure* on an oriented Riemannian  $n$ -manifold is a lift of its oriented orthonormal frame bundle through  $\text{Spin}(n) \rightarrow \text{SO}(n)$ . Its complex spin representation defines the *spinor bundle*  $S$ , with Clifford action  $c$  and spin connection. The *Dirac operator* is  $D = c \circ \nabla^S$ . In even dimensions the chirality operator splits  $S = S^+ \oplus S^-$ . In formal curvature roots,

$$\hat{A}(TM) = \prod_j \frac{x_j/2}{\sinh(x_j/2)}.$$

**Definition 12.15** (Chirality and instanton number). Let  $M$  be an oriented Riemannian spin four-manifold with the spinor splitting above. For a Hermitian vector bundle  $E$  with unitary connection  $A$ , the coupled Dirac operator is odd and has a *chiral part*

$$D_A^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E).$$

For an instanton or anti-instanton from Exercise 6.8, its *instanton number* is

$$k = -\frac{1}{8\pi^2} \int_M \text{Tr}(F_A \wedge F_A) \in \mathbb{Z}.$$

With the Chern–Weil convention  $\text{ch}(E) = \text{Tr} \exp(iF_A/2\pi)$  used above,  $k = \int_M \text{ch}_2(E)$ . Thus for an  $SU(r)$  bundle,  $\text{ch}_2(E) = -c_2(E)$ ; conventions that call  $\int_M c_2(E)$  the instanton charge use the opposite sign.

**Exercise 12.10** (Index theory and invertible free-fermion phases). Let  $H(k)$  be a continuous family of finite-dimensional self-adjoint free-fermion Hamiltonians over a compact momentum space  $X$ , with a uniform gap at zero. Flatten its spectrum to

$$Q(k) = \text{sgn } H(k) = H(k)|H(k)|^{-1}, \quad P(k) = \frac{1 - Q(k)}{2},$$

and prove that the occupied spaces  $E_k = \text{im } P(k)$  form a vector bundle  $E \rightarrow X$ . Show that its stable class  $[E] \in K^0(X)$  is invariant under gap-preserving deformations.

Prove that stacking Hamiltonians gives direct sum and addition in  $K^0(X)$ . After passing to reduced  $K$ -theory, show that spectral reversal replaces  $E$  by  $E^\perp$  and gives the additive inverse because  $E \oplus E^\perp$  is trivial. Explain why an invertible phase can have protected boundary modes or symmetry defects but no nontrivial intrinsic bulk fusion sectors. Contrast this group-valued classification with the sector-valued noninvertible phases in the next section.

**Exercise 12.11** (The free-fermion periodic table of topological insulators and superconductors). Let  $H(k)$  be a spectrally flattened Bloch or Bogoliubov–de Gennes Hamiltonian on the compactified momentum space  $S^d$ . Time-reversal, particle-hole, and chiral symmetries act by

$$\mathsf{T}H(k)\mathsf{T}^{-1} = H(-k), \quad \mathsf{C}H(k)\mathsf{C}^{-1} = -H(-k), \quad \mathsf{S}H(k)\mathsf{S}^{-1} = -H(k),$$

where  $\mathsf{T}$  and  $\mathsf{C}$  are antiunitary, with square  $+1$  or  $-1$  when present, and  $\mathsf{S}$  is unitary, with its phase chosen so that  $\mathsf{S}^2 = 1$ . Organize the complex classes A, AIII and the real classes in the order

| $s$            | 0  | 1   | 2  | 3    | 4   | 5   | 6  | 7  |
|----------------|----|-----|----|------|-----|-----|----|----|
| class          | AI | BDI | D  | DIII | AII | CII | C  | CI |
| $\mathsf{T}^2$ | +1 | +1  | 0  | -1   | -1  | -1  | 0  | +1 |
| $\mathsf{C}^2$ | 0  | +1  | +1 | +1   | 0   | -1  | -1 | -1 |

and verify that when both antiunitary symmetries occur their product gives a chiral symmetry, up to an irrelevant phase.

For this exercise, define the complex strong-classification entries in dimension  $d$  by

$$\pi_0(C_{-d}) \quad \text{for A}, \quad \pi_0(C_{1-d}) \quad \text{for AIII},$$

and define the entry in real class  $s$  by

$$\pi_0(R_{s-d}) \cong KO^{d-s}(\text{pt}).$$

Indices are periodic, and their coefficient groups are

$$\pi_0(C_q) = \begin{cases} \mathbb{Z} & q \equiv 0 \pmod{2}, \\ 0 & q \equiv 1 \pmod{2}, \end{cases} \quad \begin{array}{c|cccccccc} q \bmod 8 & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline \pi_0(R_q) & \mathbb{Z} & \mathbb{Z}/2 & \mathbb{Z}/2 & 0 & \mathbb{Z} & 0 & 0 & 0. \end{array}$$

Use these data, rather than a memorized chart, to fill every entry of the periodic table:

| class | $T^2$ | $C^2$ | $S$ | $d = 0$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|-------|-------|-------|-----|---------|---|---|---|---|---|---|---|
| A     | 0     | 0     | 0   |         |   |   |   |   |   |   |   |
| AIII  | 0     | 0     | 1   |         |   |   |   |   |   |   |   |
| AI    | +1    | 0     | 0   |         |   |   |   |   |   |   |   |
| BDI   | +1    | +1    | 1   |         |   |   |   |   |   |   |   |
| D     | 0     | +1    | 0   |         |   |   |   |   |   |   |   |
| DIII  | -1    | +1    | 1   |         |   |   |   |   |   |   |   |
| AII   | -1    | 0     | 0   |         |   |   |   |   |   |   |   |
| CII   | -1    | -1    | 1   |         |   |   |   |   |   |   |   |
| C     | 0     | -1    | 0   |         |   |   |   |   |   |   |   |
| CI    | +1    | -1    | 1   |         |   |   |   |   |   |   |   |

Recover in particular the integer quantum Hall phase in class A, the one-dimensional Majorana chain in class D, the two-dimensional quantum spin Hall phase in class AII, and the three-dimensional class-DIII topological superconductor. Explain why an abstract  $\mathbb{Z}$  entry coming from  $R_4$  is often printed as  $2\mathbb{Z}$  when its generator maps to twice the corresponding complex invariant; in dimension zero this is the even complex dimension forced by Kramers degeneracy. Finally, explain why this is only the table of *strong* phases: a lattice Brillouin torus can also carry weak invariants. Relate the symmetry-free case to the earlier decomposition of  $K^*(\mathbb{T}^d)$ , and state why antiunitary symmetries require real, equivariant, or twisted  $K$ -theory, following Freed–Moore, *Annales Henri Poincaré* **14** (2013). State why this stable noninteracting classification need not remain unchanged in the presence of strong interactions.

## 12.6 Spectral triples, ultraviolet cutoffs, and operator systems

**Definition 12.16** (Spectral triples and spectral distance). A *spectral triple*  $(A, H, D)$  consists of a unital  $*$ -algebra  $A$  represented faithfully by bounded operators on a Hilbert space  $H$  and a self-adjoint operator  $D$  with dense domain such that  $(1 + D^2)^{-1/2}$  is compact, and each commutator  $[D, a]$ , initially defined on  $\text{Dom}(D)$ , extends to a bounded operator. Its Lipschitz seminorm is

$$L_D(a) = \|[D, a]\|.$$

On the state space of the  $C^*$ -closure of  $A$ , the associated extended *spectral distance* is

$$d_D(\varphi, \psi) = \sup\{|\varphi(a) - \psi(a)| : a = a^*, L_D(a) \leq 1\}.$$

The value may be infinite if the commutator fails to distinguish enough observables.

**Exercise 12.12** (Recovering Riemannian distance). Let  $M$  be a compact connected Riemannian spin manifold, let  $A = C^\infty(M)$  act by multiplication on  $H = L^2(M, S)$ , and let  $D$  be its Dirac operator. Prove

$$[D, f] = c(df), \quad \|[D, f]\| = \sup_{x \in M} \|df_x\|.$$

Using the characterization of geodesic distance by 1-Lipschitz functions, show for the evaluation states  $\delta_x, \delta_y$  that

$$d_D(\delta_x, \delta_y) = d_M(x, y).$$

Thus the metric can be recovered from how the Dirac operator fails to commute with the observable algebra.

**Definition 12.17** (Spectral truncation). For  $\Lambda > 0$ , let

$$P_\Lambda = \mathbf{1}_{[-\Lambda, \Lambda]}(D), \quad H_\Lambda = P_\Lambda H, \quad D_\Lambda = P_\Lambda D P_\Lambda.$$

Compactness of the resolvent makes  $H_\Lambda$  finite-dimensional. The compressed observables form

$$S_\Lambda = P_\Lambda A P_\Lambda = \{P_\Lambda a P_\Lambda : a \in A\} \subseteq B(H_\Lambda).$$

As in equivariant compression, this is an operator system with identity  $P_\Lambda$ , inherited adjoint, and inherited matrix-order cones. A state on  $S_\Lambda$  is a positive linear functional  $\omega$  with  $\omega(P_\Lambda) = 1$ . The truncated commutator seminorm and spectral distance are

$$L_\Lambda(x) = \|[D_\Lambda, x]\|, \quad d_\Lambda(\omega, \eta) = \sup_{\substack{x=x^* \\ L_\Lambda(x) \leq 1}} |\omega(x) - \eta(x)|.$$

**Exercise 12.13** (Metric geometry of the truncated circle). Return to the Toeplitz operator system  $S_N$  from Exercise 11.6, restrict the algebra to  $A = C^\infty(S^1)$ , and set

$$D = -i \frac{d}{d\theta}, \quad D_N = P_N D P_N.$$

Verify

$$[D_N, P_N M_f P_N]_{jk} = (j - k) \hat{f}(j - k)$$

and deduce that the kernel of  $L_N$  on  $S_N$  consists only of scalars.

Show that rotations commute with  $D_N$  and hence preserve  $L_N$  and the spectral distance on the state space of  $S_N$ . For

$$v_{N,\theta} = (2N + 1)^{-1/2} \sum_{k=-N}^N e^{-ik\theta} e_k,$$

identify the expectation  $\langle v_{N,\theta}, P_N M_f P_N v_{N,\theta} \rangle$  with convolution by a Fejér kernel and show that it tends to  $f(\theta)$  as  $N \rightarrow \infty$ . Interpret these vector states as points blurred by finite spectral resolution; see Connes–van Suijlekom, *Spectral Truncations in Noncommutative Geometry and Operator Systems*, Communications in Mathematical Physics **383** (2021).

## 13 Quantum Topology

### 13.1 Subfactors, Temperley–Lieb algebras, and the Jones polynomial

**Definition 13.1** (Subfactors, conditional expectations, and index). A *subfactor* is an inclusion  $N \subseteq M$  of factors with common unit. Conditional expectations  $E : M \rightarrow N$  are understood in the sense of Definition 8.9. For finite factors with normalized trace, the *Jones index*  $[M : N]$  is the Murray–von Neumann dimension of  $L^2(M)$  as a left  $N$ -module, and  $E_N : M \rightarrow N$  denotes the unique trace-preserving normal conditional expectation.

For arbitrary factors and a faithful normal conditional expectation  $E$ , set

$$[M : N]_E = \lambda_E^{-1}, \quad \lambda_E = \sup(\{0\} \cup \{\lambda > 0 : E(x) \geq \lambda x \text{ for every } x \in M_+\}).$$

The *minimal index* is  $[M : N]_{\min} = \inf_E [M : N]_E$ , with value  $+\infty$  if no finite-index expectation exists. For the type-III inclusions below,  $[M : N]$  denotes this minimal index.

**Definition 13.2** (Basic construction). For finite factors  $N \subseteq M$ , the *Jones projection*  $e_N$  on  $L^2(M)$  is the orthogonal projection onto  $L^2(N)$ . The *basic construction* is  $M_1 = \langle M, e_N \rangle''$ . Iteration gives the *Jones tower*  $N \subseteq M \subseteq M_1 \subseteq M_2 \subseteq \dots$ .

**Exercise 13.1** (Jones tower and Temperley–Lieb relations). For a finite-index inclusion  $N \subseteq M$  of finite factors, derive  $e_N x e_N = E_N(x) e_N$  and construct the canonical trace on the basic construction. Prove that the normalized Jones projections in the tower satisfy

$$e_i^2 = e_i, \quad e_i e_j = e_j e_i \quad (|i - j| > 1), \quad e_i e_{i \pm 1} e_i = [M : N]^{-1} e_i.$$

**Definition 13.3** (Braid groups and Markov traces). The *braid group*  $B_n$  has generators  $\sigma_i$  with relations  $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$  and  $\sigma_i \sigma_j = \sigma_j \sigma_i$  for  $|i - j| > 1$ . A *Markov trace* on a tower of braid-group quotients is a compatible trace satisfying a fixed stabilization rule.

**Exercise 13.2** (Jones polynomial). Construct a braid-group representation from the Temperley–Lieb generators. Use the Markov trace and Markov moves to obtain an invariant of oriented links, normalize it to the Jones polynomial, and derive its skein relation. Compute it for the unknot, trefoil, and Hopf link.

## 13.2 Conformal nets and superselection sectors

**Definition 13.4** (Local conformal nets). Let  $\mathcal{I}$  be the proper open intervals of  $S^1$ . A *local conformal net* assigns a von Neumann algebra  $\mathcal{A}(I) \subset B(H)$  to each  $I \in \mathcal{I}$  such that

$$I \subset J \implies \mathcal{A}(I) \subset \mathcal{A}(J), \quad I \cap J = \emptyset \implies [\mathcal{A}(I), \mathcal{A}(J)] = 0.$$

The assignment is covariant under orientation-preserving diffeomorphisms of  $S^1$ , represented projectively on  $H$ , and the rotation generator is positive. Its vacuum vector  $\Omega$  is invariant under the Möbius subgroup and cyclic for the local algebras, and  $\bigvee_I \mathcal{A}(I) = B(H)$ . Physically,  $S^1$  is the compactified chiral light ray and  $\mathcal{A}(I)$  consists of observables localized in  $I$ . The norm-closed  $C^*$ -algebra

$$\mathfrak{A} = C^* \left( \bigcup_{I \in \mathcal{I}} \mathcal{A}(I) \right)$$

generated by the local algebras is the *quasilocal algebra*.

**Definition 13.5** (Duality and regularity for conformal nets). Write  $I'$  for the interior of the complement of  $I$ . A net has *Haag duality* when  $\mathcal{A}(I) = \mathcal{A}(I')'$ . It has the *split property* when  $\bar{I} \subset J$  implies  $\mathcal{A}(I) \subset F \subset \mathcal{A}(J)$  for some type-I factor  $F$ . It is *strongly additive* when removing an interior point from  $I$  to obtain  $I_1 \cup I_2$  gives  $\mathcal{A}(I) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$ . A representation of the quasilocal algebra is *locally normal* when its restriction to every  $\mathcal{A}(I)$  is normal.

**Exercise 13.3** (The two-interval subfactor). For  $E = I_1 \cup I_2$  a union of disjoint intervals, set  $\mathcal{A}(E) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$ . First use locality and vacuum cyclicity for  $\mathcal{A}(I')$  to prove that the vacuum is separating for  $\mathcal{A}(I)$ . Use the split property and the spatial tensor product to identify separated local algebras as independent subsystems. Then use locality to obtain the inclusion

$$\mathcal{A}(E) \subseteq \mathcal{A}(E')'.$$

Under the split and irreducibility hypotheses, show that this is a subfactor. Define the  $\mu$ -index  $\mu(\mathcal{A}) = [\mathcal{A}(E')' : \mathcal{A}(E)]$ , and prove that it is independent of the two-interval configuration by conformal covariance. Contrast single-interval Haag duality  $\mathcal{A}(I) = \mathcal{A}(I')'$  with the possible failure of duality for  $E$ , measured by this Jones index.

**Definition 13.6** (DHR sectors). A *DHR sector* of  $\mathcal{A}$  is represented by a locally normal, transportable endomorphism  $\rho$  of the quasilocal algebra that acts identically on  $\mathcal{A}(I')$  for some interval  $I$ ; transportability means that an equivalent representative can be localized in any interval. Composition is the tensor product. For a finite-index irreducible sector localized in  $I$ , its statistical dimension satisfies

$$d(\rho)^2 = [\mathcal{A}(I) : \rho(\mathcal{A}(I))].$$

**Exercise 13.4** (Critical Ising chain, Rydberg spectroscopy, and the Ising net). For  $J, h > 0$ , consider the transverse-field Ising chain

$$H_I = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i.$$

Use the Jordan–Wigner transformation and a Bogoliubov rotation to obtain

$$\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh \cos k}.$$

Show that the gap is  $2|J - h|$  and closes at  $h = J$ , where the long-wavelength modes have linear dispersion. For open chains of lengths  $L = 8, 10, 12, 14$ , diagonalize the finite Hamiltonian at  $h = J$ , separate the  $\mathbb{Z}/2\mathbb{Z}$  parity sectors, and rescale gaps by the lowest even-parity gap.

The Rydberg experiment realizes a different microscopic chain,

$$H_R = \sum_i \left( \frac{\Omega}{2} P_{i-1} X_i P_{i+1} - \Delta n_i \right) + \sum_{j-i \geq 2} V_{ij} n_i n_j,$$

in the nearest-neighbor blockade regime, where  $n_i$  projects onto the Rydberg state and  $P_i = 1 - n_i$ . At finite  $L$ , numerically diagonalize the blockaded Hilbert space, separate reflection and parity sectors, and compare the rescaled gaps with the ratios  $2 : 4 : 6 : 8$  and  $3 : 5 : 7$  reported in the modulation-spectroscopy peaks of Sun et al., *Experimental observation of conformal field theory spectra* (2026). Plot the deviations against  $1/L$  and distinguish a finite-size extrapolation from an exact finite-chain identity.

### 13.3 Fusion, braiding, and modular tensor categories

**Definition 13.7** (Monoidal and rigid categories). A *monoidal category* is a category  $\mathcal{C}$  with a tensor functor, unit object, and coherent associativity and unit isomorphisms. A *dual* of  $X$  is an object  $X^*$  with evaluation and coevaluation morphisms satisfying the snake identities. A monoidal category is *rigid* if every object has left and right duals.

**Definition 13.8** (Fusion, braided, and modular categories). A *fusion category* over  $\mathbb{C}$  is a finite semisimple rigid  $\mathbb{C}$ -linear monoidal category with simple unit. It is *unitary* when it is a  $C^*$ -category and its tensor product and structural isomorphisms are compatible with the adjoint. A *braiding* is a natural family  $c_{X,Y} : X \otimes Y \rightarrow Y \otimes X$  satisfying the hexagon identities. A *unitary modular tensor category* is a unitary ribbon fusion category with unitary nondegenerate braiding.

**Exercise 13.5** (Fusion rules and braiding). Prove that simple-object classes of a fusion category form a based ring under tensor product. Derive the pentagon and hexagon equations for associativity and braiding data. Compute quantum dimensions and braid-group representations for the Fibonacci fusion rule  $\tau \otimes \tau \cong 1 \oplus \tau$ .

### 13.4 TQFT, Reshetikhin–Turaev theory, and Chern–Simons theory

**Definition 13.9** (Bordisms and topological field theories). The *bordism category* has closed oriented  $(n - 1)$ -manifolds as objects, oriented  $n$ -dimensional bordisms as morphisms, and disjoint union as tensor product. An  *$n$ -dimensional topological field theory* is a symmetric monoidal functor from this category to vector spaces. An extended theory may assign categories to codimension-two manifolds. A TQFT is *invertible* when its state spaces are tensor-invertible (hence one-dimensional over a field) and every bordism acts invertibly; noninvertible TQFTs may instead carry nontrivial superselection and fusion sectors.

**Definition 13.10** (The Reshetikhin–Turaev  $SU(2)_k$  theory). For  $k \in \mathbb{N}$ , let  $\mathcal{C}_k = SU(2)_k$  be the unitary modular tensor category with simple objects  $V_0, \dots, V_k$ , where  $V_j$  has spin  $j/2$ . The Reshetikhin–Turaev construction uses the colored ribbon-graph calculus of  $\mathcal{C}_k$  and surgery to define a symmetric monoidal  $(2 + 1)$ -dimensional TQFT

$$Z_k^{\text{RT}} = Z_{\mathcal{C}_k}$$

on bordisms equipped with the extra framing data needed to cancel the framing anomaly. It assigns invariants to closed framed three-manifolds and to framed links colored by the objects  $V_j$ . We call  $Z_k^{\text{RT}}$  the Reshetikhin–Turaev model of quantum  $SU(2)$  Chern–Simons theory at level  $k$ ; this terminology does not define it by a functional integral.

**Definition 13.11** (Chern–Simons theory and Wilson loops). Let  $G$  have an invariant inner product on  $\mathfrak{g}$ , and fix a trivialization of a principal  $G$ -bundle over a closed oriented three-manifold. For the resulting  $\mathfrak{g}$ -valued connection potential  $A$ , the *Chern–Simons functional* at level  $k$  is

$$S_{CS}(A) = \frac{k}{4\pi} \int_M \left( \langle A \wedge dA \rangle + \frac{1}{3} \langle A \wedge [A \wedge A] \rangle \right).$$

For a representation  $V$  and closed curve  $K$ , the *Wilson loop* is  $W_V(K) = \text{Tr}_V(\text{Hol}_A(K))$ .

**Exercise 13.6** (Gauge invariance, level, and knot invariants). For compact, connected, simply connected simple  $G$ , normalize the inner product so that the winding term of a gauge transformation lies in  $2\pi\mathbb{Z}$ . Compute the change of  $S_{CS}$  and derive  $k \in \mathbb{Z}$  as the condition for  $e^{iS_{CS}}$  to be gauge invariant. Prove that its critical points are flat connections.

**Exercise 13.7** (Reshetikhin–Turaev links and Jones evaluations). Use the Temperley–Lieb realization of  $\mathcal{C}_k$  and its ribbon functor to show that the  $Z_k^{\text{RT}}$  invariant of a framed link in  $S^3$ , with a component colored by  $V_j$ , is the corresponding  $j$ -colored Jones evaluation at

$$q = e^{2\pi i/(k+2)}.$$

In particular, identify  $V_1$  with the fundamental color giving the ordinary Jones polynomial. Compute the normalized invariants of the unknot, Hopf link, and trefoil, state the normalization of the colored unknot, and track the framing factor separately from the ambient-isotopy invariant.

### 13.5 Anyons, topological order, and quantum Hall phases

**Definition 13.12** (Anyons and topological degeneracy). In a two-dimensional gapped phase, an *anyon type* is a simple object of its braided fusion category. It is *abelian* if its quantum dimension is 1 and *nonabelian* otherwise. The *topological degeneracy* on a closed surface is the dimension of the state space assigned by the associated topological field theory.

**Exercise 13.8** (The toric code: error correction from topological order). Give an  $L \times L$  square cellulation of the torus, with  $L \geq 2$ , a qubit on every edge, and denote the resulting physical Hilbert space by  $\mathcal{H}$ . Let  $X$  and  $Z$  be self-adjoint unitaries satisfying  $X^2 = Z^2 = 1$  and  $XZ = -ZX$  on one qubit. Operators on distinct edges commute. For each vertex  $v$  and plaquette  $p$ , set

$$A_v = \prod_{e \ni v} X_e, \quad B_p = \prod_{e \in \partial p} Z_e, \quad H = - \sum_v A_v - \sum_p B_p.$$

Prove that all  $A_v$  and  $B_p$  commute and that the ground space is their common  $+1$  spectral subspace. Establish the two relations  $\prod_v A_v = 1$  and  $\prod_p B_p = 1$ , prove that the remaining stabilizers are independent, find  $E_0 = -2L^2$ , and compute

$$\dim \ker(H - E_0) = 2^{2L^2 - (L^2 + L^2 - 2)} = 4.$$

Show that an open  $Z$ -string creates violated vertex terms only at its endpoints, while an open dual  $X$ -string creates violated plaquette terms only at its endpoints. Prove that contractible closed strings are products of stabilizers. Show that strings around the two noncontractible cycles give two pairs of logical operators, with a primal  $Z$ -loop and a dual  $X$ -loop anticommuting exactly when their intersection number is odd. Deduce that the code encodes two logical qubits and has distance  $L$ .

Let  $\mathcal{S}$  be the stabilizer group generated by the  $A_v$  and  $B_p$ , let  $\mathcal{S}' \subseteq B(\mathcal{H})$  be its commutant, and let  $P$  be the ground space projection. Prove that the logical observable algebra is

$$\mathcal{A}_{\log} = P\mathcal{S}'P = \text{span}\{P\bar{X}_1^{i_1}\bar{Z}_1^{j_1}\bar{X}_2^{i_2}\bar{Z}_2^{j_2}P : i_r, j_r \in \{0, 1\}\} \cong M_2(\mathbb{C}) \otimes M_2(\mathbb{C}),$$

where the barred operators are noncontractible logical loops. Equivalently, identify the logical Pauli group as the Pauli centralizer of  $\mathcal{S}$  modulo  $\mathcal{S}$ . If an error channel has Kraus operators  $E_a$  for which every  $E_a^*E_b$  is supported on fewer than  $L$  edges, prove that

$$PE_a^*E_bP \in \mathbb{C}P = \mathcal{A}'_{\log}$$

where, as in the definition of protected observable algebras, the prime is taken inside the code corner  $PB(\mathcal{H})P$ , and hence verify the exact OAQEC condition for  $\mathcal{A}_{\log}$ . Deduce in particular that the toric code corrects arbitrary errors on at most  $\lfloor (L-1)/2 \rfloor$  edges.

Interpret the endpoint excitations as anyons  $e$  and  $m$ . Derive  $e^2 = m^2 = 1$ , set  $\varepsilon = em$ , and show that taking  $e$  once around  $m$  produces the phase  $-1$ . Explain how local error syndromes, global logical Wilson loops, and the fourfold TQFT degeneracy express the same topological structure; see Kitaev, *Fault-tolerant quantum computation by anyons*, and Dauphinais–Kribs–Vasmer, *Stabilizer Formalism for Operator Algebra Quantum Error Correction*, Quantum **8** (2024).

**Exercise 13.9** (Quantum Hall phases and measured topological order). Use flux insertion for a Laughlin state at filling  $\nu = 1/m$  to derive quasiparticle charge  $e/m$ , exchange phase  $\pi/m$ , Hall conductance  $\nu e^2/h$ , and genus- $g$  degeneracy  $m^g$ . Compare these consequences separately with conductance plateaus, fractional shot-noise charge measurements, and interferometric phase data, stating which topological predictions each observation tests and which it does not.

## 14 Topological Structures in Holography

### 14.1 Ryu–Takayanagi surfaces and entanglement wedges

**Definition 14.1** (Relative homology and the Ryu–Takayanagi surface). Let  $\Sigma$  be a static bulk time slice with conformal boundary  $B$ , and let  $A \subset B$  be a spatial region. A codimension-one surface



$\gamma \subset \Sigma$  satisfies the *RT homology constraint* for  $A$  if

$$\partial\gamma = \partial A \quad \text{and} \quad \partial r_A = A \cup \gamma$$

with the appropriate orientations for some bulk region  $r_A \subset \Sigma$ . Equivalently, after regulating the conformal boundary, the cycle  $A - \gamma$  is null-homologous in the bulk.

In a static state with a semiclassical asymptotically anti-de Sitter dual, the *Ryu–Takayanagi prescription* at leading order is

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G_N},$$

where  $\gamma_A$  has least area among surfaces satisfying the anchoring and homology conditions. The cutoff used to regulate its area is the geometric counterpart of the field-theory ultraviolet cutoff.

**Exercise 14.1** (An interval in AdS<sub>3</sub>). On the time-zero slice of Poincaré AdS<sub>3</sub>, use

$$ds^2 = L^2 \frac{dz^2 + dx^2}{z^2}, \quad z > 0,$$

and regulate the boundary at  $z = \epsilon$ . Show that the geodesic anchored at the endpoints of an interval of length  $\ell$  is a semicircle orthogonal to  $z = 0$ , and compute

$$\text{Length}(\gamma_A) = 2L \log \frac{\ell}{\epsilon} + O(\epsilon^2/\ell^2).$$

Using the Brown–Henneaux relation  $c = 3L/(2G_3)$ , derive

$$S(A) = \frac{c}{3} \log \frac{\ell}{\epsilon} + O(1).$$

Compare this with the replica calculation for a vacuum interval in a two-dimensional CFT. State clearly that the equality is a test of the holographic dictionary in its semiclassical regime, not a derivation of AdS/CFT for arbitrary CFTs; see Ryu–Takayanagi, *Physical Review Letters* **96** (2006) and Brown–Henneaux, *Communications in Mathematical Physics* **104** (1986).

**Definition 14.2** (Entanglement wedges, HRT surfaces, and quantum corrections). For a static RT surface, the *entanglement wedge* of  $A$  is the domain of dependence of the bulk region  $r_A$  bounded by  $A \cup \gamma_A$ . In a time-dependent spacetime, the *Hubeny–Rangamani–Takayanagi surface* is instead an extremal codimension-two spacetime surface anchored at  $\partial A$  and homologous to  $A$ , with least area among the appropriate extremal candidates.

Beyond leading classical order, a *quantum extremal surface* extremizes the generalized entropy

$$S_{\text{gen}}(\gamma) = \frac{\text{Area}(\gamma)}{4G_N} + S_{\text{bulk}}(r_A),$$

and the entropy is obtained from the least admissible extremum. Thus RT is the static classical limit, HRT is its covariant classical extension, and the quantum-extremal prescription includes bulk entanglement corrections.

**Exercise 14.2** (Topology change for two intervals). On a constant-time slice of Poincaré AdS<sub>3</sub>, let  $A = [x_1, x_2]$  and  $B = [x_3, x_4]$  with  $x_1 < x_2 < x_3 < x_4$ . The two admissible pairings of boundary endpoints have regulated lengths

$$L_{\text{disc}} = 2L \log \frac{x_2 - x_1}{\epsilon} + 2L \log \frac{x_4 - x_3}{\epsilon},$$

$$L_{\text{conn}} = 2L \log \frac{x_4 - x_1}{\epsilon} + 2L \log \frac{x_3 - x_2}{\epsilon}.$$

Show that the RT surface changes topology when these two expressions cross, and derive

$$I(A : B) = \max \left\{ 0, \frac{c}{3} \log \frac{(x_2 - x_1)(x_4 - x_3)}{(x_4 - x_1)(x_3 - x_2)} \right\}.$$

Explain why vanishing leading-order mutual information corresponds to a disconnected classical entanglement wedge, while finite- $G_N$  bulk correlations need not vanish exactly.

**Exercise 14.3** (Entropy inequalities from cutting and regluing). For disjoint  $A$  and  $B$  in a static bulk slice, draw minimal representatives for the regions needed in subadditivity and strong subadditivity. Cut the surfaces where they intersect and reglue the pieces into admissible, not necessarily minimal, surfaces for the opposite sides of

$$S(A) + S(B) \geq S(A \cup B),$$

and

$$S(AB) + S(BC) \geq S(B) + S(ABC).$$

Use minimality to prove the inequalities. Identify where the relative homology constraint is needed to ensure that the reglued surfaces are admissible. Explain why this argument is geometric and classical and why the bulk-entropy term must be included at quantum order.

## 14.2 Black-hole topology and entropy

**Definition 14.3** (The BTZ horizon and thermofield-double topology). The nonrotating BTZ black hole has metric

$$ds^2 = -\frac{r^2 - r_+^2}{L^2} dt^2 + \frac{L^2}{r^2 - r_+^2} dr^2 + r^2 d\phi^2, \quad \phi \sim \phi + 2\pi,$$

temperature  $T = r_+/(2\pi L^2)$ , and horizon entropy

$$S_{\text{BH}} = \frac{2\pi r_+}{4G_3}.$$

Its maximally extended geometry has two asymptotic boundaries and is dual, in the standard semiclassical dictionary, to a thermofield-double state of two CFT copies. The bifurcation circle is homologous to either complete boundary separately, but not to the union of both boundaries.

**Exercise 14.4** (Homology, horizons, and black-hole entropy). For the pure thermofield-double state on  $H_L \otimes H_R$ , prove for its reduced states that  $S(\rho_L) = S(\rho_R)$  while the joint entropy vanishes. Apply the RT homology constraint to show that the surface for one complete boundary is the bifurcation circle, while the empty surface is admissible for the union of both boundaries. Recover  $S(H_L) = 2\pi r_+/(4G_3)$  and interpret the result as entanglement between the two boundary theories.

For a boundary circle of circumference  $2\pi L$ , use  $\beta = 2\pi L^2/r_+$  and  $c = 3L/(2G_3)$  to verify that the high-temperature CFT expression

$$S_{\text{thermal}} = \frac{\pi c}{3} \frac{2\pi L}{\beta}$$

equals  $S_{\text{BH}}$ . In the Arithmetic part this thermal formula will be recovered from modular invariance and asymptotic state counting. Consult Hubeny–Rangamani–Takayanagi, *Journal of High Energy Physics* **07** (2007) for the covariant prescription and Faulkner–Lewkowycz–Maldacena, *Journal of High Energy Physics* **11** (2013) for the bulk quantum correction, and Engelhardt–Wall, *Journal of High Energy Physics* **01** (2015) for quantum extremal surfaces.

## 15 Gauge Duality and the Geometric Langlands Program

### 15.1 Supersymmetry and electric–magnetic duality

**Definition 15.1** ( $\mathcal{N} = 4$  super Yang–Mills). Let  $G$  be a compact Lie group and let  $P \rightarrow M$  be a principal  $G$ -bundle over an oriented Riemannian four-manifold. The bosonic fields of  $\mathcal{N} = 4$  super Yang–Mills theory are a connection  $A$  and six adjoint-valued scalar fields  $X_1, \dots, X_6$ . With an invariant positive inner product on  $\mathfrak{g}$ , the bosonic action has the form

$$\frac{1}{g^2} \int_M \left( \frac{1}{2} |F_A|^2 + \frac{1}{2} \sum_I |d_A X_I|^2 + \frac{1}{2} \sum_{I < J} |[X_I, X_J]|^2 \right) \text{vol} - i\theta k(A),$$

supplemented by fermions and Yukawa couplings fixed by supersymmetry. The six scalars transform under the  $R$ -symmetry group  $\text{Spin}(6)_R \cong SU(4)_R$ , and the theory has sixteen real supercharges. Its gauge coupling and theta angle combine into

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}.$$

The potential is minimized when the  $X_I$  commute, so scalar expectation values can break  $G$  to a maximal torus and make electric and magnetic charges visible simultaneously.

**Definition 15.2** (Langlands dual group). Let  $T \subset G$  be a maximal torus. Its character and cocharacter lattices are

$$X^*(T) = \text{Hom}(T, U(1)), \quad X_*(T) = \text{Hom}(U(1), T).$$

The *Langlands dual group*  ${}^L G$  is the compact form whose complexified root datum is dual to that of  $G$ : characters and cocharacters are exchanged, and roots are exchanged with coroots. Thus

$$X^*({}^L T) = X_*(T), \quad X_*({}^L T) = X^*(T).$$

Ignoring global-form refinements,  $U(1)$  is self-dual,  ${}^L SU(n) = PSU(n)$ , and the root systems  $B_n$  and  $C_n$  are exchanged. The complex reductive group used in geometric Langlands is  ${}^L G_{\mathbb{C}}$ .

**Definition 15.3** (Electric–magnetic duality transformation). The basic  $S$ -transformation on gauge-theory parameters is

$$(G, \tau) \mapsto ({}^L G, -1/\tau).$$

It exchanges character and cocharacter lattices, and hence electric and magnetic charge labels. For a non-simply-laced root system define the transformation by  $\tau \mapsto -1/(n_G \tau)$ , where  $n_G$  is the ratio of squared long- and short-root lengths.

**Exercise 15.1** (Charge lattices and the dual group). For a torus  $T = U(1)^r$ , prove that a representation restricts to characters  $\lambda : T \rightarrow U(1)$  with  $\lambda \in X^*(T)$ . A magnetic monopole defines a transition map around its equator, hence a cocharacter  $\mu : U(1) \rightarrow T$  with  $\mu \in X_*(T)$ . Show that composition gives an integer pairing

$$\langle \lambda, \mu \rangle \in \mathbb{Z},$$

which is the Dirac quantization condition. Verify directly that exchanging  $X^*(T)$  and  $X_*(T)$  exchanges electric and magnetic labels. For  $G = SU(2)$ , compare the weight and coweight lattices and explain why the global dual group is adjoint rather than simply connected.

## 15.2 The geometric Langlands twist

**Definition 15.4** (Cohomological twist data). In Euclidean signature the rotation group is locally  $\text{Spin}(4) = SU(2)_+ \times SU(2)_-$ . A *topological twist* replaces it by a diagonal combination with a  $\text{Spin}(4)$  subgroup of  $\text{Spin}(6)_R$ . The *cohomological twist data* include scalar odd operators  $Q_\ell, Q_r$  whose projective linear combinations

$$Q_t = Q_\ell + tQ_r, \quad t \in \mathbb{P}^1,$$

satisfy  $Q_t^2 = 0$  on gauge-invariant quantities. The associated observables are classes

$$H_{Q_t} = \ker Q_t / \text{im } Q_t.$$

**Definition 15.5** (Canonical parameter). For  $(\tau, t) \in \mathbb{H} \times \mathbb{P}^1$ , define the *canonical parameter*

$$\Psi = \frac{\tau + \bar{\tau}}{2} + \frac{\tau - \bar{\tau}}{2} \frac{t - t^{-1}}{t + t^{-1}} = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2} \frac{t - t^{-1}}{t + t^{-1}}.$$

Under the  $S$ -transformation, define the twist-parameter change by  $t \mapsto \pm t\tau/|\tau|$ ; the sign is immaterial because  $t$  and  $-t$  define equivalent twists. This rule and  $\tau \mapsto -1/(n_G\tau)$  give  $\Psi \mapsto -1/(n_G\Psi)$ , where  $n_G$  is the ratio of the squared lengths of long and short roots and equals one in the simply laced case.

**Definition 15.6** (Kapustin–Witten equations). For a connection  $A$ , an adjoint-valued one-form  $\phi$ , and  $t \in \mathbb{P}^1$ , the *Kapustin–Witten equations* are

$$\begin{aligned} (F_A - \phi \wedge \phi + t d_A \phi)^+ &= 0, \\ (F_A - \phi \wedge \phi - t^{-1} d_A \phi)^- &= 0, \\ d_A^* \phi &= 0. \end{aligned} \tag{KW}$$

At  $t = 0$  and  $t = \infty$ , multiply the appropriate equation by  $t$  or  $t^{-1}$  before taking the limit.

**Exercise 15.2** ( $Q$ -cohomology and the Kapustin–Witten parameter). Let  $Q^2 = 0$ . Prove that adding  $QV$  to a  $Q$ -closed element does not change its class in  $\ker Q / \text{im } Q$ , and determine why the same statement is undefined outside  $\ker Q$ . Next substitute  $t = i$  into (KW) and verify directly that its first two lines combine into

$$F_{A+i\phi} = F_A - \phi \wedge \phi + i d_A \phi = 0.$$

Finally, compute  $\Psi$  at  $t = \pm 1$  and at  $t = \pm i$ , and verify algebraically that  $S$ -duality acts by  $\Psi \mapsto -1/(n_G\Psi)$ .

## 15.3 Dimensional reduction and Hitchin moduli space

**Definition 15.7** (Hitchin equations and Higgs bundles). For a connection  $A$  on a principal  $G$ -bundle over  $C$  and  $\phi \in \Omega^1(C, \text{ad } P)$ , the *Hitchin equations* are

$$F_A - \phi \wedge \phi = 0, \quad d_A \phi = 0, \quad d_A^* \phi = 0. \tag{H}$$

Writing the  $(1, 0)$  part of  $\phi$  as  $\varphi$ , these become, up to a conventional normalization of  $\varphi$ ,

$$\bar{\partial}_A \varphi = 0, \quad F_A + [\varphi, \varphi^\dagger] = 0.$$

Thus  $(E, \varphi)$  is a  $G_{\mathbb{C}}$ -Higgs bundle:  $E$  is a holomorphic principal  $G_{\mathbb{C}}$ -bundle and

$$\varphi \in H^0(C, K_C \otimes \text{ad } E).$$

The quotient of solutions to (H) by compact gauge transformations is the *Hitchin moduli space*  $\mathcal{M}_H(G, C)$ . Equivalently, after the appropriate stability condition is imposed, it is the moduli space of polystable  $G_{\mathbb{C}}$ -Higgs bundles.

**Exercise 15.3** (Hitchin equations from four-dimensional instantons). On  $\mathbb{R}^4$  take a connection invariant under translations in  $x^3, x^4$  and write  $X = A_3, Y = A_4$ . Decompose the anti-self-duality equations into

$$F_{12} + [X, Y] = 0, \quad D_1 X - D_2 Y = 0, \quad D_1 Y + D_2 X = 0,$$

up to the sign determined by the chosen orientation. With  $z = x^1 + ix^2$  and a suitable choice of  $\varphi = (X + iY) dz$ , show that the last two equations give  $\bar{\partial}_A \varphi = 0$  and the first gives the real Hitchin equation. This calculation is not the full supersymmetric compactification, but it exhibits the same dimensional-reduction mechanism using the instantons already studied in Exercise 6.8.

## 15.4 From gauge theory to branes

**Definition 15.8** (Sigma-model reduction). A two-dimensional sigma model on  $\Sigma$  with target  $X$  has fields  $\Phi : \Sigma \rightarrow X$ . In the adiabatic limit of the twisted gauge theory on  $\Sigma \times C$ , its low-energy configurations are slowly varying solutions of (H), hence maps

$$\Phi : \Sigma \longrightarrow \mathcal{M}_H(G, C).$$

The twist determines which topological sigma model survives. At  $\Psi = \infty$  it is the  $B$ -model in complex structure  $J$ ; at  $\Psi = 0$  it is the  $A$ -model with symplectic form  $\omega_K$ .

**Definition 15.9** (Mirror-dual Hitchin targets). The Hitchin targets for  $G$  and  ${}^L G$  are *mirror-dual at the canonical parameters* when there is an equivalence

$$B_J(\mathcal{M}_H(G, C)) \simeq A_K(\mathcal{M}_H({}^L G, C)), \quad (\text{M})$$

where the subscripts record the complex or symplectic structure.

## 15.5 Line operators, Hecke modifications, and eigensheaves

**Definition 15.10** (Wilson and 't Hooft lines). A Wilson line measures the holonomy of a connection in a representation  $R$ :

$$W_R(\gamma) = \text{Tr}_R \text{Hol}_\gamma(\mathcal{A}).$$

At  $\Psi = \infty$ , the topological line uses the flat complex connection  $\mathcal{A} = A + i\phi$  and is labeled by a representation of  $G$ . A *'t Hooft line* is defined instead by prescribing a magnetic singularity transverse to the line. Its charge is a cocharacter of  $G$ , equivalently a weight of  ${}^L G$ . The  $S$  transformation exchanges these two kinds of probes. When a line lies along a curve in  $\Sigma$  at a point  $x \in C$ , it acts on branes in the reduced sigma model.

**Definition 15.11** (Hecke modification). Let  $x \in C$ . A *Hecke modification* of a holomorphic principal  $G_{\mathbb{C}}$ -bundle  $E$  at  $x$  is another bundle  $E'$  together with an isomorphism

$$E|_{C \setminus \{x\}} \cong E'|_{C \setminus \{x\}}.$$

Its relative position at  $x$  is labeled by a dominant coweight of  $G$ . A *Hecke functor*  $H_V$  is the correspondence on bundles and sheaves obtained by summing modifications with dominant coweight labeled by the representation  $V$  of  ${}^L G$ .

**Definition 15.12** ( $D$ -modules and Hecke eigensheaves). Let  $\text{Bun}_G(C)$  denote the moduli stack of holomorphic  $G_{\mathbb{C}}$ -bundles on  $C$ ; a stack is needed because bundles can have automorphisms. A  $D$ -module on it is, schematically, a sheaf carrying a compatible action of differential operators. Let  $\mathcal{E}$  be a flat  ${}^L G_{\mathbb{C}}$ -bundle on  $C$ . A *Hecke eigensheaf* with eigenvalue  $\mathcal{E}$  is a  $D$ -module  $\mathcal{F}_{\mathcal{E}}$  equipped, for every representation  $V$  of  ${}^L G$ , with compatible isomorphisms

$$H_V(\mathcal{F}_{\mathcal{E}}) \cong V_{\mathcal{E}} \boxtimes \mathcal{F}_{\mathcal{E}}, \quad (\text{HE})$$

where  $V_{\mathcal{E}}$  is the flat vector bundle obtained from  $\mathcal{E}$  through  $V$ . Compatibility with tensor products of representations is part of the definition.

**Definition 15.13** (The Kapustin–Witten dictionary). The *Kapustin–Witten dictionary* is the chain of labels

$$\begin{array}{c} \text{flat } {}^L G_{\mathbb{C}}\text{-connection } \mathcal{E} \\ \updownarrow \\ \text{point } B\text{-brane on } \mathcal{M}_H({}^L G, C) \\ \updownarrow \\ \text{Hecke eigenbrane on } \mathcal{M}_H(G, C) \\ \updownarrow \\ \text{Hecke eigensheaf } \mathcal{F}_{\mathcal{E}} \text{ on } \text{Bun}_G(C). \end{array}$$

The line-operator labels on the two middle entries are exchanged by character–cocharacter duality.

**Exercise 15.4** (From an eigenbrane to the Hecke condition). Suppose topological line operators  $L_V(x)$  obey the representation-ring law

$$L_V(x)L_W(x) \cong L_{V \otimes W}(x)$$

and a brane  $\mathcal{B}$  obeys  $L_V(x)\mathcal{B} \cong \mathcal{B} \otimes E_{V,x}$ . Show that the spaces  $E_{V,x}$  must satisfy  $E_{V \otimes W,x} \cong E_{V,x} \otimes E_{W,x}$ . Explain why topological motion of  $x$  makes  $E_V$  a local system on  $C$ . After Wilson–’t Hooft label exchange, rewrite this statement as (HE).

## Part V

# Arithmetic

## 16 Partitions and Generating Functions

### 16.1 Partition generating functions and finite boxes

**Definition 16.1** (Partitions and  $q$ -Pochhammer symbols). An *integer partition* of  $n \geq 0$  is a finite nonincreasing sequence  $\lambda = (\lambda_1, \dots, \lambda_r)$  of positive integers with  $|\lambda| = \sum_i \lambda_i = n$ . Let  $p(n)$  be the

number of partitions, with  $p(0) = 1$ . For a formal variable  $q$ , set

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

Analytically, the infinite product converges for  $|q| < 1$ . Euler's partition generating function is

$$P(q) = \sum_{n \geq 0} p(n)q^n = \frac{1}{(q; q)_\infty}.$$

**Exercise 16.1** (Bosonic and fermionic occupation numbers). Consider oscillator modes with one-particle energies  $\hbar\omega, 2\hbar\omega, 3\hbar\omega, \dots$ . Show that a bosonic Fock state of excitation energy  $n\hbar\omega$  is a partition of  $n$ , and derive

$$\prod_{m \geq 1} \frac{1}{1 - q^m} = \sum_{n \geq 0} p(n)q^n.$$

For fermionic occupation numbers in  $\{0, 1\}$ , show that  $\prod_{m \geq 1} (1 + q^m)$  counts partitions into distinct parts. Construct a bijection, by separating powers of two in the multiplicities, between partitions into distinct parts and partitions into odd parts, and conclude

$$\prod_{m \geq 1} (1 + q^m) = \prod_{m \geq 1} \frac{1}{1 - q^{2m-1}}.$$

Interpret the two products as different quasiparticle descriptions with the same exact degeneracies.

**Definition 16.2** (Gaussian binomial coefficients). For  $0 \leq r \leq n$ , the *Gaussian binomial coefficient* is

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_r (q; q)_{n-r}},$$

and it is zero outside this range. Despite its quotient presentation, it is a polynomial in  $q$  with nonnegative integer coefficients.

## 16.2 Rogers–Ramanujan identities and exclusion rules

**Definition 16.3** (Rogers–Ramanujan sum and product series). For  $|q| < 1$ , define

$$G_{\text{sum}}(q) = \sum_{r \geq 0} \frac{q^{r^2}}{(q; q)_r}, \quad H_{\text{sum}}(q) = \sum_{r \geq 0} \frac{q^{r(r+1)}}{(q; q)_r},$$

$$G_{\text{prod}}(q) = \frac{1}{(q; q^5)_\infty (q^4; q^5)_\infty}, \quad H_{\text{prod}}(q) = \frac{1}{(q^2; q^5)_\infty (q^3; q^5)_\infty}.$$

**Exercise 16.2** (Difference conditions and Rogers–Ramanujan counting). Fix the number  $r$  of parts. Subtract the staircase  $(2r - 1, 2r - 3, \dots, 1)$  to prove that  $q^{r^2}/(q; q)_r$  generates partitions with exactly  $r$  parts whose successive parts differ by at least two. Use the staircase  $(2r, 2r - 2, \dots, 2)$  for the same difference condition with smallest part at least two. Interpret the product sides as partitions into parts congruent to 1 or 4 modulo 5, and to 2 or 3 modulo 5, respectively. Compute all four series through  $q^{30}$  using finite products and finite sums, and verify coefficientwise that  $G_{\text{sum}} = G_{\text{prod}}$  and  $H_{\text{sum}} = H_{\text{prod}}$  to that order. Translate the coefficient equalities into finite tests of the two partition identities and identify which further step would be required for a proof in all degrees.

## 17 Theta Functions and Modular Forms

### 17.1 Lattice theta series and Poisson duality

**Definition 17.1** (Integral lattices and theta series). A *Euclidean lattice*  $L \subset \mathbb{R}^d$  is the integer span of a basis of  $\mathbb{R}^d$ . Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y \rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

It is *integral* if  $L \subseteq L^*$ , *unimodular* if  $L = L^*$ , and *even* if  $\|x\|^2 \in 2\mathbb{Z}$  for every  $x \in L$ . For  $\tau$  in the upper half-plane and  $q = e^{2\pi i\tau}$ , its theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{\|x\|^2/2}.$$

Its coefficient of  $q^n$  counts lattice vectors of squared length  $2n$ .

**Exercise 17.1** (Poisson duality for lattice theta series). Apply Exercise 4.3 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\text{vol}(L)} \Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives  $\Theta_L(\tau+1) = \Theta_L(\tau)$ . Interpret the first identity as momentum–winding duality for a compact free boson and as the reciprocal-lattice relation behind diffraction. Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

### 17.2 Modular forms, theta functions, and eta products

**Definition 17.2** (The modular group and modular forms). The modular group  $SL_2(\mathbb{Z})$  acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by  $S : \tau \mapsto -1/\tau$  and  $T : \tau \mapsto \tau + 1$ . A *modular form of integral weight  $k$*  for  $SL_2(\mathbb{Z})$  is a holomorphic function  $f$  on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion  $f(\tau) = \sum_{n \geq 0} a_n q^n$  has no negative powers. It is a *cusp form* if  $a_0 = 0$ .

**Definition 17.3** (Jacobi theta functions and the Dedekind eta function). Set

$$\begin{aligned} \vartheta_2(\tau) &= \sum_{n \in \mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n \in \mathbb{Z}} q^{n^2/2}, & \eta(\tau) &= q^{1/24}(q; q)_\infty. \\ \vartheta_4(\tau) &= \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2/2}, \end{aligned}$$

These functions have weight 1/2 transformation laws with phases or permutations, rather than being integral-weight scalar modular forms for the full modular group. In particular,

$$\eta(\tau + 1) = e^{\pi i/12} \eta(\tau), \quad \eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau).$$



**Exercise 17.2** (Theta transformations and the Jacobi triple product). Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2}\vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2}\vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2}\vartheta_2(\tau).$$

Prove the  $T$ -transformation laws directly from the series. Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n \in \mathbb{Z}} z^n q^{n^2/2} = \prod_{m \geq 1} (1 - q^m)(1 + zq^{m-1/2})(1 + z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

### 17.3 Eisenstein series and arithmetic applications

**Definition 17.4** (Eisenstein series and the discriminant). Let  $\sigma_r(n) = \sum_{d|n} d^r$ . The normalized Eisenstein series of weights four and six are

$$E_4(\tau) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n, \quad E_6(\tau) = 1 - 504 \sum_{n \geq 1} \sigma_5(n) q^n.$$

The modular discriminant is the weight-twelve cusp form

$$\Delta(\tau) = \eta(\tau)^{24} = q \prod_{n \geq 1} (1 - q^n)^{24} = \frac{E_4(\tau)^3 - E_6(\tau)^2}{1728}.$$

**Exercise 17.3** (Eisenstein series and the  $E_8$  shell). Construct Eisenstein series by summing  $(m\tau + n)^{-k}$  over nonzero lattice points for even  $k > 2$ , prove their modular transformation law, and obtain the displayed Fourier expansions for  $E_4$  and  $E_6$ . For a positive-definite even unimodular lattice, first use consistency of the  $S$  and  $T$  laws to prove that its dimension  $d$  is divisible by eight, and then show that its theta series is a modular form of weight  $d/2$ . Construct the  $E_8$  lattice as

$$\{x \in \mathbb{Z}^8 : \sum_i x_i \equiv 0 \pmod{2}\} \cup \{x \in (\mathbb{Z} + \tfrac{1}{2})^8 : \sum_i x_i \equiv 0 \pmod{2}\}.$$

Enumerate its vectors of squared length two and obtain 240, matching the coefficient of  $q$  in  $E_4$ .

**Exercise 17.4** (Partition asymptotics from modular inversion). Put  $q = e^{-t}$  and use the modular transformation of  $\eta$  to prove

$$P(e^{-t}) \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \quad (t \downarrow 0).$$

Apply a saddle-point argument, or an appropriate Tauberian theorem with its hypotheses stated, to derive the Hardy–Ramanujan formula

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Interpret  $\log p(n)$  as the microcanonical entropy of the bosonic modes in the first exercise of this part. Distinguish this asymptotic result from an exact coefficient formula; see Hardy–Ramanujan, *Proceedings of the London Mathematical Society* **17** (1918).

## 18 Automorphic Harmonic Analysis and Arithmetic Langlands

### 18.1 Places, local fields, and adeles

**Definition 18.1** ( $p$ -adic completions). For a prime  $p$  and nonzero  $x \in \mathbb{Q}$ , write  $x = p^{v_p(x)}a/b$  with  $p \nmid ab$  and define

$$|x|_p = p^{-v_p(x)}, \quad |0|_p = 0.$$

This is a non-Archimedean absolute value:

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

The completion is the locally compact field  $\mathbb{Q}_p$ , and its compact open subring of integers is

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}.$$

The ordinary absolute value gives the Archimedean completion  $\mathbb{R}$ . A *number field* is a finite extension  $F/\mathbb{Q}$ . It has completions  $F_v$  indexed by its Archimedean and non-Archimedean places  $v$ ; at a finite place,  $\mathcal{O}_v \subset F_v$  is the ring of integers and the residue field has some cardinality  $q_v$ .

**Definition 18.2** (Adeles and ideles). The *adele ring* of  $\mathbb{Q}$  is the restricted product

$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p' \mathbb{Q}_p = \{(x_{\infty}, x_2, x_3, \dots) : x_p \in \mathbb{Z}_p \text{ for all but finitely many } p\}.$$

The restriction makes  $\mathbb{A}_{\mathbb{Q}}$  a locally compact ring. Its group of units is the *idele group*

$$\mathbb{A}_{\mathbb{Q}}^{\times} = \mathbb{R}^{\times} \times \prod_p' \mathbb{Q}_p^{\times},$$

where the finite components lie in  $\mathbb{Z}_p^{\times}$  for almost every  $p$ . For a number field  $F$ , define  $\mathbb{A}_F$  and  $\mathbb{A}_F^{\times}$  in the same way. The diagonal embedding of  $F^{\times}$  allows one to define the *idele class group*

$$C_F = F^{\times} \backslash \mathbb{A}_F^{\times}.$$

**Exercise 18.1** (The product formula). Compute the  $p$ -adic absolute values of  $x = 12/25$ . For arbitrary  $x \in \mathbb{Q}^{\times}$ , use unique factorization to prove

$$|x|_{\infty} \prod_p |x|_p = 1,$$

noting that only finitely many factors differ from one. Show that the diagonal image of  $x$  is an idele, and prove that the idelic norm

$$|(x_v)|_{\mathbb{A}} = |x_{\infty}|_{\infty} \prod_p |x_p|_p$$

is trivial on  $\mathbb{Q}^{\times}$ . Explain why it therefore descends to the idele class group.

**Definition 18.3** (Hecke characters and their  $L$ -functions). A *Hecke character* is a continuous character

$$\chi : F^{\times} \backslash \mathbb{A}_F^{\times} \longrightarrow \mathbb{C}^{\times}$$

with the usual unitary or quasicharacter growth condition. It factors into local characters  $\chi_v : F_v^\times \rightarrow \mathbb{C}^\times$ . At an unramified finite place its local Euler factor is

$$L_v(s, \chi_v) = \frac{1}{1 - \chi_v(\varpi_v)q_v^{-s}},$$

where  $\varpi_v$  is a uniformizer, characterized up to a unit by  $|\varpi_v|_v = q_v^{-1}$ . The global  $L$ -function is the product of these factors, supplemented at ramified and Archimedean places by the appropriate local factors. For the trivial character of  $\mathbb{Q}$ , its finite part is the Riemann zeta function.

**Definition 18.4** (Schwartz–Bruhat functions). A *Schwartz–Bruhat function* on  $\mathbb{A}_F$  is a finite linear combination of restricted tensor products

$$\Phi = \bigotimes_v \Phi_v,$$

where  $\Phi_v$  is an ordinary Schwartz function at each Archimedean place, is locally constant with compact support at each finite place, and equals the indicator of  $\mathcal{O}_v$  for almost every finite  $v$ .

## 18.2 Automorphic representations and the trace formula

**Definition 18.5** (Automorphic forms and representations). Let  $G$  be a reductive group over a number field  $F$ . An *automorphic form* is, roughly, a smooth function on

$$G(F) \backslash G(\mathbb{A}_F)$$

that is finite under a maximal compact subgroup and under the center of the Archimedean universal enveloping algebra, has moderate growth, and satisfies an appropriate central-character condition. It is *cuspidal* when its constant term

$$\int_{N_P(F) \backslash N_P(\mathbb{A}_F)} \varphi(ng) \, dn$$

vanishes for every proper parabolic subgroup  $P$  with unipotent radical  $N_P$ .

Right translation by  $G(\mathbb{A}_F)$  acts on automorphic forms. An *automorphic representation* is an irreducible constituent  $\pi$  of this representation.

**Exercise 18.2** (Poisson summation as an abelian trace formula). For  $f \in \mathcal{S}(\mathbb{R})$ , use the Fourier normalization

$$\hat{f}(m) = \int_{\mathbb{R}} f(x) e^{-2\pi i m x} \, dx$$

and define on  $L^2(\mathbb{R}/\mathbb{Z})$  the integral operator with kernel

$$K_f(x, y) = \sum_{n \in \mathbb{Z}} f(x - y + n).$$

Diagonalize it in the Fourier basis  $e^{2\pi i m x}$ . Compute its trace once as the sum of its eigenvalues and once as  $\int_0^1 K_f(x, x) \, dx$ , obtaining

$$\sum_{m \in \mathbb{Z}} \hat{f}(m) = \sum_{n \in \mathbb{Z}} f(n).$$

Identify the two sides as the spectral and geometric expansions. Explain which features of this calculation depend on commutativity and which survive for an operator defined by convolution on a nonabelian quotient.

### 18.3 Modular eigenforms as automorphic representations

**Definition 18.6** (Classical Hecke operators). Let  $f(\tau) = \sum_{n \geq 1} a_n q^n$  be a cusp form of weight  $k$  for  $SL_2(\mathbb{Z})$ . For a prime  $p$ , define

$$(T_p f)(\tau) = p^{k-1} f(p\tau) + \frac{1}{p} \sum_{a=0}^{p-1} f\left(\frac{\tau+a}{p}\right).$$

Then

$$T_p f = \sum_{n \geq 1} (a_{pn} + p^{k-1} a_{n/p}) q^n,$$

where  $a_{n/p} = 0$  if  $p \nmid n$ . For  $n \geq 1$ , define  $T_n$  on Fourier coefficients by

$$T_n f = \sum_{m \geq 1} \left( \sum_{d|(m,n)} d^{k-1} a_{mn/d^2} \right) q^m.$$

**Definition 18.7** (Normalized Hecke eigenform). A *normalized Hecke eigenform* has  $a_1 = 1$  and  $T_n f = a_n f$  for every  $n$ .

**Exercise 18.3** (Hecke recurrences and the Euler product). Derive the displayed Fourier-coefficient formula for  $T_p$ . If  $f$  is a normalized eigenform, compare coefficients to prove

$$a_{p^{r+1}} = a_p a_{p^r} - p^{k-1} a_{p^{r-1}} \quad (r \geq 1).$$

Derive  $T_m T_n = \sum_{d|(m,n)} d^{k-1} T_{mn/d^2}$  from the coefficient formula, and use it to prove  $a_{mn} = a_m a_n$  for  $(m, n) = 1$ , and then show

$$L(f, s) = \sum_{n \geq 1} \frac{a_n}{n^s} = \prod_p \frac{1}{1 - a_p p^{-s} + p^{k-1-2s}}$$

in a right half-plane. Factor the denominator as  $(1 - \alpha_p p^{-s})(1 - \beta_p p^{-s})$  and derive  $\alpha_p + \beta_p = a_p$  and  $\alpha_p \beta_p = p^{k-1}$ .

**Exercise 18.4** (The  $GL_2$  Hecke correspondence at  $p$ ). Let  $K_p = GL_2(\mathbb{Z}_p)$ . Show that index- $p$  sublattices of  $\mathbb{Z}_p^2$  are parametrized by  $\mathbb{P}^1(\mathbb{F}_p)$  and use this to obtain a decomposition of

$$K_p \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} K_p$$

into  $p+1$  one-sided cosets. Relate these representatives to the  $p+1$  terms in the formula for  $T_p$ . Explain why simultaneous Hecke diagonalization resembles a complete family of commuting observables, and why this analogy does not make Hecke symmetry generic in quantum mechanics.

### 18.4 Satake parameters and the dual group

**Definition 18.8** (Spherical Hecke algebra). Let  $F_v$  be a non-Archimedean local field, let  $G$  be split reductive over  $F_v$ , and choose a hyperspecial compact subgroup  $K_v = G(\mathcal{O}_v)$ . The *spherical Hecke algebra* is

$$\mathcal{H}_v(G, K_v) = C_c(K_v \backslash G(F_v) / K_v)$$

with convolution. An irreducible representation  $\pi_v$  with a nonzero  $K_v$ -fixed vector is *unramified*. A one-dimensional simultaneous eigenspace for  $\mathcal{H}_v(G, K_v)$  defines a *spherical Hecke character*.

## 18.5 Galois representations and matching $L$ -functions

**Definition 18.9** ( $\ell$ -adic representations and Frobenius). Let

$$G_F = \text{Gal}(\overline{F}/F)$$

be the absolute Galois group. An  $\ell$ -adic representation is a continuous homomorphism

$$\rho_\ell : G_F \longrightarrow GL_n(\overline{\mathbb{Q}}_\ell).$$

At a finite place  $v \nmid \ell$  where inertia acts trivially, the representation is *unramified*; a Frobenius element  $\text{Frob}_v$  is then defined up to conjugacy. Fix the Frobenius convention by declaring the local polynomial to be

$$P_v(X) = \det(1 - X\rho_\ell(\text{Frob}_v)).$$

The Galois local factor is  $P_v(q_v^{-s})^{-1}$ .

**Exercise 18.5** (Local zeta function of an elliptic curve). Let  $\alpha_p, \beta_p$  be roots of  $X^2 - a_pX + p$  and define a sequence  $N_r$  by

$$N_r = p^r + 1 - \alpha_p^r - \beta_p^r,$$

use  $\sum_{r \geq 1} z^r / r = -\log(1 - z)$  to prove

$$\exp\left(\sum_{r \geq 1} N_r \frac{T^r}{r}\right) = \frac{1 - a_p(E)T + pT^2}{(1 - T)(1 - pT)}.$$

For the curve  $E : y^2 = x^3 - x$  at  $p = 5$ , count  $E(\mathbb{F}_5)$ , determine  $a_5 = 5 + 1 - \#E(\mathbb{F}_5)$ , and compute the first three  $N_r$ . Compare the resulting degree-two factor with any weight-two Hecke eigenform having the same coefficient  $a_5$ .

## 18.6 The Langlands correspondence and its status

**Definition 18.10** (Langlands matching and functoriality). An automorphic representation  $\pi$  of  $GL_n(\mathbb{A}_F)$  and an  $n$ -dimensional  $\ell$ -adic representation  $\rho$  are *matched* at an unramified place  $v$  when their local polynomials agree:

$$L_v(s, \pi_v) = \det(1 - q_v^{-s}\rho(\text{Frob}_v))^{-1}.$$

They are *globally matched* when this holds at all but finitely many places. For reductive groups, an  $L$ -homomorphism

$${}^L H \longrightarrow {}^L G$$

defines *functorial transfer data* when it maps almost all unramified conjugacy classes compatibly and preserves their associated local  $L$ -factors.

## 19 CFT Characters and Integrable Models

### 19.1 Virasoro representations and character formulas

**Definition 19.1** (The Virasoro algebra and its characters). The *Virasoro algebra* has generators  $L_n$  for  $n \in \mathbb{Z}$  and a central element acting by  $c$ , with

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

A highest-weight vector  $|h\rangle$  satisfies  $L_0|h\rangle = h|h\rangle$  and  $L_n|h\rangle = 0$  for  $n > 0$ . For a positive-energy module  $H_a$  with finite-dimensional  $L_0$  eigenspaces, its *character* is

$$\chi_a(\tau) = \text{Tr}_{H_a} q^{L_0 - c/24}.$$

The shift by  $-c/24$  is the Casimir energy produced by mapping the plane to the cylinder.

**Exercise 19.1** (Descendants, partitions, and null states). Use the Poincaré–Birkhoff–Witt theorem to order Virasoro descendants as

$$L_{-\lambda_1} \cdots L_{-\lambda_r} |h\rangle, \quad \lambda_1 \geq \cdots \geq \lambda_r \geq 1.$$

Before quotienting by null vectors, deduce the Verma-module character

$$\chi_{c,h}^{\text{Verma}}(\tau) = \frac{q^{h-c/24}}{(q; q)_\infty}.$$

For the vacuum, impose  $L_{-1}|0\rangle = 0$  and find the resulting generic vacuum descendant product. Explain why additional null vectors subtract states and turn unrestricted partition numbers into the more structured  $q$ -series of minimal models.

## 19.2 Modular invariance and exact state counting

**Definition 19.2** (Rational characters and modular invariants). A finite family of positive-energy sectors  $H_a$  is a *rational modular character system* when its characters span a representation of the modular group,

$$\chi_a(-1/\tau) = \sum_b S_{ab} \chi_b(\tau), \quad \chi_a(\tau + 1) = \sum_b T_{ab} \chi_b(\tau).$$

A full torus partition function has the form

$$Z(\tau, \bar{\tau}) = \sum_{a,b} M_{ab} \chi_a(\tau) \overline{\chi_b(\tau)}$$

with nonnegative integer multiplicities. It is modular invariant when  $M$  commutes with  $S$  and  $T$ .

**Exercise 19.2** (Exact state counting in the Ising CFT). For the Ising theory with  $c = 1/2$  and primary weights  $0, 1/2, 1/16$ , take

$$\begin{aligned} \chi_0 &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/2} &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/16} &= \frac{1}{\sqrt{2}} \sqrt{\frac{\vartheta_2}{\eta}}. \end{aligned}$$

Expand each as  $q^{h-c/24}$  times a power series and interpret its first coefficients as exact descendant degeneracies. Derive

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \quad T_{aa} = e^{2\pi i(h_a - c/24)}.$$

Verify the modular invariance of  $|\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2$  and recover the Ising fusion rules from the Verlinde formula. Relate these towers to the critical Ising spectroscopy exercise in Quantum Topology.

**Exercise 19.3** (The Cardy density of states). Let a modular-invariant unitary CFT have discrete spectrum, unique vacuum, and left and right central charges  $c, \bar{c}$ . Set  $\tau = i\beta/(2\pi)$  and use  $S$ -invariance to show that the high-temperature partition function is controlled by the vacuum at inverse temperature  $4\pi^2/\beta$ . A saddle-point inverse Laplace transform then gives, for large left and right excitation numbers,

$$\log \rho(N, \bar{N}) \sim 2\pi \sqrt{\frac{cN}{6}} + 2\pi \sqrt{\frac{\bar{c}\bar{N}}{6}}.$$

State how  $c$  is replaced by  $c_{\text{eff}} = c - 24h_{\text{min}}$  in a nonunitary theory. For a nonrotating state with  $c = \bar{c}$  and  $N = \bar{N}$ , recover the thermal entropy used in the BTZ exercise and hence reproduce the horizon entropy from boundary-state counting; see Cardy, *Nuclear Physics B* **270** (1986).

### 19.3 Integrable lattice models

**Definition 19.3** (Transfer matrices and integrability). For a two-dimensional lattice model on a cylinder, a *row transfer matrix*  $T_N(u)$  propagates one width- $N$  row to the next and depends on a spectral parameter  $u$ . The model is *integrable* when it admits a nontrivial commuting family

$$[T_N(u), T_N(v)] = 0 \quad \text{for every } u, v,$$

usually obtained from a Yang–Baxter or star–triangle relation. Commuting transfer matrices permit simultaneous spectral analysis, but integrability alone does not guarantee that every finite-size degeneracy has a simple closed formula.

**Exercise 19.4** (Hard hexagons and exact finite-volume counting). In the hard-hexagon model, occupied vertices of a triangular lattice may not be adjacent. On an  $N \times M$  torus define

$$Z_{N,M}(z) = \sum_{I \text{ independent}} z^{|I|}.$$

Index a row transfer matrix by cyclic binary rows and set its matrix element to zero when two consecutive rows violate nearest-neighbor exclusion, and otherwise to  $z^{(|a|+|b|)/2}$ . Prove  $Z_{N,M}(z) = \text{Tr}(T_N(z)^M)$  and compute the polynomial for several small sizes. Show that the coefficients are exact state counts, whereas the largest eigenvalue controls the thermodynamic free energy.

Verify the algebraic identity

$$z_c = \frac{11 + 5\sqrt{5}}{2} = \left( \frac{1 + \sqrt{5}}{2} \right)^5,$$

and, for the finite matrices already constructed, evaluate the largest eigenvalue and its first two  $z$ -derivatives near  $z_c$ . Plot their change with  $N$  and explain why finite matrices remain analytic even when their large-size limit may develop a singularity.

## 20 Root Systems, DAHA, and Macdonald Operators

### 20.1 Root systems, scattering, and Hecke algebras

**Definition 20.1** (Root systems and affine Weyl groups). Let  $V$  be a finite-dimensional Euclidean space. A finite set  $R \subset V \setminus \{0\}$  is a *reduced crystallographic root system* if it spans  $V$ , is preserved

by the reflections

$$s_\alpha(v) = v - \langle v, \alpha^\vee \rangle \alpha, \quad \alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle},$$

has  $R \cap \mathbb{R}\alpha = \{\alpha, -\alpha\}$ , and satisfies  $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$  for all  $\alpha, \beta \in R$ . The group  $W = \langle s_\alpha : \alpha \in R \rangle$  is its *Weyl group*, and the *coroot lattice* is  $Q^\vee = \text{span}_{\mathbb{Z}}\{\alpha^\vee : \alpha \in R\}$ .

The hyperplanes

$$H_{\alpha, m} = \{v \in V : \langle v, \alpha \rangle = m\}, \quad \alpha \in R, \quad m \in \mathbb{Z},$$

are the *affine reflection hyperplanes*. Reflections in them generate the *affine Weyl group*  $W_{\text{aff}} = Q^\vee \rtimes W$ . Its connected chambers are called *alcoves*.

**Definition 20.2** (Hecke deformation of reflections). Let  $(W, \{s_i\})$  be a Coxeter system, and let  $m_{ij}$  be the order of  $s_i s_j$ . Over  $\mathbb{C}(t^{1/2})$ , its *Hecke algebra*  $\mathcal{H}_t(W)$  is generated by  $T_i$  with the same braid relations as the  $s_i$  and with

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

At  $t = 1$  this becomes  $T_i^2 = 1$ , so the algebra specializes to the group algebra of  $W$ . For generic  $t$ , a reflection has been replaced by an operator with two allowed eigenvalues. This algebraic fact matches a two-channel exchange matrix only in models whose scattering actually obeys the displayed quadratic relation; it is not a claim about generic particle collisions.

**Exercise 20.1** (From factorized exchange to a Hecke algebra). Let  $R_i$  be operators describing adjacent exchanges such that distant exchanges commute, three-particle consistency gives  $R_i R_{i+1} R_i = R_{i+1} R_i R_{i+1}$ , and every  $R_i$  has the same two nonzero eigenvalues  $a$  and  $b$ . Rescale the  $R_i$  so that the eigenvalues become  $t^{1/2}$  and  $-t^{-1/2}$ , when this is possible. Prove that the resulting operators define a representation of  $\mathcal{H}_t(S_n)$ . Explain exactly which assumption would fail for a two-body scattering matrix with three distinct eigenchannels.

## 20.2 Quantum groups and the Yang–Baxter equation

**Definition 20.3** (Quantized enveloping algebras and  $R$ -matrices). For generic  $q$ , the Hopf algebra  $U_q(\mathfrak{sl}_2)$  is generated by  $E, F, K, K^{-1}$  with

$$KEK^{-1} = q^2 E, \quad KFK^{-1} = q^{-2} F, \quad [E, F] = \frac{K - K^{-1}}{q - q^{-1}},$$

and coproduct

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

It is paired algebraically with the dense coordinate Hopf algebra of  $SU_q(2)$  introduced in Operator Space Theory: one side acts on states, while the other is made from their matrix coefficients.

A quasitriangular completion of a Hopf algebra has an invertible *universal  $R$ -matrix*  $\mathcal{R}$  satisfying

$$\mathcal{R}\Delta(x) = \Delta^{\text{op}}(x)\mathcal{R}, \quad (\Delta \otimes \text{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\text{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

On finite-dimensional modules, the braid operator  $\check{R} = P(\pi \otimes \pi)(\mathcal{R})$ , with  $P$  exchanging tensor factors, satisfies

$$\check{R}_{12}\check{R}_{23}\check{R}_{12} = \check{R}_{23}\check{R}_{12}\check{R}_{23}.$$

When  $\check{R}$  has exactly two eigenvalues, a normalization may turn its minimal polynomial into the Hecke quadratic relation. This implication requires the two-eigenvalue hypothesis; a general quantum-group  $R$ -matrix need not factor through a Hecke algebra.



**Exercise 20.2** (Coproduct consistency and the Yang–Baxter equation). Check that the displayed coproduct preserves every defining relation of  $U_q(\mathfrak{sl}_2)$  and is coassociative. Derive the quantum Yang–Baxter equation

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

from the two coproduct identities, and then derive the braid relation for  $\check{R}$ . Interpret the two sides as the two factorizations of a three-body scattering process. Prove that a two-eigenvalue  $\check{R}$  can be rescaled to a Hecke generator exactly under the compatibility condition found in the preceding exercise.

**Exercise 20.3** (A direct optical Yang–Baxter test). Consider the unitary polarization transformations

$$A(\theta) = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \quad B(\theta) = \begin{pmatrix} \cos \theta & -i \sin \theta \\ -i \sin \theta & \cos \theta \end{pmatrix}.$$

Multiply the matrices and prove that, away from singular boundary cases,

$$A(\theta_1)B(\theta_2)A(\theta_3) = B(\theta_3)A(\theta_2)B(\theta_1)$$

exactly when

$$\tan \theta_2 = \frac{\sin(\theta_1 + \theta_3)}{\cos(\theta_1 - \theta_3)}.$$

For  $\theta_1 = 56^\circ$  and  $\theta_3 = 23^\circ$ , obtain  $\theta_2 \approx 49.49^\circ$ . For an arbitrary input qubit  $|\psi\rangle$ , show that the output-state overlap has absolute value one when the relation holds and is generally smaller away from it. Compare these predictions with the wave-plate experiment of Zheng et al., *Journal of the Optical Society of America B* **30** (2013). State clearly that this experiment tests a two-dimensional Temperley–Lieb reduction of the Yang–Baxter equation, not the full representation theory of a quantum group or DAHA.

### 20.3 Quantum spherical functions and qKZ transport

**Definition 20.4** (Quantum spherical functions). Let  $H = U_q(\mathfrak{g})$  and let  $B \subseteq H$  be a right coideal subalgebra, so  $\Delta(B) \subseteq B \otimes H$ . It plays the role of the stabilizer algebra of a quantum homogeneous space. If an  $H$ -module  $V$  contains a vector  $v$  satisfying

$$bv = \varepsilon(b)v \quad (b \in B),$$

then the matrix coefficient

$$\varphi_V(x) = \langle v, \pi_V(x)v \rangle$$

is a *quantum zonal spherical function*. It belongs to the coordinate Hopf dual of  $H$  and is invariant under the left and right quantum stabilizer actions.

**Exercise 20.4** (Flatness, qKZ transport, and affine Hecke monodromy). Let a vector-valued function satisfy the adjacent exchange rule

$$\Psi(\dots, z_i, z_{i+1}, \dots) = \check{R}_i(z_i/z_{i+1})\Psi(\dots, z_{i+1}, z_i, \dots).$$

Show that consistency of the two routes reversing three adjacent variables is exactly the spectral Yang–Baxter equation. Add a periodic rule that returns  $z_1$  at the end as  $qz_1$ , and explain why translations enlarge the finite braid action to an affine one. If the  $R$ -matrices satisfy a Hecke quadratic, prove that the exchange monodromy factors through an affine Hecke algebra. Write the resulting reflection and translation operators in the same generators used in the polynomial representation below.

## 20.4 DAHA and Macdonald spectral theory

**Definition 20.5** (The  $GL_n$  form of type- $A$  DAHA). Let

$$\mathcal{P}_n = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_n^{\pm 1}],$$

let  $X_j$  act by multiplication by  $x_j$ , and let  $s_i$  exchange  $x_i$  and  $x_{i+1}$ . Define the Demazure–Lusztig operators

$$T_i = t^{1/2}s_i + (t^{1/2} - t^{-1/2})\frac{s_i - 1}{x_i/x_{i+1} - 1}, \quad 1 \leq i < n,$$

and the affine cyclic shift

$$(\pi f)(x_1, \dots, x_n) = f(x_2, \dots, x_n, qx_1).$$

The quotients in the formula for  $T_i$  are Laurent polynomials because  $s_i f - f$  vanishes when  $x_i = x_{i+1}$ . For generic  $q, t$ , the operator algebra generated by  $X_j^{\pm 1}$ ,  $T_i$ , and  $\pi^{\pm 1}$  is called the polynomial realization of the  $GL_n$  form of the type- $A_{n-1}$  *double affine Hecke algebra*, or DAHA. Passing to the semisimple  $A_{n-1}$  version removes the common center-of-mass coordinate.

The first affine lattice is visible in the commuting multiplication operators  $X_j$ . Words in the  $T_i$  and  $\pi$  produce a second commuting family  $Y_j$  of  $q$ -difference-reflection operators. The Hecke generators mediate between the two. This  $X$ – $Y$  symmetry is the meaning of “double affine.”

**Definition 20.6** (Macdonald operators and polynomials). Let  $T_{q,x_i}$  shift one variable,

$$(T_{q,x_i} f)(x_1, \dots, x_n) = f(x_1, \dots, qx_i, \dots, x_n).$$

The first type- $A$  *Macdonald difference operator* is

$$D_{q,t} = \sum_{i=1}^n \left( \prod_{j \neq i} \frac{tx_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

Although its summands are rational, their sum preserves symmetric Laurent polynomials. In DAHA it comes from a Weyl-invariant symmetric function of the commuting  $Y$ -operators; higher symmetric functions give a full commuting family.

For generic  $q, t$ , a *Macdonald polynomial*  $P_\lambda(x; q, t)$  is a symmetric polynomial triangular in the monomial-symmetric basis and satisfying

$$D_{q,t} P_\lambda = \left( \sum_{i=1}^n q^{\lambda_i} t^{n-i} \right) P_\lambda, \quad \lambda_1 \geq \dots \geq \lambda_n \geq 0.$$

**Exercise 20.5** (The Calogero–Sutherland limit). Write  $x_i = e^{z_i}$ , set  $q = e^\epsilon$  and  $t = e^{g\epsilon}$ , and expand  $D_{q,t}$  through order  $\epsilon^2$ . After subtracting its scalar value on 1 and removing the center-of-mass drift, show that the nontrivial differential operator is proportional to

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i < j} \coth\left(\frac{z_i - z_j}{2}\right) (\partial_{z_i} - \partial_{z_j}).$$

On the circle, put  $z_i = i\theta_i$  and conjugate by

$$\Psi_0(\theta) = \prod_{i < j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g.$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = - \sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i < j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

For the degree-zero, degree-one, and two-particle degree-two eigenfunctions computed above, take the same limit explicitly and identify the resulting polynomials and Sutherland eigenvalues.

## 20.5 Root-system integrals and elliptic Yang–Baxter identities

**Definition 20.7** (Askey–Wilson and Gustafson weights). For a list of parameters write  $(a_1, \dots, a_r; q)_\infty = \prod_s (a_s; q)_\infty$ , and set  $f(z^{\pm 1}) = f(z)f(z^{-1})$  and  $f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon, \delta = \pm 1} f(z_i^\epsilon z_j^\delta)$ . For  $|q|, |t|, |a_r| < 1$ , the *Askey–Wilson weight* is

$$\Delta_{AW}(z) = \frac{(z^{\pm 2}; q)_\infty}{\prod_{r=1}^4 (a_r z^{\pm 1}; q)_\infty},$$

and the  $BC_n$  *Gustafson weight* is

$$\begin{aligned} \Delta_{BC_n}(z) &= \prod_{i=1}^n \frac{(z_i^{\pm 2}; q)_\infty}{\prod_{r=1}^4 (a_r z_i^{\pm 1}; q)_\infty} \\ &\quad \times \prod_{1 \leq i < j \leq n} \frac{(z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}. \end{aligned}$$

A *Gustafson integral* is a torus integral of such a root-system weight with the normalized measure  $(2^n n!)^{-1} \prod_i dz_i / (2\pi i z_i)$ .

**Exercise 20.6** (Root systems in the Gustafson integral). Prove directly that  $\Delta_{BC_n}$  is invariant under permutations and independent inversions of the  $z_i$ . Identify its factors with the roots  $\pm e_i$ ,  $\pm 2e_i$ , and  $\pm e_i \pm e_j$ . Set  $n = 1$  and recover the Askey–Wilson weight and the order-two Weyl normalization. For an integer  $g \geq 0$ , set  $t = q^g$ , cancel the quotient of  $q$ -products into a finite product, and take  $q \rightarrow 1^-$  after writing  $z_i = e^{i\theta_i}$ . Identify the resulting power-law pair repulsion and its  $BC_n$  reflection symmetry.

**Definition 20.8** (Elliptic weights and star–triangle data). For  $|p| < 1$ , define

$$\theta(z; p) = (z; p)_\infty (p/z; p)_\infty, \quad (a; q, p)_m = \prod_{j=0}^{m-1} \theta(aq^j; p),$$

and

$$\Gamma(z; p, q) = \prod_{j, k \geq 0} \frac{1 - p^{j+1} q^{k+1} / z}{1 - p^j q^k z}.$$

A family of edge weights  $W_u(x, y)$  and one-spin weights  $S(x)$  is *star–triangle data* when

$$\int S(x) W_u(a, x) W_v(b, x) W_w(c, x) dx = R(u, v, w) W_{u'}(b, c) W_{v'}(a, c) W_{w'}(a, b)$$

for its stated parameter constraint and contour. Elliptic star–triangle data are those for which the weights are finite products and ratios of  $\Gamma(\cdot; p, q)$ .

In additive notation define the *elliptic number*

$$[x]_{h,\tau} = \frac{\vartheta_{11}(hx, \tau)}{\vartheta_{11}(h, \tau)}.$$

A terminating series whose successive-term ratio is a balanced product of elliptic numbers is a *modular hypergeometric series*; termination makes it a finite algebraic expression in theta values.

**Exercise 20.7** (The rational–trigonometric–elliptic hierarchy). Use the Jacobi triple product to prove

$$[x]_{h,\tau} \longrightarrow \frac{\sin(\pi hx)}{\sin(\pi h)} \quad (\text{Im } \tau \rightarrow \infty), \quad \frac{\sin(\pi hx)}{\sin(\pi h)} \longrightarrow x \quad (h \rightarrow 0).$$

Prove  $\theta(z; 0) = 1 - z$  and  $(a; q, p)_m \rightarrow (a; q)_m$  as  $p \rightarrow 0$ . Hence organize the three levels as

$$\begin{array}{c} \text{elliptic hypergeometric identities and elliptic Yang–Baxter data} \\ \downarrow p \rightarrow 0 \\ \text{basic hypergeometric identities, Gustafson integrals, and trigonometric } R\text{-matrices} \\ \downarrow q \rightarrow 1 \\ \text{ordinary hypergeometric identities, Selberg integrals, and rational data.} \end{array}$$

Explain why the integral/star–triangle branch and the  $6j$ -symbol/recoupling branch are complementary realizations of Yang–Baxter consistency rather than two names for the same construction. Use the star–triangle identity twice on a three-row lattice strip to derive the equality of the two local reorderings that underlies commuting transfer matrices. Finally relate the trigonometric middle level to the Macdonald–Ruijsenaars operators above.

## 21 Cyclotomic and Aperiodic Structure

### 21.1 Algebraic numbers, roots of unity, and cyclotomic fields

**Definition 21.1** (Algebraic numbers and cyclotomic fields). A complex number  $\alpha$  is *algebraic* over  $\mathbb{Q}$  if it is a root of a nonzero polynomial in  $\mathbb{Q}[x]$ , and an *algebraic integer* if it is a root of a monic polynomial in  $\mathbb{Z}[x]$ . Its *minimal polynomial* is the monic irreducible polynomial in  $\mathbb{Q}[x]$  that it satisfies. A *Galois conjugate* is another root of that polynomial.

For  $n \geq 1$ , let  $\zeta_n = e^{2\pi i/n}$ . The  $n$ th *cyclotomic field* is  $\mathbb{Q}(\zeta_n)$ , and the  $n$ th *cyclotomic polynomial*  $\Phi_n(x)$  is the minimal polynomial of  $\zeta_n$ . It satisfies

$$x^n - 1 = \prod_{d|n} \Phi_d(x), \quad [\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n).$$

**Exercise 21.1** (Roots of unity in topological data). Prove the cyclotomic factorization recursively and show that  $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$  by  $\zeta_n \mapsto \zeta_n^a$ . For the  $SU(2)_k$  theory from Quantum Topology, put  $\xi = e^{\pi i/(k+2)}$  and prove that the quantum dimensions

$$d_j = \frac{\sin((j+1)\pi/(k+2))}{\sin(\pi/(k+2))}, \quad 0 \leq j \leq k,$$

belong to the real subfield of  $\mathbb{Q}(\xi)$  and are algebraic integers. At  $k = 3$ , identify the nontrivial dimension  $2\cos(\pi/5) = (1 + \sqrt{5})/2$  and recover its polynomial  $x^2 - x - 1$ . Show directly that the topological twists in  $SU(2)_k$  are roots of unity up to the common framing phase. Explain what this proves for this family without asserting that arbitrary algebraic numbers are physical topological data.

## 21.2 Diophantine structure and substitutions

**Definition 21.2** (Continued fractions and Pisot numbers). For irrational  $\alpha$ , its simple continued fraction is

$$\alpha = [a_0; a_1, a_2, \dots] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}},$$

with convergents  $p_n/q_n = [a_0; \dots, a_n]$ . A real algebraic integer  $\beta > 1$  is a *Pisot number* if every other Galois conjugate has absolute value less than one.

**Definition 21.3** (Substitutions and the Fibonacci word). A *substitution* on a finite alphabet replaces each letter by a finite word and extends by concatenation. Its incidence matrix records how many copies of each letter occur in every substituted letter. The Fibonacci substitution is

$$\sigma(a) = ab, \quad \sigma(b) = a, \quad M_\sigma = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Its one-sided fixed word is the limit of  $a, ab, aba, abaab, \dots$

**Exercise 21.2** (Frequencies and aperiodicity of the Fibonacci chain). Diagonalize  $M_\sigma$ , prove that its Perron–Frobenius eigenvalue is  $\phi$ , and derive letter frequencies

$$\text{freq}(a) = \frac{1}{\phi}, \quad \text{freq}(b) = \frac{1}{\phi^2}.$$

Prove that every finite subword recurs with bounded gaps. Show that the infinite word cannot be periodic because a periodic word has rational letter frequencies. If tiles of lengths  $\phi$  and 1 replace  $a$  and  $b$ , show that substitution followed by rescaling by  $\phi$  reproduces the tiling. Separate exact inflation symmetry from ordinary translation symmetry.

## 21.3 Cut-and-project order and diffraction

**Definition 21.4** (Cut-and-project sets). Let  $\mathcal{L}$  be a lattice in  $\mathbb{R}^d \times \mathbb{R}^m$ , with projections  $\pi_\parallel$  and  $\pi_\perp$  onto physical and internal space. A *cut-and-project scheme* requires  $\pi_\parallel$  to be injective on  $\mathcal{L}$  and  $\pi_\perp(\mathcal{L})$  to be dense. For a compact window  $W \subset \mathbb{R}^m$  with nonempty interior, define the *model set*

$$\Lambda(W) = \{\pi_\parallel(x) : x \in \mathcal{L}, \pi_\perp(x) \in W\}.$$

It is *regular* when the boundary of  $W$  has measure zero. The higher-dimensional lattice supplies long-range order, while the irrational orientation of physical space generally destroys translation periodicity.

**Exercise 21.3** (The Fibonacci chain by cut and project). Let  $\phi' = -1/\phi$  and embed  $\mathbb{Z}^2$  by

$$(m, n) \mapsto (m + n\phi, m + n\phi') \in \mathbb{R}_\parallel \times \mathbb{R}_\perp.$$

Prove the injectivity and density conditions in the definition. Choose a generic half-open interval window of length  $1 + 1/\phi$ , with boundary disjoint from the projected lattice, and order the resulting physical points. Determine the two possible nearest-neighbor spacings and show, after an overall rescaling and a possible exchange of letters, that their sequence is a Fibonacci tiling. Track how translating the window changes the local pattern without changing its set of finite patches. Interpret the internal coordinate as an arithmetic phase invisible in physical space.

**Definition 21.5** (Autocorrelation and diffraction). For a uniformly discrete point set  $\Lambda \subset \mathbb{R}^d$ , its Dirac comb is  $\omega = \sum_{x \in \Lambda} \delta_x$ . When the vague limit exists, its *autocorrelation* is

$$\gamma = \lim_{R \rightarrow \infty} \frac{\omega|_{B_R} * \widetilde{\omega|_{B_R}}}{\text{vol}(B_R)}, \quad \tilde{\mu}(f) = \overline{\mu(\tilde{f})}, \quad \tilde{f}(x) = \overline{f(-x)}.$$

The positive measure  $\hat{\gamma}$  is its *diffraction measure*; its atoms are Bragg peaks.

**Exercise 21.4** (Sharp diffraction without periodicity). For a finite sample  $\Lambda_R$ , show that the kinematic scattering intensity is

$$I_R(k) = \left| \sum_{x \in \Lambda_R} e^{-ik \cdot x} \right|^2$$

and relate its volume-normalized limit to  $\hat{\gamma}$ . Recover the ordinary reciprocal lattice for a periodic crystal. For Fibonacci approximants obtained from the first  $F_n$  tiles, compute  $I_R(k)$  on a grid of wave numbers and track the largest peaks as  $n$  increases. Express the observed peak locations as integer combinations of 1 and  $1/\phi$ , and verify directly that no finite reciprocal lattice contains all of them.

Compare this mechanism with the sharp noncrystallographic diffraction reported in Shechtman et al., *Physical Review Letters* **53** (1984). State what diffraction establishes—long-range coherent order and its symmetry—and what it does not establish without a structural model.

## 21.4 Quasicrystal spectra and gap labelling

**Definition 21.6** (The Fibonacci Hamiltonian and integrated density of states). Let  $\alpha = 1/\phi$  and

$$v_\theta(n) = \mathbf{1}_{[1-\alpha, 1)}(\theta + n\alpha \bmod 1).$$

The *Fibonacci Hamiltonian* on  $\ell^2(\mathbb{Z})$  is

$$(H_\theta \psi)_n = \psi_{n+1} + \psi_{n-1} + \lambda v_\theta(n) \psi_n.$$

For the uniquely ergodic rotation of  $\theta$ , its *integrated density of states* is

$$N(E) = \int_0^1 \langle \delta_0, \mathbf{1}_{(-\infty, E]}(H_\theta) \delta_0 \rangle d\theta.$$

It is the limiting number of states below  $E$  per lattice site.

Let  $\Omega$  be the closure of the shift orbit of a Fibonacci sequence in  $\{0, 1\}^{\mathbb{Z}}$ , and let  $S : \Omega \rightarrow \Omega$  be the shift. The associated observable algebra is the crossed product  $\mathcal{A}_\alpha = C(\Omega) \rtimes_S \mathbb{Z}$ , generated by  $C(\Omega)$  and a unitary  $U$  satisfying  $UfU^* = f \circ S^{-1}$ . The unique invariant measure  $\mu$  gives the canonical trace

$$\tau \left( \sum_n f_n U^n \right) = \int_\Omega f_0 d\mu.$$

The Hamiltonians above are the covariant representations of one element of this algebra.

**Exercise 21.5** (Diophantine trace labels). Prove that  $N(E)$  is nondecreasing and is constant on every open spectral gap. Approximate  $\alpha$  by Fibonacci convergents and interpret the finite periodic approximants as large-unit-cell crystals. Define the Fibonacci label group

$$\mathcal{L}_\alpha = (\mathbb{Z} + \alpha\mathbb{Z}) \cap [0, 1].$$

Use the substitution recurrence to show that the measures of one-letter and two-letter cylinder sets belong to  $\mathbb{Z} + \alpha\mathbb{Z}$ . Prove the same for every finite disjoint union of cylinder sets. For the corresponding projections in  $C(\Omega) \subset \mathcal{A}_\alpha$ , compute their traces and identify them with elements of  $\mathcal{L}_\alpha$ . Compare these values with the plateaus of the finite-approximant integrated density of states, without asserting that every allowed label is an open spectral gap.